

CKAA: Cross-subspace Knowledge Alignment and Aggregation for Robust Continual Learning

Lingfeng He, De Cheng, Zhiheng Ma, Huaijie Wang, Dingwen Zhang, Nannan Wang, *Senior Member, IEEE*, Xinbo Gao, *Fellow, IEEE*

Abstract—Continual Learning (CL) empowers AI models to continuously learn from sequential task streams. Recently, parameter-efficient fine-tuning (PEFT)-based CL methods have garnered increasing attention due to their superior performance. They typically allocate a unique sub-module for learning each task, with a task recognizer to select the appropriate sub-modules for testing images. However, due to the feature subspace misalignment from independently trained sub-modules, these methods tend to produce ambiguous decisions under misleading task-ids. To address this, we propose Cross-subspace Knowledge Alignment and Aggregation (CKAA), a novel framework that enhances model robustness against misleading task-ids through two key innovations: (1) Dual-level Knowledge Alignment (DKA): By aligning intra-class feature distributions across different subspaces and learning a robust global classifier through a feature simulation process, DKA enables the model to distinguish features from both correct and incorrect subspaces during training. (2) Task-Confidence-guided Mixture of Adapters (TC-MoA): A robust inference scheme that adaptively aggregates task-specific knowledge from relevant sub-modules based on task-confidence scores, avoiding overconfidence in misleading task-id predictions. Extensive experiments demonstrate that CKAA outperforms existing PEFT-based CL methods. Code is available at https://github.com/HHHLF/CKAA_for_CIL.

Index Terms—Continual learning, parameter-efficient, cross-subspace, knowledge alignment

I. INTRODUCTION

Continual Learning (CL) requires the model to continuously learn from sequentially arriving data streams, which is essential for AI models to adapt to the ever-evolving real-world scenarios [1]–[3]. CL presents a major challenge: as the model is updated to accommodate new categories, they often suffer from catastrophic forgetting [4]–[6], wherein knowledge of old

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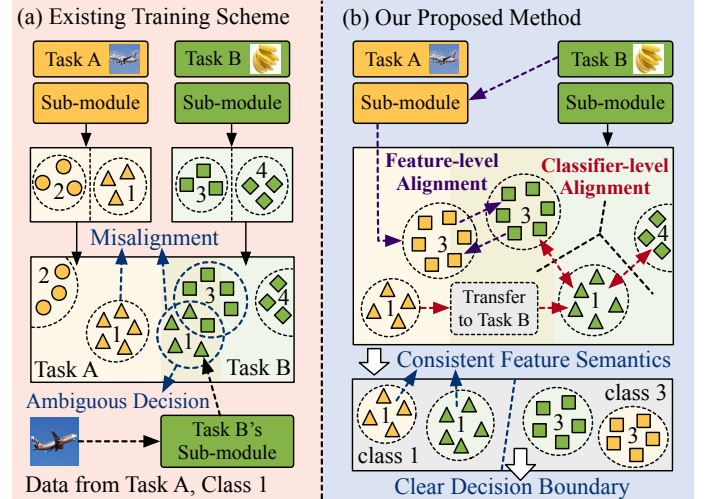


Fig. 1: Illustration of our motivation. Existing methods (a) encounter ambiguous decisions for features projected into the incorrect subspace due to the misalignment between subspaces. Our CKAA (b) addresses this by semantically aligning feature distributions across subspaces and learning robust decision boundaries, effectively distinguishing wrongly projected features.

classes is diminished. There are two common scenarios in CL: Task-Incremental Learning (TIL) [7] and Class-Incremental Learning (CIL) [8]. We investigate the more challenging CIL setting with unknown task-ids during inference [9].

With the advancement of pre-trained models, Parameter Efficient Fine-Tuning (PEFT)-based methods [10]–[12] exhibit superior performance in vision tasks. These methods keep the pre-trained backbone frozen while inserting lightweight modules for training [13]–[16] rather than learning a network from scratch [17]–[19]. PEFT-based methods have been investigated in the field of CL [20]–[22] and have proven highly effective in mitigating catastrophic forgetting by maintaining the strong generalizability of pre-trained models throughout sequential training.

Several methods [20], [23]–[25] are proposed to harness the potential of PEFT methods in the field of CIL. They typically allocate a dedicated sub-module for each task, effectively uncovering the unique knowledge for each task and ensuring model plasticity for new tasks. Since task-ids are unavailable during CIL inference, a task recognizer is designed to identify task-ids and select the appropriate sub-module for testing images. Unfortunately, task-id misidentification is inevitable, leading to features being erroneously projected into incorrect subspaces, which in turn results in ambiguous and unreliable

CKAA：面向鲁棒持续学习的跨子空间知识对齐与聚合

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摘要——持续学习（CL）使人工智能模型能够从序列任务流中持续学习。近年来，基于参数高效微调（PEFT）的持续学习方法因其卓越的性能而受到越来越多的关注。这类方法通常为每个任务分配一个独立的子模块进行学习，并配备一个任务识别器，用于为待测图像选择合适的子模块。然而，由于独立训练的子模块导致的特征子空间对齐问题，这些方法在面对误导性任务ID时往往会产生模糊的决策结果。为解决这一问题，我们提出了一种名为“跨子空间知识对齐与聚合”（CKAA）的新框架；该框架通过两项关键创新来增强模型对误导性任务ID的鲁棒性：(1) 双层知识对齐（DKA）：通过对不同子空间内的类内特征分布进行对齐，并借助特征模拟过程学习一个鲁棒的全局分类器，DKA使模型在训练过程中能够区分来自正确与错误子空间的特征。(2) 任务置信度引导的适配器混合模型（TCMoA）：一种鲁棒的推理方案，该方案基于任务置信度分数自适应地聚合来自相关子模块的任务特定知识，从而避免在误导性任务ID预测中产生过高的置信度。大量实验表明，CKAA的性能优于现有的基于PEFT的CL方法。代码可获取自https://github.com/HHHLF/CKAA_适用于_CIL。

索引词——持续学习、参数高效、交叉子空间、知识对齐

1. 引言

持续学习（CL）要求模型持续从依次到达的数据流中进行学习，这对于人工智能模型适应不断演变的现实世界场景至关重要[1]–[3]。CL面临着项重大挑战：当模型为适配新类别而进行更新时，它们往往会出现“灾难性遗忘”现象[4]–[6]，即对旧知识的遗忘。

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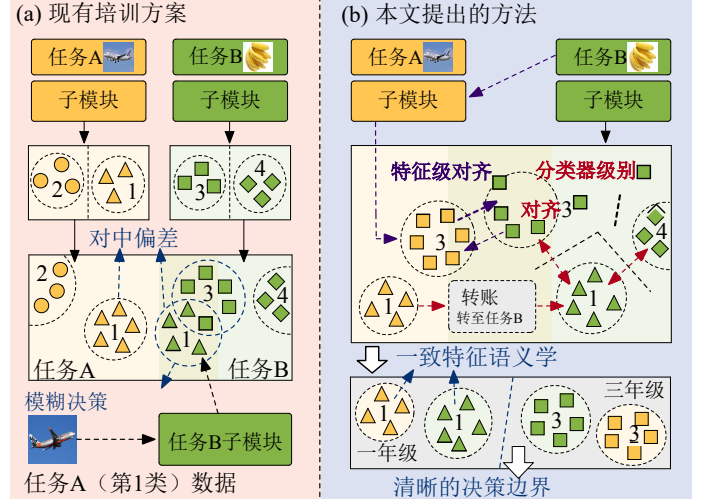


图 1：本文研究动机的示意图。现有方法(a)由于子空间之间存在错位问题，当特征被投影到错误的子空间时，会面临模糊决策问题；而我们的 CKAA (b)通过对不同子空间中的特征分布进行语义对齐并学习鲁棒的决策边界，从而有效区分出被错误投影的特征。

类别数量减少。在连续学习（CL）中存在两种常见场景：任务增量学习（TIL）[7]与类别增量学习（CIL）[8]。我们针对在推理阶段任务ID未知的更具挑战性的CIL场景展开研究[9]。

随着预训练模型的不断发展，基于参数高效微调（PEFT）的方法[10]–[12]在视觉任务中展现出卓越的性能。这些方法在保持预训练主干网络冻结状态的同时，插入轻量级模块进行训练[13]–[16]，而非从头开始学习整个网络[17]–[19]。基于PEFT的方法已在持续学习（CL）领域得到研究[20]–[22]，并已被证明在缓解灾难性遗忘问题方面非常有效——即在序列训练过程中始终维持预训练模型强大的泛化能力。

多种方法[20]，[23]–[25]旨在充分发挥PEFT方法在CIL领域中的潜力。这些方法通常为每个任务分配一个专用子模块，从而有效揭示每个任务的独特知识，并确保模型对新任务具备适应性。由于在CIL推理过程中无法获取任务ID，因此设计了一个任务识别器来识别任务ID并为待测图像选择合适的子模块。然而，任务ID的误识别不可避免，这会导致特征被错误地投影到不正确的子空间

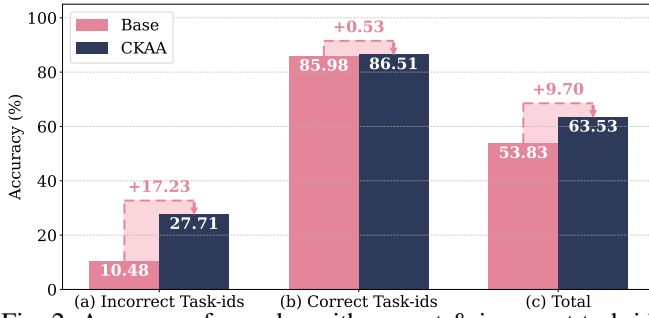


Fig. 2: Accuracy of samples with correct & incorrect task-ids.

decisions. As illustrated in Fig.1(a), when an image from [Task A, Class 1] selects an improper [sub-module B], the model delivers an ambiguous decision between Class 1 and Class 3 for the feature within task B’s subspace, resulting in prediction errors. Such errors stem from the misaligned feature subspaces caused by the independent training and diverse optimization objectives of the sub-modules, where the classifier fails to distinguish wrongly projected features within incorrect subspaces, despite their meaningful semantic information.

As shown in Fig.2, we visualize the classification accuracy of samples predicted with (a) correct and (b) incorrect task-ids on 10S-ImageNetA. The term **Base** refers to performance obtained using task-specific modules trained independently. It suffers from the misalignment problem, resulting in poor accuracy (10.48%). These observations highlight the vulnerability of the model’s robustness when provided with incorrect task-ids, leading to performance degradation. We further consider that the misalignment issue is mainly reflected in two aspects: (1) The feature semantics of the same image from the correct and incorrect sub-modules remain inconsistent; (2) The decision boundaries trained solely with correct task-ids cannot classify features within incorrect subspaces.

Motivated by these analysis, we propose a novel Cross-subspace Knowledge Alignment and Aggregation (CKAA) framework. First, we build a visual prompt-based task recognizer, alongside dedicated task-specific sub-modules to maintain model plasticity. Building on this architecture, we propose a Dual-level Knowledge Alignment (DKA, Fig.1(b)) approach, including feature and classifier-level alignment. Our Feature-level Alignment (FA) groups intra-class features across subspaces together via contrastive learning. Unlike standard contrastive learning [26], [27] in CL, the positive pairs in FA are defined across task-specific subspaces to model the feature variation caused by incorrect task-IDs. Moreover, rather than directly minimizing cross-subspace feature discrepancies through L_2 - or KL-based regularization [4], [5], FA is a relative objective that encourages cross-subspace within-class features to be more compact than negative samples, preserving plasticity of task-specific modules. Building upon these semantically unified feature subspaces, we further train a robust global classifier capable of distinguishing features projected into both correct and incorrect subspaces. To achieve this, we propose an affinity-guided dense feature simulation to simulate wrongly projected features, and train a task-adaptive classifier for each task to distinguish features of all seen classes

projected into its subspace. Such dense simulation better preserves complex intra-class structures and generates diverse features for classifier training. The task-adaptive classifiers are aggregated into a global classifier for robust inference. Consequently, DKA mitigates the misalignment in both two aforementioned aspects.

To further enhance model robustness against misleading task-ids during inference, we propose a Task-Confidence-guided Mixture of Adapters (TC-MoA) scheme, which dynamically aggregates task-specific knowledge from semantically unified sub-modules. Specifically, the task recognizer computes task-confidence scores for each test image, reflecting its semantic relevance to each task. These scores serve as routing weights to adaptively combine intermediate features from task-specific sub-modules, forming a Mixture of Adapters (MoA) architecture. Different from existing module selection strategies [20], [24], our soft aggregation mechanism avoids overconfidence in misleading task-ids by adaptively integrating knowledge from potentially relevant sub-modules, resulting in a more effective and robust inference framework for CIL frameworks with task-specific design.

Unlike existing CIL methods with task-specific designs that mainly focus on improving module selection [20], [23] or aggregation [28], [29] for inference, CKA identifies the underlying cause of prediction failure under misleading task-IDs as cross-subspace semantic and decision-boundary misalignment, and improves robustness from both training and inference perspectives. Through the proposed DKA and TC-MoA approaches, CKA preserves the plasticity of independently adapted task-specific modules while establishing compatible predictions across different subspaces.

The main contributions can be summarized as follows:

- To enhance the model’s robustness against misleading task-ids, we propose a novel CKA framework, which introduces Dual-level Knowledge Alignment (DKA) to align feature semantics and learn robust global decision boundaries across task-specific subspaces, effectively mitigating the misalignment issue.
- A Task-Confidence-guided Mixture of Adapters (TC-MoA) scheme is designed to adaptively aggregate knowledge from potentially relevant sub-modules during inference, avoiding overconfidence in misleading task-ids.
- Extensive experiments on four mainstream CL benchmarks demonstrate that the proposed CKA outperforms existing state-of-the-art methods.

II. RELATED WORKS

A. Continual Learning

Continual Learning aims to update the model with new data while mitigating catastrophic forgetting [6], [30], [31] of previously acquired knowledge. Traditional methods can be categorized into replay-based [17], [18], [32], [33], regularization-based [34]–[37], and expansion-based methods [38]–[41]. Replay-based methods maintain a memory buffer to store the instances of previous tasks. Regularization-based methods introduce penalty terms to prevent substantial changes in the parameters important for old tasks. Expansion-based methods

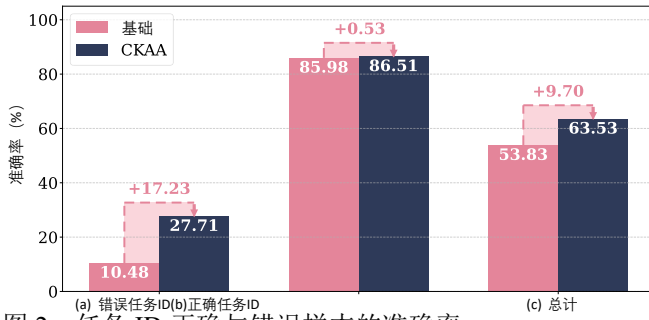


图2: 任务ID正确与错误样本的准确率

中,进而导致模糊且不可靠的决策。如图.1(a)所示,当来自[任务A,类别1]的图像被错误地分配至[子模块B]时,模型会对位于子模块B子空间中的特征做出在类别1与类别3之间存在歧义的决策,从而引发预测错误。此类错误源于子模块独立训练及多样化优化目标所导致的特征子空间错位——在这种情况下,分类器无法区分被错误投影到不正确子空间中的特征,尽管这些特征蕴含着有意义的语义信息。

如图.2所示,我们可视化了在10S-ImageNetA数据集上使用(a)正确任务ID和(b)错误任务ID预测样本时的分类准确率。术语Base指的是使用独立训练的任务专用模块所获得的性能;该模型存在任务对齐问题,导致准确率较低(10.48%)。这些观察结果揭示了当输入错误任务ID时,模型鲁棒性所存在的脆弱性,从而导致性能下降。我们进一步认为,这种任务对齐问题主要体现在两个方面:(1)来自正确与错误子模块的同一幅图像所对应的特征语义仍存在不一致现象;(2)仅使用正确任务ID训练的决策边界无法对位于错误子空间内的特征进行分类。

基于上述分析,我们提出了一种全新的跨子空间知识对齐与聚合(CKAA)框架。首先,我们构建了一个基于视觉提示的任务识别器,并配套设计了专门针对特定任务的子模块,以保持模型的可塑性。基于这一架构,我们提出了一种双层知识对齐(DKA,图.1(b))方法,该方法涵盖特征层与分类器层的对齐机制。我们的特征层对齐(FA)通过对比学习将不同子空间中的同类特征进行聚类。与标准的对比学习方法[26]不同,[27]在CL中,FA中的正样本对是在任务特定子空间上定义的,旨在建模由错误任务ID所引起的特征变化;此外,该方法并非直接通过基于 L_2 或KL的正则化[4]来最小化跨子空间的特征差异,[5]FA是一个相对客观指标,旨在促使类内特征比负样本更具紧凑性,从而保留任务特定模块的可塑性。基于这些语义上统一的特征子空间,我们进一步训练一个鲁棒的全局分类器,使其能够区分投射到正确子空间与错误子空间中的特征。为实现这一目标,我们提出一种基于亲和度引导的密集特征模拟方法,用于模拟错误投射的特征,并为每个任务训练一个任务自适应分类器,用于区分所有

已见类别的特征在该任务对应子空间中的投影。这种密集特征模拟方法能更好地保留复杂的类内结构,并为分类器训练生成多样化的特征。这些任务自适应分类器被聚合为一个全局分类器,用于实现鲁棒的推理。因此,DKA在上述两个方面均有效缓解了特征对齐不一致的问题。

为进一步增强模型在推理过程中对误导性任务ID的鲁棒性,我们提出一种基于任务置信度引导的适配器混合架构(TC-MoA)方案,该方案能够动态聚合来自语义统一子模块的任务特定知识。具体而言,任务识别器为每张测试图像计算任务置信度分数,用以反映该图像与各任务之间的语义相关性;这些分数可作为路由权重,用于自适应地组合来自各任务特定子模块的中间特征,从而构建出一种适配器混合(MoA)架构。这一方法与现有的模块选择策略[20]有所不同。[24]我们的软聚合机制通过自适应地整合来自潜在相关子模块的知识,避免了对误导性任务ID产生过度置信的情况,从而为具有任务特定设计的CIL框架构建了一个更高效、更鲁棒的推理框架。

与现有的基于任务特定设计的CIL方法(这些方法主要侧重于优化模块选择[20])不同,[23]或聚合[28],[29]在推理阶段,CKAA将预测失败(在存在误导性任务ID的情况下)的潜在原因归因于跨子空间语义与决策边界对齐不良,并从训练与推理两个维度提升了模型的鲁棒性。通过所提出的DKA与TC-MoA方法,CKAA在保持各任务特定模块独立适应性的同时,实现了不同子空间之间兼容的预测结果。

主要贡献可归纳如下:

- 为增强模型对误导性任务ID的鲁棒性,我们提出一种新型的CKAA框架,该框架引入了“双层知识对齐”(Dual-level Knowledge Alignment, DKA)机制,用于对齐特征语义并在各任务特定子空间内学习鲁棒的全局决策边界,从而有效缓解特征对齐偏差问题。
- 本文设计了一种基于任务置信度引导的适配器混合方案(TCMoA),该方案可在推理过程中自适应地聚合来自潜在相关子模块的知识,从而避免因对误导性任务ID产生过高的置信度而引发的问题。
- 在四个主流CL基准测试集上开展的大量实验表明,所提出的CKAA方法优于现有的最先进方法。

II. 相关研究

A. 持续学习

持续学习旨在利用新数据对模型进行更新,同时缓解灾难性遗忘问题[6]。[30],[31]基于先前获取的知识。传统方法可归类为基于回放的[17],[18],[32],[33]基于正则化的[34]–[37]方法,以及基于扩展的方法[38]–[41]。基于回放的方法通过维护一个记忆缓冲区来存储先前任务的实例;基于正则化的方法引入惩罚项,以防止对旧任务至关重要的参数发生显著变化;基于扩展的方法特征提取。

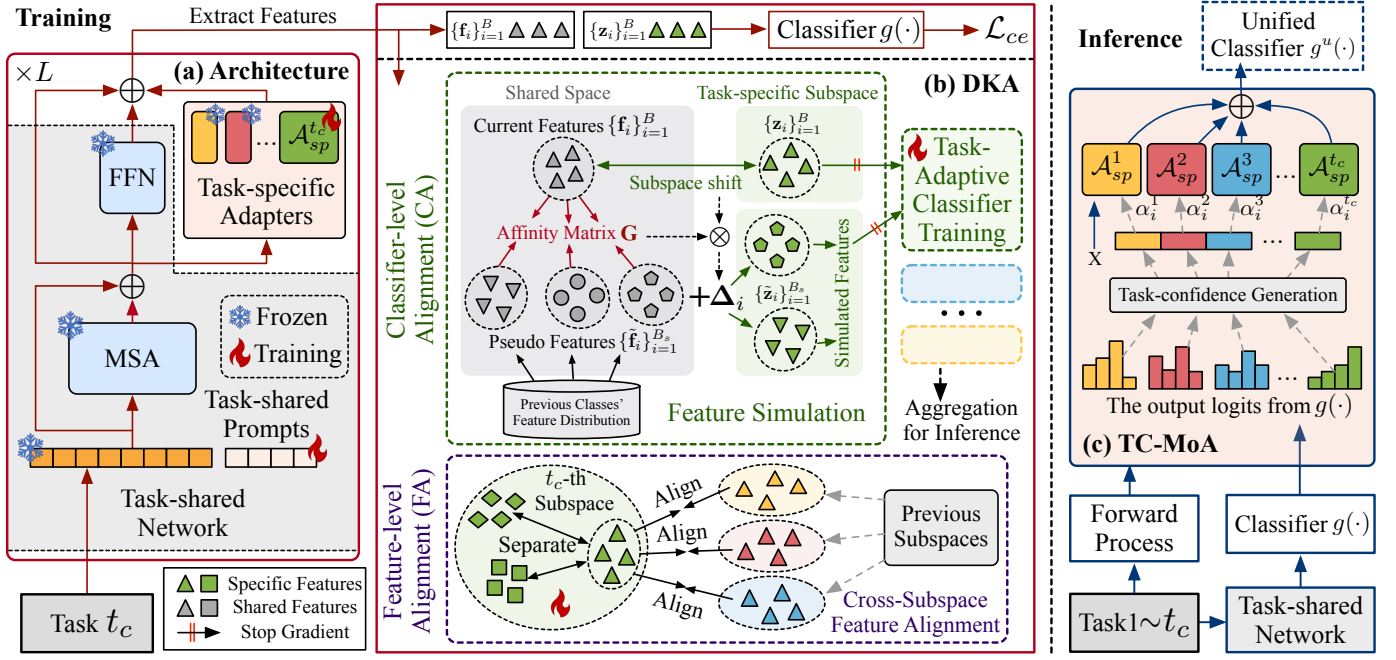


Fig. 3: The framework of our proposed method. Our network architecture (a) is composed of a visual prompt-based task-shared network and adapter-based task-specific sub-modules. To enhance the model’s robustness against incorrect task-ids, we propose Dual-level Knowledge Alignment approach (b, DKA), which aligns task-specific distributions at the feature and classifier levels. During inference, the Task-Confidence-guided Mixture of Adapters (c, TC-MoA) effectively aggregates task-specific knowledge into the task-shared backbone.

freeze specific parameters for old tasks while allocating new parameters for subsequent tasks.

Recently, continual learning with pre-trained models [20], [23], [24], [42]–[46] have gained growing attention, owing to the strong representation capacity of pre-trained models. Some methods fully fine-tune the pre-trained models [47], [48], which is highly time-consuming. Other methods incorporate PEFT methods into continual learning. Prompt pool-based methods [20], [23], [24], [49] maintain a prompt pool for all tasks and select appropriate prompts for each specific task. Some adapter-based methods [22], [25], [50] build expandable adapter-based architecture while reducing catastrophic forgetting through regularization terms or semantic adjustment. PGP [51], VPT-NSP [3], InfLoRA [21] and LoDA [16] theoretically integrate gradient projection into the prompt-based and LoRA-based CL, achieving impressive performance.

B. Parameter-Efficient Fine-Tuning

Parameter-Efficient Fine-Tuning (PEFT) is a paradigm to fine-tune pre-trained models by inserting sub-modules with a small number of parameters while keeping the pre-trained parameters frozen. PEFT methods have shown superior performance in NLP [10], [52], [53]. Recently, PEFT methods have been proposed for Vision Transformer (ViT) [54]–[56]. Prompt Tuning [12], [57], [58] and Prefix-Tuning [53] prepend a set of learnable tokens to the inputs before the transformer blocks. Adapter [11] inserts lightweight modules into the Feed Forward Network (FFN) and Low-Rank Adaptation (LoRA) [10] injects learnable rank decomposition matrices to re-parameterize the pre-trained weights. Mixture-of-Experts

(MoE) [59] aggregates multiple sub-networks as experts via a routing network. The above PEFT methods have demonstrated comparable performance to full fine-tuning.

III. METHODOLOGY

Problem Definition. In Continual Learning, there is a sequence of tasks with non-overlapping classes. We define the sequential dataset as $\mathcal{D} = \{\mathcal{D}_1, \dots, \mathcal{D}_T\}$. $\mathcal{D}_t = \{\mathbf{x}_i^t, y(\mathbf{x}_i^t)\}_{i=1}^{|\mathcal{D}_t|}$ is the dataset associated with the t -th task of size $|\mathcal{D}_t|$, and $y(\mathbf{x}_i^t) \in \mathcal{Y}^t$ denotes the corresponding class of $\mathbf{x}_i^t \in \mathbb{R}^{H \times W \times C}$. $\mathcal{Y}^t$ is the label space of task t , and $\mathcal{Y}^t \cap \mathcal{Y}^{t'} = \emptyset$ for $t \neq t'$. The objective of continual learning is to train a model sequentially on T tasks and perform well on testing images from all seen classes $\mathcal{Y}^1 \cup \dots \cup \mathcal{Y}^T$.

We follow existing PEFT-based methods [20], [21], [23] and assume that a pre-trained Vision Transformer (ViT) [60] is available for model initialization. The model is decoupled into a feature extractor $\mathcal{F} : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^d$ and a classifier $g : \mathbb{R}^d \rightarrow \mathbb{R}^{\sum_{t=1}^T |\mathcal{Y}^t|}$. Similar to existing work [20], we focus on the Class-Incremental Learning (CIL) setting, where task-ids are unknown during inference.

Overall Framework. The overall framework of our CKAA is illustrated in Fig.3. Our network architecture (Fig.3(a)) consists of a task-shared network to function as a task recognizer, and task-specific sub-modules to capture unique knowledge for individual tasks. To enhance the model’s robustness against incorrect task-ids, we propose a Dual-level Knowledge Alignment (DKA, Fig.3(b)) approach for training these sub-modules. Our DKA consists of two components: (1) Feature-level Alignment (FA), which groups the same class features

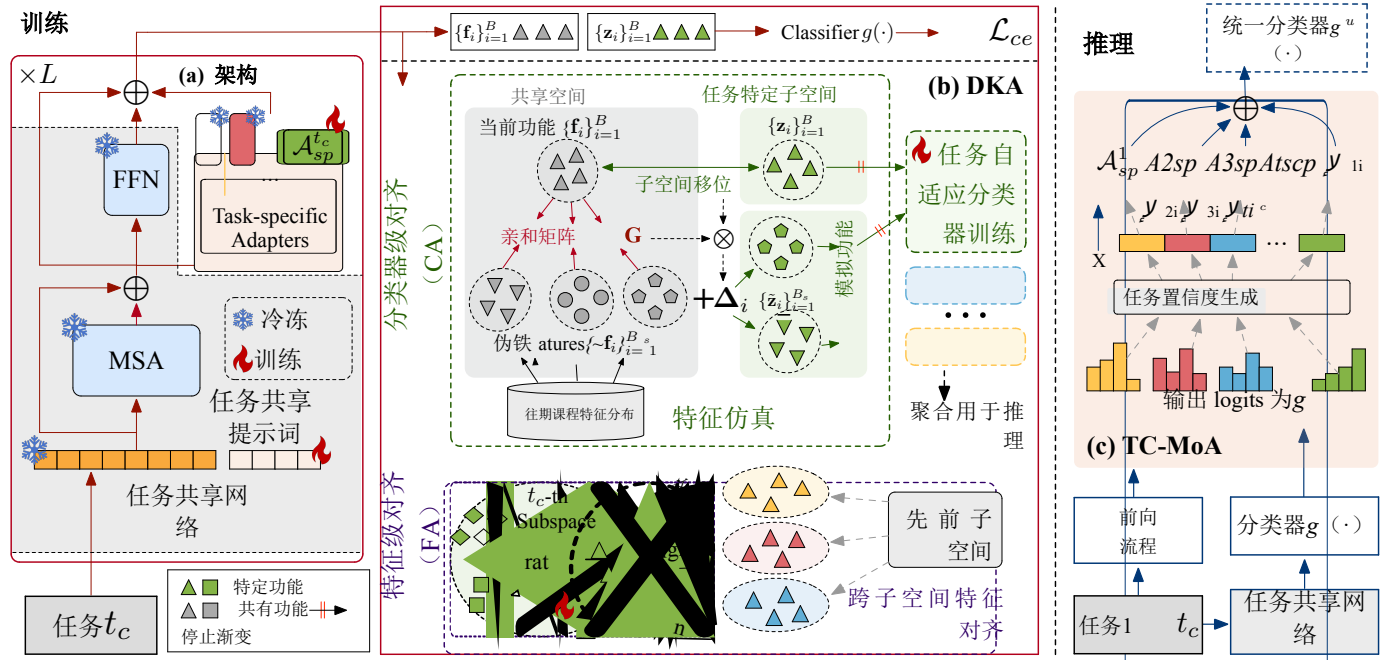


图 3: 本文所提方法的总体架构。我们的网络架构(a)由一个基于视觉提示的任务共享网络与基于适配器的任务特定子模块组成。为增强模型对错误任务ID的鲁棒性, 我们提出了“双层知识对齐”方法 (b, DKA), 该方法在特征层和分类器层对任务特定分布进行对齐。在推理阶段, 采用“任务置信度引导的适配器混合机制” (c, TC-MoA) 将任务特定知识有效地聚合到任务共享主干网络中。

为旧任务冻结特定参数, 同时为后续任务分配新参数。

近期, 基于预训练模型的持续学习[20]、[23], [24], [42]–[46]因预训练模型强大的表征能力而日益受到关注。部分方法对预训练模型进行全量微调[47], [48]这种方法非常耗时。其他方法将 PEFT 方法融入到持续学习中。基于提示池的方法[20], [23], [24], [49] 为所有任务维护一个提示词池, 并为每个具体任务选择合适的提示词。部分基于适配器的方法[22], [25], [50] 构建基于可扩展适配器的架构, 同时通过正则化项或语义调整来缓解灾难性遗忘问题。PGP[51]、VPT - NSP[3]、InfLoRA[21]以及 LoDA [16]在理论上将梯度投影整合到基于提示的学习策略和基于 LoRA 的微调方法中, 取得了令人瞩目的性能表现。

B. 参数高效微调

参数高效微调 (PEFT) 是一种微调预训练模型的方法: 通过插入参数量较少的子模块, 同时保持预训练模型的参数冻结。PEFT 方法在自然语言处理领域已展现出卓越的性能[10], [52], [53]近期, 研究人员提出了适用于 Vision Transformer (ViT)[54]–[56] 的 PEFT 方法。Prompt Tuning [12], [57], [58] Prefix-Tuning[53]在输入数据进入Transformer模块之前, 为其添加一组可学习的标记; Adapter[11]将轻量级模块插入 Feed Forward Network (FFN) 中; 而 Low-Rank Adaptation (LoRA)[10]则通过注入可学习的秩分解矩阵来对预训练权重进行重新参数化; Mixture-of-Experts (MoE)

[59]则通过路由网络将多个子网络聚合为多个“专家”网络。上述 PEFT 方法所展现的性能与全量微调方法相当。

III. 研究方法

问题定义。在持续学习中, 存在一系列类别互不重叠的任务。我们将该序列数据集定义为 $D = \{D_1, \dots, D_T\}$, $D_t = \{\mathbf{x}_i^t, \mathbf{y}(\mathbf{x}_i^t)\}_{i=1}^{|D_t|}$ 表示与第 t 个任务相关的数据集, 其规模为 $|D_t|$; 而 $\mathbf{y}(\mathbf{x}_i^t) \in Y^t$ 表示 $\mathbf{x}_i^t \in \mathbb{R}^{H \times W \times C}$ 对应的类别。 Y^t 是任务 t 的标签空间, 且对于 t , 有 $Y^t \cap Y^{t'} = \emptyset, t \neq t'$ 。持续学习的目标是在 T 个任务上对模型进行顺序训练, 并在来自所有已见类别 $Y^1 \cup \dots \cup Y^T$ 的测试图像上取得良好性能。

我们采用现有的基于 PEFT 的方法[20], [21], [23]并假设可使用预训练的 Vision Transformer (ViT)[60]用于模型初始化。该模型被解耦为一个特征提取器 $F: \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^d$ 和一个分类器 $g: \mathbb{R}^d \rightarrow \sum_{t=1}^T |Y^t|$ 。与现有研究[20]类似, 我们聚焦于类增量学习 (CIL) 场景, 即在推理过程中任务ID未知。

总体框架。我们提出的 CKAA 总体框架如图.3所示。我们的网络架构 (图.3(a)) 包含一个用于充当任务识别器的任务共享网络, 以及用于捕获特定任务专属知识的任务专用子模块。为增强模型对错误任务ID的鲁棒性, 我们提出一种双层知识对齐 (DKA, 图.3(b)) 方法用于训练这些子模块。我们的 DKA 包含两个组成部分: (1) 特征级对齐 (FA), 将来自不同特征子空间的同一类别特征进行聚

from different feature subspaces together; (2) Classifier-level Alignment (CA), which learns a task-adaptive classifier during each session to address cases where previous images are incorrectly assigned with the current task-id, and aggregates them to build a robust global classifier for inference. During inference, we introduce a Task-Confidence-guided Mixture of Adapters (TC-MoA, Fig.3(c)) approach, which generates task-confidence scores as routing weights to adaptively aggregate task-specific knowledge without task-ids.

A. Network Architecture

Our network architecture is composed of: (1) A trainable task-shared network, which serves as a task recognizer and captures the common knowledge from the downstream task; (2) Task-specific sub-modules, which explore the unique knowledge for individual tasks and ensure model plasticity.

Task-shared Network. We develop our task-shared component on top of Visual Prompt Tuning (VPT) [54]. Since the visual prompts can inject knowledge before each transformer block as a global operation, they are well-suited to capturing the common knowledge of downstream tasks with strong generalizability. Specifically, a set of task-shared learnable vectors denoted as $\mathcal{P}_{sh} = \{\mathbf{P}_l\}_{l=1}^L$, are inserted into the original input at each transformer block, where $\mathbf{P}_l \in \mathbb{R}^{N_p \times d}$ is the prompt corresponding to the l -th block. At l -th block, the prompt $\mathbf{P}_l$ is concatenated with the patch representations $\mathbf{X}_l$ to serve as the input to the MSA layer:

$$[\hat{\mathbf{X}}_l, \hat{\mathbf{P}}_l] = \text{MSA}([\mathbf{X}_l; \mathbf{P}_l]); \quad \mathbf{X}_{l+1} = \text{FFN}(\hat{\mathbf{X}}_l), \quad (1)$$

where $\text{MSA}(\cdot)$ and $\text{FFN}(\cdot)$ denotes the Multi-head Self-Attention (MSA) layer and the Feed Forward Network (FFN) layer in ViT. The pre-trained feature extractor $\mathcal{F}$ and the prompts $\mathcal{P}_{sh}$ work together to serve as the task-shared network, the shared feature representation of input $\mathbf{x}$ is denoted as $\mathbf{f} = \mathcal{F}(\mathbf{x}; \mathcal{P}_{sh})$. Since the task-shared network should adapt to all downstream tasks, it is essential to mitigate its forgetting when training new tasks. To achieve this objective, we update $\mathcal{P}_{sh}$ orthogonally to the previous feature space following [3], [51], [61]. These prompts $\mathcal{P}_{sh}$ are updated throughout all sequential tasks, encapsulating the common knowledge among them. Such orthogonally optimized prompts establish a shared backbone and a powerful task recognizer, tailored for downstream tasks.

Task-specific Sub-modules. Due to the stringent orthogonal constraints, the task-shared network is insufficient to capture fine-grained knowledge specific to each individual task, thereby limiting the model's plasticity. Therefore, we allocate a task-specific sub-module for each task to capture its unique knowledge and generate more discriminative features. We design task-specific sub-modules based on an adapter architecture [11], denoted as $\mathcal{A}_{sp}^t = \{\mathcal{A}_l^t\}_{l=1}^L$. Since the residual connection nature of adapter-tuning, it can effectively enable the model to aggregate unique knowledge from different tasks independently. For the adapter $\mathcal{A}_l^t \in \mathcal{A}_{sp}^t$ in l -th block, it is composed of a downsampling matrix $\mathbf{W}_{down} \in \mathbb{R}^{d \times \hat{d}}$ and an upsampled matrix $\mathbf{W}_{up} \in \mathbb{R}^{\hat{d} \times d}$. Denote the input as $\mathbf{X}$, the forward process can be formulated as $\mathcal{A}(\mathbf{X}) =$

$\text{ReLU}(\mathbf{X} \cdot \mathbf{W}_{down}) \cdot \mathbf{W}_{up}$. At l -th block, $\mathcal{A}_l^t$ incorporate task-specific knowledge in the FFN layer through a residual connection:

$$\mathbf{X}_{l+1} = \text{FFN}(\hat{\mathbf{X}}_l) + \mathcal{A}_l^t(\hat{\mathbf{X}}_l). \quad (2)$$

The final task-specific feature of input $\mathbf{x}$ is denoted as $\mathbf{z} = \mathcal{F}(\mathbf{x}; \mathcal{P}_{sh}, \mathcal{A}_{sp}^t)$.

Basic loss. During t_c -th session, given a batch $\{\mathbf{x}_i\}_{i=1}^B$ of size B (with the superscript t_c omitted for simplicity), the features $\{\mathbf{f}_i\}_{i=1}^B$ and $\{\mathbf{z}_i\}_{i=1}^B$ from the shared network and the t_c -th sub-module are extracted, where $\mathbf{f}_i = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh})$ and $\mathbf{z}_i = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh}, \mathcal{A}_{sp}^t)$. These features are trained with the classifier g and a commonly-used cross-entropy loss:

$$\mathcal{L}_{ce} = \frac{1}{B} \sum_{i=1}^B \mathcal{L}(\sigma(g(\mathbf{f}_i)), y(\mathbf{f}_i)) + \mathcal{L}(\sigma(g(\mathbf{z}_i)), y(\mathbf{z}_i)), \quad (3)$$

where $y(\mathbf{f})$ represents the ground-truth class label of feature $\mathbf{f}$. $\mathcal{L}$ denotes the standard cross-entropy loss, and $\sigma(\cdot)$ denotes the softmax operation. Following existing methods [3], [20], [24], g is trained separately on the classes of each task, and the weights are concatenated for inference.

B. Dual-level Knowledge Alignment

Given our task-specific designs, it is essential to select appropriate sub-modules by estimating task-ids through the task recognizer during inference. However, there are always inevitable incorrect task-ids predictions. To enhance the model's robustness against these incorrect task-ids, we propose a Dual-level Knowledge Alignment (DKA) training approach.

Feature-level Alignment (FA). For a robust feature extractor capable of generating meaningful features despite incorrect task-ids, the same image should achieve consistent semantics when projected into the correct and the incorrect feature subspaces. Therefore, we semantically unify features across different feature subspaces through FA. Specifically, we propose a Cross-Subspace Feature Alignment (CSFA) loss, which encourages features of the same class in the current subspace to align with those in the previous subspace. We first extract the features of the current dataset $\mathcal{D}_{t_c}$ in the t_p -th previous subspace for $t_p < t_c$, denoted as $\tilde{\mathbf{z}}_i^{t_c \rightarrow t_p} = \mathcal{F}(\mathbf{x}_i^{t_c}; \mathcal{P}_{sh}, \mathcal{A}_{sp}^{t_p})$ and $\mathbf{x}_i^{t_c} \in \mathcal{D}_{t_c}$. During training, we randomly sample a feature set $\tilde{\mathcal{C}}_{sp} = \{\tilde{\mathbf{z}}_j^{t_c \rightarrow t_p}\}_{j=1}^B$ in previous subspaces. Given an input batch $\mathcal{C}_{sp} = \{\mathbf{z}_i\}_{i=1}^B$, for a feature $\mathbf{z}_k \in \mathcal{C}_{sp}$ with label $y(\mathbf{z}_k)$ in the current subspace, we build its positive set by features from the sampled set with the same ground-truth, where $\mathcal{P}(\mathbf{z}_k) = \{\tilde{\mathbf{z}}_j^{t_c \rightarrow t_p} | y(\tilde{\mathbf{z}}_j^{t_c \rightarrow t_p}) = y(\mathbf{z}_k), \tilde{\mathbf{z}}_j^{t_c \rightarrow t_p} \in \tilde{\mathcal{C}}_{sp}\}$. Meanwhile, the features from the current subspace with different labels are regarded as the negative samples, where $\mathcal{N}(\mathbf{z}_k) = \{\mathbf{z}_j | y(\mathbf{z}_j) \neq y(\mathbf{z}_k), \mathbf{z}_j \in \mathcal{C}_{sp}\}$. Then CSFA loss for an input batch $\{\mathbf{z}_i\}_{i=1}^B$ is formulated as follows:

$$\mathcal{L}_{csfa} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\sum_{\mathbf{z}^+ \in \mathcal{P}(\mathbf{z}_i)} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}^+)/\tau_f)}{\sum_{\mathbf{z}_j \in \mathcal{P}(\mathbf{z}_i) \cup \mathcal{N}(\mathbf{z}_i)} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau_f)}, \quad (4)$$

where τ_f is a temperature factor for contrastive learning. The CSFA loss groups the same class features together in different subspaces, alleviating feature-level misalignment.

类：(2) 分类器级对齐 (CA)，在每个会话期间学习一个任务自适应分类器，以解决先前图像被错误地分配到当前任务ID的情况，并对这些图像进行聚合，构建一个用于推理的鲁棒全局分类器。在推理阶段，我们引入了一种基于任务置信度引导的适配器混合模型 (TC-MoA，图.3(c)) 方法，该方法通过生成任务置信度分数作为路由权重，从而在无需任务ID的情况下自适应地聚合任务特定知识。

A. 网络架构

我们的网络架构由以下部分组成：(1) 一个可训练的任务共享网络，该网络充当任务识别器，并从下游任务中捕获通用知识；(2) 任务特定子模块，用于探索各任务独有的知识并确保模型的可塑性。**任务共享网络**。我们在 Visual Prompt Tuning (VPT)[54]的基础上开发了这一任务共享组件。由于视觉提示可在每个 Transformer 模块之前作为全局操作注入知识，因此它们非常适合捕获具有强泛化能力的下游任务的通用知识。具体而言，一组任务共享的可学习向量记作 $\mathcal{P}_{sh} = \{\mathbf{P}_l\}_{l=1}^L$ 这些向量被插入到每个 Transformer 模块的原始输入中，其中 $\mathbf{P}_l \in \mathbb{R}^{N_P \times d}$ 是对应于第 l 个模块的提示词。在第 l 个模块中，将提示词 $\mathbf{P}_l$ 与 Patch 表示 $\mathbf{X}_l$ 进行拼接，作为 MSA 层的输入：

$$[\hat{\mathbf{X}}_l, \hat{\mathbf{P}}_l] = \text{MSA}([\mathbf{X}_l; \mathbf{P}_l]); \quad \mathbf{X}_{l+1} = \text{FFN}(\hat{\mathbf{X}}_l), \quad (1)$$

其中，MSA ($\cdot$) 和 FFN ($\cdot$) 分别表示 ViT 中的多头自注意力层 (MSA) 与前馈网络层 (FFN)。预训练特征提取器 F 与提示词 $\mathcal{P}_{sh}$ 共同构成任务共享网络；输入 $\mathbf{x}$ 的共享特征表示记作 $\mathbf{f} = F(\mathbf{x}; \mathcal{P}_{sh})$ 。由于该任务共享网络需适配所有下游任务，在训练新任务时需减轻其遗忘效应。为实现这一目标，我们按照[3]的方法，将 $\mathcal{P}_{sh}$ 正交更新至先前的特征空间。[51], [61]这些提示词 $\mathcal{P}_{sh}$ 在所有序列任务中持续更新，从而封装了各任务之间的共性知识。这种正交优化的提示词构建了一个共享的底层框架以及一个强大的任务识别器，专为下游任务而设计。

任务特定子模块。由于存在严格的正交约束，任务共享网络不足以捕捉各独立任务特有的细粒度知识，从而限制了模型的可塑性。因此，我们为每个任务分配一个任务特定子模块，以捕捉其独特知识并生成更具判别力的特征。我们基于适配器架构[11]设计了这些任务特定子模块，记作 $\text{Atsp} = \{\text{Atl}\}_{l=1}^L$ 由于适配器调优具有残差连接特性，它能够有效使模型独立地聚合来自不同任务的独特知识。对于第 l 个模块中的适配器 $\text{Atl} \in \text{Atsp}$,

它由一个下采样矩阵 $\mathbf{W}_{down} \in \mathbb{R}^{d \times d}$ 组成。

以及一个上采样矩阵 $\mathbf{W}_{up} \in \mathbb{R}^{d \times d}$ 。设输入为 $\mathbf{X}$ ，则前向过

程可表示为 $A(\mathbf{X}) = \text{ReLU}(\mathbf{X} \cdot \mathbf{W}_{down}) \cdot \mathbf{W}_{up}$ 。在第 l 个模块中， Atl 通过残差连接将任务特定知识融入到 FFN 层中：

$$\mathbf{X}_{l+1} = \text{FFN}(\hat{\mathbf{X}}_l) + \mathcal{A}_l^t(\hat{\mathbf{X}}_l). \quad (2)$$

输入 $\mathbf{x}$ 的最终任务特定特征记作 $\mathbf{z} = F(\mathbf{x}; \mathcal{P}_{sh}, \text{Atsp})$ 。

基础损失函数。在第 t_c 次训练轮次中，给定一个批次 $\{\mathbf{x}_i\}_{i=1}^B$ 尺寸为 B (为简化起见，省略上标 t_c)，特征集 $\{\mathbf{f}_i\}_{i=1}^B$ 以及 $\{\mathbf{z}_i\}_{i=1}^B$ 从共享网络及第 t_c 个子模块中提取特征，其中 $\mathbf{f}_i = F(\mathbf{x}_i; \mathcal{P}_{sh})$ 且 $\mathbf{z}_i = F(\mathbf{x}_i; \mathcal{P}_{sh}, \text{Atsp})$ 。这些特征通过分类器 g 及常用的交叉熵损失函数进行训练：

$$\mathcal{L}_{ce} = \frac{1}{B} \sum_{i=1}^B \mathcal{L}(\sigma(g(\mathbf{f}_i)), y(\mathbf{f}_i)) + \mathcal{L}(\sigma(g(\mathbf{z}_i)), y(\mathbf{z}_i)), \quad (3)$$

其中 $y(\mathbf{f})$ 表示特征 $\mathbf{f}$ 的真实类别标签； L 表示标准交叉熵损失函数， $\sigma(\cdot)$ 表示 softmax 激活函数。遵循现有方法[3], [20][24]， g 在每个任务对应的类别上分别进行训练，随后将这些权重进行拼接以用于推理。

B. 双层知识对齐

鉴于我们针对特定任务所设计的架构，在推理阶段通过任务识别器估算任务ID以选择合适的子模块至关重要。然而，始终存在不可避免的错误任务ID预测情况。为增强模型对这些错误任务ID的鲁棒性，我们提出一种双层知识对齐 (DKA) 训练方法。**特征级对齐 (FA)**。为了构建一个即使在存在错误任务ID的情况下仍能生成有意义特征的鲁棒特征提取器，同一张图像在投影到正确与错误特征子空间时应具有一致的语义含义。因此，我们通过FA对不同特征子空间中的特征进行语义统一。具体而言，我们提出一种跨子空间特征对齐 (CSFA) 损失函数，该函数鼓励当前子空间中同一类别的特征与先前子空间中的相应特征进行对齐。我们首先为 $t_p < t_c$ 情况，从当前数据集 $\mathcal{D}_{t_c}$ 中提取第 t_p 个先前子空间中的特征，记为 $\tilde{\mathbf{z}}_{t_p} = \mathcal{F}(\mathbf{x}_{t_p}^{t_c}; \mathcal{P}_{sh}, \text{Atsp})$ 且 $\tilde{\mathbf{z}}_{t_p} \in \mathcal{D}_{t_c}$ 。在训练过程中，我们随机采样一个特征 $\{\tilde{\mathbf{z}}_{t_p}^{t_c}\}_{j=1}^B$ 在集合 $\mathcal{C}_{sp} = \mathcal{C}_{sp}^{t_p}$ 先前的子空间中。给定输入 $\mathbf{z} = \{\mathbf{z}_i\}_{i=1}^B$ 对于具有标签的特征 $\mathbf{z}_k \in \mathcal{C}_{sp}$ 在当前子空间中，我们利用来自采样集且具有相同基础属性的特征构建其正集，其中 $\mathcal{P}(\mathbf{z}_k) = \{\tilde{\mathbf{z}}_{t_p}^{t_c} | y(\tilde{\mathbf{z}}_{t_p}^{t_c}) = y(\mathbf{z}_k), \text{ 且 } \tilde{\mathbf{z}}_{t_p}^{t_c} \in \mathcal{C}_{sp}\}$ 。同时，来自当前子空间且标签不同的特征被视为负样本，其中 $\mathcal{N}(\mathbf{z}_k) = \{\mathbf{z}_j | y(\mathbf{z}_j) \neq y(\mathbf{z}_k), \mathbf{z}_j \in \mathcal{C}_{sp}\}$ 。则对于输入批次 $\{\mathbf{z}_i\}_{i=1}^B$ 其公式如下：

$$\mathcal{L}_{csfa} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\sum_{\mathbf{z}^+ \in \mathcal{P}(\mathbf{z}_i)} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}^+) / \tau_f)}{\sum_{\mathbf{z}_j \in \mathcal{P}(\mathbf{z}_i) \cup \mathcal{N}(\mathbf{z}_i)} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j) / \tau_f)}, \quad (4)$$

其中 τ_f 为对比学习中的温度因子。CSFA 损失函数将同一类别特征聚类到不同的子空间中，从而缓解特征层面的错配问题。

Classifier-level Alignment (CA). Building upon the semantically unified subspaces enhanced by FA, CA aims to train a robust global classifier capable of distinguishing features across all subspaces even under incorrect task-ids. Specifically, in t_c -th session, we train a task-adaptive classifier g^{t_c} , which seeks to avoid ambiguous predictions when previous data $\mathbf{x}_i^{t_c} \in \mathcal{D}_1 \cup \dots \cup \mathcal{D}_{t_c-1}$ are projected into the current feature subspace. To achieve this objective, we first simulate the features of the previous data in the current subspace, then the simulated features are used to jointly train g^{t_c} alongside the current features. We further decompose the simulation process into *three steps*: (1) *Modeling the affinities* between current and previous data within the shared feature space through the task-shared network; (2) *Modeling the subspace shift* within current data between the current subspace and the shared ones; (3) *Transferring the subspace shift* of the current shared features to the previous ones through the affinity matrix.

Since modeling the affinities requires the previous data distributions but there are no exemplars saved for training, we address this issue through Gaussian sampling. After the t -th session, for the k -th class in the t -th task, we approximate its feature distribution in the shared feature space as a Gaussian distribution $\mathcal{G}_{k,t}$. During the t_c -th session, for each arriving batch, we sample pseudo representations $\mathcal{S}_{sh} = \{\tilde{\mathbf{f}}_i, y(\tilde{\mathbf{f}}_i)\}_{i=1}^{B_s}$ with size B_s from previous shared feature distributions $\cup_{t \leq t_c-1} \mathcal{G}_{k,t}$. In conjunction with the current features $\mathcal{C}_{sh} = \{\mathbf{f}_i, y(\mathbf{f}_i)\}_{i=1}^B$ and $\mathcal{C}_{sp} = \{\mathbf{z}_i, y(\mathbf{z}_i)\}_{i=1}^B$, the affinity matrix $\mathbf{G} \in \mathbb{R}^{B_s \times B}$ between the previous data and current data in the shared feature space are formulated as follows:

$$\mathbf{G}_{ij} = \begin{cases} \exp(\text{sim}(\tilde{\mathbf{f}}_i, \mathbf{f}_j)/\tau_g), & \text{if } \mathbf{f}_j \in \mathcal{K}(\tilde{\mathbf{f}}_i, K_g), \\ 0, & \text{otherwise,} \end{cases} \quad (5)$$

where τ_g is a temperature factor. $\text{sim}(\mathbf{a}, \mathbf{b})$ denotes the cosine similarity between vector $\mathbf{a}$ and $\mathbf{b}$, where $\text{sim}(\mathbf{a}, \mathbf{b}) = \mathbf{a}^\top \mathbf{b} / (\|\mathbf{a}\| \cdot \|\mathbf{b}\|)$. $\mathcal{K}(\tilde{\mathbf{f}}_i, K_g)$ denotes the K_g nearest neighbors in $\mathcal{C}_{sh}$ of $\tilde{\mathbf{f}}_i$. $\mathbf{G}_{ij}$ represents the affinity between sampled feature $\tilde{\mathbf{f}}_i$ and current feature $\mathbf{f}_j$.

The subspace shift of current data between the current subspace and the shared feature space can be simply modeled by the residual between the specific feature $\{\mathbf{z}_i\}_{i=1}^B$ and the shared feature $\{\mathbf{f}_i\}_{i=1}^B$ of current data. Subsequently, we transfer the subspace shift from the current features $\{\mathbf{f}_i\}_{i=1}^B$ to the sampled pseudo features $\{\tilde{\mathbf{f}}_i\}_{i=1}^{B_s}$. The simulated features $\{\tilde{\mathbf{z}}_i\}_{i=1}^{B_s}$ are derived as follows:

$$\tilde{\mathbf{z}}_i = \tilde{\mathbf{f}}_i + \Delta_i, \quad \Delta_i = \frac{\sum_{j=1}^B \mathbf{G}_{ij}(\mathbf{z}_j - \mathbf{f}_j)}{\sum_{j=1}^B \mathbf{G}_{ij}}. \quad (6)$$

Through the above process, we derive the simulated feature set $\mathcal{S}_{sp} = \{\tilde{\mathbf{z}}_i, y(\tilde{\mathbf{z}}_i)\}_{i=1}^{B_s}$. Then we obtain the unified feature set $\mathcal{U} = \mathcal{S}_{sh} \cup \mathcal{S}_{sp} \cup \mathcal{C}_{sh} \cup \mathcal{C}_{sp}$, which includes features of images from all seen tasks in the current subspace. We utilize it to train the task-adaptive classifier g^{t_c} :

$$\mathcal{L}_{ca} = \frac{1}{|\mathcal{U}|} \sum_{\{\mathbf{f}, y(\mathbf{f})\} \in \mathcal{U}} \mathcal{L}(\sigma(g^{t_c}(\mathbf{f})), y(\mathbf{f})). \quad (7)$$

After training t_c -th session, to reduce computational overhead for inference, we aggregate the parameters from all the

Algorithm 1 Training pipeline for task t_c

Inputs: Dataset $\mathcal{D}_{t_c}$, backbone $\mathcal{F}$, shared prompts $\mathcal{P}_{sh}$, adapter $\mathcal{A}_{sp}^{t_c}$.

Outputs: Updated $\mathcal{P}_{sh}$, $\mathcal{A}_{sp}^{t_c}$, classifier g^{t_c} .

- 1: Initialize $\mathcal{A}_{sp}^{t_c}$ and g^{t_c} ; **if** $t_c = 1$, initialize $\mathcal{P}_{sh}$;
 - 2: **for** $\{\mathbf{x}_i, y_i\} \in \mathcal{D}_{t_c}$ **do**
 - 3: $\mathbf{f}_i = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh})$; // Shared features
 - 4: $\mathbf{z}_i = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh}, \mathcal{A}_{sp}^{t_c})$; // Specific features
 - 5: Compute $\mathcal{L}_{ce}$ (Eq.3);
 - 6: **if** $t_c > 1$ **then**
 - 7: Extract $\tilde{\mathbf{z}}_i^{t_c \rightarrow t_p} = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh}, \mathcal{A}_{sp}^{t_p})$;
 - 8: Compute $\mathcal{L}_{csfa}$ (Eq.4); // Feature-level alignment
 - 9: Sample $\mathcal{S}_{sh}$ from $\cup_{t < t_c} \mathcal{G}_{k,t}$;
 - 10: Compute affinities $\mathbf{G}$ (Eq.5);
 - 11: Simulate $\mathcal{S}_{sp}$ (Eq.6);
 - 12: Compute $\mathcal{L}_{cu}$ (Eq.7); // Classifier-level alignment
 - 13: **end if**
 - 14: Compute total loss $\mathcal{L}$ (Eq.9);
 - 15: Update $\mathcal{P}_{sh}$, $\mathcal{A}_{sp}^{t_c}$, g^{t_c} ;
 - 16: **end for**
 - 17: **for** class $y \in \mathcal{Y}^{t_c}$ **do**
 - 18: Compute $\mathcal{G}_{y,t_c}$ via feature statistics;
 - 19: **end for**
 - 20: Build unified classifier g^u (Eq.8);
-

task-adaptive classifiers, i.e., $g^1, \dots, g^{t_c}$, to build a global classifier g^u that fits all the task-specific feature subspaces. For the parameter corresponding to the k -th class in the t_p -th task, i.e., $k \in \mathcal{Y}^{t_p}$, the parameter aggregation can be formulated as follows:

$$\mathbf{W}^u[k] = \frac{\sum_{t=t_p}^{t_c} \mathbf{W}^t[k]}{t_c - t_p + 1}, \quad \mathbf{b}^u[k] = \frac{\sum_{t=t_p}^{t_c} \mathbf{b}^t[k]}{t_c - t_p + 1}, \quad (8)$$

where $\mathbf{W}^u[k]$ ($\mathbf{b}^u[k]$) is the k -th class's weight (bias) in g^u , and $\mathbf{W}^t[k]$ ($\mathbf{b}^t[k]$) is the k -th class's weight (bias) in g^t . Such an aggregation strategy enables the global classifier to differentiate features projected into all subspaces, even when confronted with misleading task-ids.

C. Optimization

The overall framework is trained by minimizing:

$$\mathcal{L} = \mathcal{L}_{ce} + \lambda_{fa} \mathcal{L}_{csfa} + \lambda_{ca} \mathcal{L}_{ca}. \quad (9)$$

For simplicity, the trade-off parameters λ_{fa} and λ_{ca} for loss terms are set to 1.0. During each training session, our framework can be optimized in an end-to-end fashion. The overall training pipeline is outlined in Alg.1.

D. Task-Confidence-guided Mixture of Adapters

Given the task-specific nature of our architecture, it is essential to aggregate the task-specific knowledge from relevant sub-modules during inference without task-ids. To achieve this objective, we leverage the task-shared knowledge from the task recognizer to perform an instance-adaptive aggregation for each testing image. Given an image $\mathbf{x}_i$ and its shared feature $\mathbf{f}_i$, we denote the output of the classifier g before softmax as

分类器级对齐 (CA)。基于通过 FA 增强的语义统一子空间, CA 旨在训练一个鲁棒的全局分类器, 使其能够即使在任务 ID 错误的情况下也区分所有子空间中的特征。具体而言, 在第 t_c 次会话中, 我们训练一个任务自适应分类器 g^{t_c} , 该分类器旨在避免当前数据 $\mathbf{x}_i \in \mathcal{D}_1 \dots \mathcal{D}_{t_c-1}$ 投影到当前特征子空间时出现歧义预测。为实现这一目标, 我们首先在当前子空间中模拟先前数据的特征, 然后利用这些模拟特征与当前特征共同训练 g^{t_c} 。我们进一步将仿真过程分解为三个步骤: (1) 通过任务共享网络对当前数据与先前数据在共享特征空间中的亲和力进行建模; (2) 对当前数据内部当前子空间与共享子空间之间的子空间偏移进行建模; (3) 通过亲和力矩阵将当前共享特征的子空间偏移量传递至先前特征。

由于建模亲和力关系需要预先获取数据分布, 但训练时并无预存样本, 我们通过高斯采样来解决这一问题。在第 t 次会话后, 对于第 t 个任务中的第 k 个类别, 我们将其在共享特征空间中的特征分布近似为高斯分布 $\mathcal{G}_{k,t}$ 。在第 t_c 次会话期间, 针对每个到达的批次, 我们采样得到伪表示 $\mathcal{S}_{sh} = \{\tilde{\mathbf{f}}_i, y(\tilde{\mathbf{f}}_i)\}_{i=1}^{B_s}$, 其大小为 B_s , 源自先前的共享特征分布 $\mathcal{G}_{k,t}$ 。结合当前特征 $\mathcal{C}_{sh} = \{\mathbf{f}_i, y(\mathbf{f}_i)\}_{i=1}^{B_s}$ 且 $\mathcal{C}_{sp} = \{\mathbf{z}_i, y(\mathbf{z}_i)\}_{i=1}^{B_s}$ 先前数据与当前数据在共享特征空间中的亲和力矩阵 $\mathbf{G} \in \mathbb{R}^{\{B_s \times B_s\}}$ 可表示为如下形式:

$$\mathbf{G}_{ij} = \begin{cases} \exp(\text{sim}(\tilde{\mathbf{f}}_i, \mathbf{f}_j)/\tau_g), & \text{if } \mathbf{f}_j \in \mathcal{K}(\tilde{\mathbf{f}}_i, K_g), \\ 0, & \text{otherwise,} \end{cases} \quad (5)$$

其中 τ_g 为温度因子; $\text{sim}(\mathbf{a}, \mathbf{b})$ 表示向量 $\mathbf{a}$ 与 $\mathbf{b}$ 之间的余弦相似度, 其中 $\text{sim}(\mathbf{a}, \mathbf{b}) = \mathbf{a}^\top \mathbf{b} / (\|\mathbf{a}\| \|\mathbf{b}\|)$; $\mathcal{K}(\tilde{\mathbf{f}}_i, K_g)$ 表示在 $\mathcal{C}_{sh}$ 中对 $\tilde{\mathbf{f}}_i$ 的 K_g 个最近邻; $\mathbf{G}_{ij}$ 表示采样特征 $\tilde{\mathbf{f}}_i$ 与当前特征 $\mathbf{f}_j$ 之间的亲和力。

当前子空间与共享特征空间之间当前数据的子空间偏移, 可简单地通过特定特征 $\{\mathbf{z}_i\}$ 之间的残差来建模。 $\}_{i=1}^{B_s}$ 共享特征 $\{\mathbf{f}_i\}_{i=1}^{B_s}$ 基于当前数据。随后, 我们将子空间偏移量从当前特征集 $\{\mathbf{f}_i\}_{i=1}^{B_s}$ 针对采样得到的伪特征集 $\{\tilde{\mathbf{f}}_i\}_{i=1}^{B_s}$ 。模拟特征集 $\{\tilde{\mathbf{z}}_i\}_{i=1}^{B_s}$ 的推导过程如下:

$$\tilde{\mathbf{z}}_i = \tilde{\mathbf{f}}_i + \Delta_i, \quad \Delta_i = \frac{\sum_{j=1}^{B_s} \mathbf{G}_{ij}(\mathbf{z}_j - \mathbf{f}_j)}{\sum_{j=1}^{B_s} \mathbf{G}_{ij}}. \quad (6)$$

通过上述过程, 我们推导出模拟特征集 $\mathcal{S}_{sp} = \{\tilde{\mathbf{z}}_i, y(\tilde{\mathbf{z}}_i)\}_{i=1}^{B_s}$ 。随后, 我们得到统一特征集 $\mathcal{U} = \mathcal{S}_{sh} \cup \mathcal{S}_{sp} \cup \mathcal{C}_{sh} \cup \mathcal{C}_{sp}$, 该集合包含了当前子空间中所有已观测任务所对应图像的特征。我们利用该集合来训练任务自适应分类器 g^{t_c} :

$$\mathcal{L}_{ca} = \frac{1}{|\mathcal{U}|} \sum_{\{\mathbf{f}, y(\mathbf{f})\} \in \mathcal{U}} \mathcal{L}(\sigma(g^{t_c}(\mathbf{f})), y(\mathbf{f})). \quad (7)$$

在完成第 t_c 次训练后, 为降低推理时的计算开销, 我们对所有训练批次的参数进行聚合。

算法 1 任务 t_c 的训练流程

Inputs: Dataset $\mathcal{D}_{t_c}$, backbone $\mathcal{F}$, shared prompts $\mathcal{P}_{sh}$, adapter $\mathcal{A}_{sp}^{t_c}$.
Outputs: Updated $\mathcal{P}_{sh}$, $\mathcal{A}_{sp}^{t_c}$, classifier g^{t_c} .
1: Initialize $\mathcal{A}_{sp}^{t_c}$ and g^{t_c} ; if $t_c = 1$, initialize $\mathcal{P}_{sh}$;
2: for $\{\mathbf{x}_i, y_i\} \in \mathcal{D}_{t_c}$ do
3: $\mathbf{f}_i = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh})$; // Shared features
4: $\mathbf{z}_i = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh}, \mathcal{A}_{sp}^{t_c})$; // Specific features
5: Compute $\mathcal{L}_{ce}$ (Eq.3);
6: if $t_c > 1$ then
7: Extract $\tilde{\mathbf{z}}_i^{t_c \rightarrow t_p} = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh}, \mathcal{A}_{sp}^{t_p})$;
8: Compute $\mathcal{L}_{csfa}$ (Eq.4); // Feature-level alignment
9: 从 $\mathcal{G}_{k,t}$ 中抽取样本 $\mathcal{S}_{sh}$;
10: 计算亲和力 $\mathbf{G}$ (等式5);
11: 模拟 $\mathcal{S}_{sp}$ (等式6);
12: 计算 $\mathcal{L}_{cu}$ (等式7); // 分类器级对齐
13: 结束 if
14: 计算总损失 L (等式9);
15: 更新 $\mathcal{P}_{sh}$, $\mathcal{A}_{sp}$, g^{t_c} ;
16: 结束
17: 对于类 $y \in Y^{t_c}$, 执行
18: 通过特征统计量计算 $\mathcal{G}_{y,t_c}$;
19: 结束
20: Build unified classifier g^u (Eq.8);

任务自适应分类器——即 $g^1, \dots, g^{t_c}$ ——用于构建一个适配所有任务特定特征子空间的全局分类器 g^u 。对于对应于第 t_p 个任务中第 k 类的参数 (即 $k \in Y^{t_p}$), 参数聚合可表述如下:

$$\mathbf{W}^u[k] = \frac{\sum_{t=t_p}^{t_c} \mathbf{W}^t[k]}{t_c - t_p + 1}, \quad \mathbf{b}^u[k] = \frac{\sum_{t=t_p}^{t_c} \mathbf{b}^t[k]}{t_c - t_p + 1}, \quad (8)$$

其中 $\mathbf{W}^u[k]$ ($\mathbf{b}^u[k]$) 表示 g^u 中第 k 类的权重 (偏置), 而 $\mathbf{W}^t[k]$ ($\mathbf{b}^t[k]$) 表示 g^t 中第 k 类的权重 (偏置)。这种聚合策略使全局分类器能够区分投射到所有子空间中的特征, 即使在面对误导性任务 ID 时亦然。

C. 最优化

整体框架通过最小化以下目标函数进行训练:

$$L = \mathcal{L}_{ce} + \lambda_{fa} \mathcal{L}_{csfa} + \lambda_{ca} \mathcal{L}_{ca}. \quad (9)$$

为简化起见, 损失项对应的权衡参数 λ_{fa} 与 λ_{ca} 均设为 1.0。在每次训练过程中, 我们的框架均可采用端到端方式进行优化。整体训练流程如算法.1 所示。

D. 任务置信度引导的适配器混合模型

鉴于本架构具有任务特定性, 因此在无任务 ID 的情况下进行推理时, 来自相关子模块的任务特定知识进行聚合至关重要。为实现这一目标, 我们利用任务识别器提供的任务共享知识, 对每张测试图像执行实例自适应聚合。给定图像 $\mathbf{x}_i$ 及其共享特征 $\mathbf{f}_i$, 我们将分类器在应用 Softmax 之前的输出记作 $\mathbf{g}$:

Algorithm 2 Inference pipeline after task t_c

Inputs: Test dataset $\{\mathbf{x}_i\}_{i=1}^N$, backbone $\mathcal{F}$, prompts $\mathcal{P}_{sh}$, adapters $\{\mathcal{A}_{sp}^t\}_{t=1}^{t_c}$, unified classifier g^u .

Outputs: Predictions $\{\hat{k}_i\}_{i=1}^N$.

- 1: **for** $\mathbf{x}_i$ in $\{\mathbf{x}_i\}_{i=1}^N$ **do**
- 2: $\mathbf{f}_i = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh});$ // Shared features
- 3: Compute task-confidence $\{P_t(t|\mathbf{f}_i)\}_{t=1}^{t_c}$ (Eq.10, 11);
- 4: $\hat{\mathbf{z}}_i = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh}, \{\mathcal{A}_{sp}^t\}, \{\alpha_i^t\});$ // TC-MoA aggregation
- 5: Predict label $\hat{k}_i$ via g^u and g (Eq.13);
- 6: **end for**

$g(\mathbf{f}_i) \in \mathbb{R}^K$, and its k -th element as $g(\mathbf{f}_i)[k]$. $K = \sum_{t=1}^{t_c} |\mathcal{Y}^t|$ is the total number of seen classes and t_c is the current task identity. To extract relevant information from each task, we only retain the classes that are semantically related to the testing image, while discarding others. Therefore, we only keep the largest K_c elements in $g(\mathbf{f}_i)$, which corresponds to the classes with the most similar semantics to the testing image:

$$\tilde{g}(\mathbf{f}_i)[k] = \begin{cases} g(\mathbf{f}_i)[k], & \text{if } g(\mathbf{f}_i)[k] \in \text{Top-}K_c(g(\mathbf{f}_i)), \\ -\infty, & \text{otherwise,} \end{cases} \quad (10)$$

where $\text{Top-}K_c(g(\mathbf{f}_i))$ denotes the top- K_c highest elements in $g(\mathbf{f}_i)$ and $K_c \ll K$. Subsequently, we derive the confidence score α_i^t of $\mathbf{f}_i$ in belonging to task t for $t \leq t_c$, representing the amount of semantic information relevant to the input contained within each task:

$$\alpha_i^t = P_t(t|\mathbf{f}_i) = \frac{\sum_{k \in \mathcal{Y}^t} \exp(\tilde{g}(\mathbf{f}_i)[k]/\tau)}{\sum_{j=1}^{J=K} \exp(\tilde{g}(\mathbf{f}_i)[j]/\tau)}, \quad (11)$$

where τ is a temperature to control the smoothness of the task-confidence scores. Afterwards, the scores $\{\alpha_i^t\}_{t=1}^{t_c}$ serves as routing weights to aggregate task-specific intermediate features, forming a Mixture-of-Adapter (MoA) architecture. At the l -th block, given the input $\mathbf{X}_{i,l}$ of $\mathbf{x}_i$, the forward process is formulated as follows:

$$\begin{cases} [\hat{\mathbf{X}}_{i,l}, \hat{\mathbf{P}}_l] = \text{MSA}([\mathbf{X}_{i,l}; \mathbf{P}_l]), \\ \mathbf{X}_{i,l+1} = \text{FFN}(\hat{\mathbf{X}}_{i,l}) + \sum_{t=1}^T \alpha_i^t \mathcal{A}_l^t(\hat{\mathbf{X}}_{i,l}). \end{cases} \quad (12)$$

Compared to assigning a single task to each test image, such a Mixture-of-Adapters (MoA) structure avoids overconfident aggregation, further enhancing the model's robustness against misleading task-ids. The aggregated feature after the L -th block is utilized for final classification, denoted as $\hat{\mathbf{z}}_i = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh}, \{\mathcal{A}_{sp}^t\}_{t=1}^{t_c}, \{\alpha_i^t\}_{t=1}^{t_c})$. The final prediction $\hat{k}_i$ is jointly determined by the global classifier g^u and the training classifier g :

$$\hat{k}_i = \arg \max_k \left[\frac{t_c}{t_c + 1} g^u(\hat{\mathbf{z}}_i)[k] + \frac{1}{t_c + 1} g(\hat{\mathbf{z}}_i)[k] \right]. \quad (13)$$

The inference of our CKAA includes deriving the task-confidence scores through the task-shared network and aggregating task-specific knowledge through the proposed TC-MoA. The overall inference pipeline is illustrated in Alg.2.

Analysis of the working mechanism of TC-MoA. We visualize the top-1 confidence score distribution of samples with correct and incorrect task-ids in Fig.4 on 10S-ImageNetA. For a test sample $\mathbf{x}_i^t$ from task t , its top-1 confidence score is denoted as $\alpha_i^{\hat{t}}$, where $\hat{t} = \arg \max_i \alpha_i^t$. For $\mathbf{x}_i^t$, if the predicted top-1 task matches its ground-truth task id, where $\hat{t} = t$, the sample is considered to have a correct prediction of task id.

As shown in Fig.4, samples with correct task-ids generally have higher top-1 scores. Therefore, for samples predicted with correct task-ids, TC-MoA assigns a single task-specific module with high confidence while neglecting others, acting as a **deterministic module selector**. For samples predicted with incorrect task-ids, TC-MoA adopts a **soft-label-like approach**, softly compensating for the incorrect modules through intermediate features from potential correct modules. Such a mechanism mitigates overconfidence in incorrect task-ids while preserving performance for correct ones, highlighting TC-MoA as a novel approach in handling incorrect task-ids in inference.

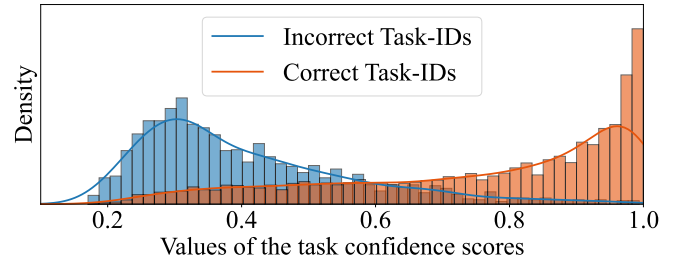


Fig. 4: Visualization of the distributions of the top-1 task-confidence scores.

IV. THEORETICAL ANALYSIS OF THE PROPOSED METHOD

We provide a theoretical analysis to explain why CKAA improves robustness against misleading task-ID predictions. The detailed derivations are provided in Appx.A. Consider the learning of task r , where $(\mathbf{x}, y) \sim \mathcal{D}_r$ denotes an input-label pair from the task distribution $\mathcal{D}_r$. Let $\mathcal{F}(\cdot)$ be the pre-trained backbone, $\mathcal{P}_{sh}$ be the task-shared prompt, and $\mathcal{A}_{sp}^t$ be the task-specific adapter of task t . The feature obtained by routing $\mathbf{x}$ to task t is denoted as $\mathbf{z}_t(\mathbf{x}) = \mathcal{F}(\mathbf{x}; \mathcal{P}_{sh}, \mathcal{A}_{sp}^t)$. Let $g(\cdot)$ be the classifier, $g_k(\cdot)$ be its logit for class k , and $h_g(\mathbf{z}) = \arg \max_k g_k(\mathbf{z})$ be the prediction function. The risk of classifying samples from task r through the subspace of task s is defined as $R_{r \rightarrow s}(g) = \Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} [h_g(\mathbf{z}_s(\mathbf{x})) \neq y]$. Therefore, the robustness objective of CKAA can be formulated as follows:

$$\mathcal{L}_r^{\text{robust}} = R_{r \rightarrow r}(g) + \frac{1}{r-1} \sum_{s=1}^{r-1} [\lambda_1 R_{r \rightarrow s}(g) + \lambda_2 R_{s \rightarrow r}(g)], \quad (14)$$

where λ_1 and λ_2 are trade-off hyper-parameters. The first term is optimized by standard classification training, while the latter two terms correspond to the new-to-old and old-to-new misrouting risks. CKAA reduces these two risks through feature-level alignment and classifier-level alignment, respectively.

Feature-level alignment. For $(\mathbf{x}, y)$, define the classification margins in the correct subspace r and an incorrect historical subspace s as $m_r(\mathbf{x}, y) = g_y(\mathbf{z}_r(\mathbf{x})) - \max_{k \neq y} g_k(\mathbf{z}_r(\mathbf{x}))$

算法 2 任务 t_c 后的推理流水线

输入: 测试数据集 $\{\mathbf{x}_i\}_{i=1}^N$ 主干网络 F 、提示词 $\mathcal{P}_{sh}$ 、适配器 $\{Atsp\}_{t=1}^c$ 统一分类器 g^u 。
输出: 预测结果 $\{\hat{k}_i\}_{i=1}^N$ 。
 1: 对于 $\mathbf{x}_i$ 属于 $\{\mathbf{x}_i\}_{i=1}^N$ 做
 2: $\mathbf{f}_i = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh});$ // 共享功能
 3: Compute task-confidence $\{P_t(t|\mathbf{f}_i)\}_{t=1}^{t_c}$ (Eq.10, 11);
 4: $\hat{\mathbf{z}}_i = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh}, \{A_{sp}^t\}, \{\alpha_i^t\});$ // TC-MoA 聚合
 5: 通过 g^u 与 g 预测标签 $\hat{k}_i$ (等式13);
 6: 结束

$g(\mathbf{f}_i)$ 耽 K ，其第 k 个元素记作 $g(\mathbf{f}_i)[k]$ 。 $K = \sum_{t=1}^c |Y^t|$ 表示已观测到的类别总数，而 t_c 表示当前任务标识。为从每个任务中提取相关信息，我们仅保留与测试图像在语义上相关的类别，其余类别则予以剔除。因此，我们仅保留 $g(\mathbf{f}_i)$ 中最大的 K_c 个元素，这些元素对应于与测试图像语义最相似的类别：

$$\tilde{g}(\mathbf{f}_i)[k] = \begin{cases} g(\mathbf{f}_i)[k], & \text{if } g(\mathbf{f}_i)[k] \in \text{Top-}K_c(g(\mathbf{f}_i)), \\ -\infty, & \text{otherwise,} \end{cases} \quad (10)$$

其中 $\text{Top-}K_c(g(\mathbf{f}_i))$ 表示 $g(\mathbf{f}_i)$ 中排名前 K_c 的元素，且 K_c 。 K 。随后，我们推导出 $\mathbf{f}_i$ 属于任务 t 的置信度评分 α_{ti} （适用于 $t \leq t_c$ ），该评分反映了每个任务中所包含的与输入相关的语义信息量：

$$\alpha_i^t = P_t(t|\mathbf{f}_i) = \frac{\sum_{k \in Y^t} \exp(\tilde{g}(\mathbf{f}_i)[k]/\tau)}{\sum_{j=1}^{j=K} \exp(\tilde{g}(\mathbf{f}_i)[j]/\tau)}, \quad (11)$$

其中 τ 为用于控制任务置信度得分平滑度的温度参数；随后，得到得分集合 $\{\alpha_{ti}\}_{t=1}^{t_c}$ 作为路由权重用于聚合任务特定的中间特征，从而构建出混合适配器（MoA）架构。在第 l 个模块中，给定输入 $\mathbf{X}_i$ （即 $\mathbf{x}_i$ 的第 l 个分量），前向过程表述如下：

$$\begin{cases} [\hat{\mathbf{X}}_{i,l}, \hat{\mathbf{P}}_l] = \text{MSA}([\mathbf{X}_{i,l}; \mathbf{P}_l]), \\ \mathbf{X}_{i,l+1} = \text{FFN}(\hat{\mathbf{X}}_{i,l}) + \sum_{t=1}^T \alpha_i^t A_l^t(\hat{\mathbf{X}}_{i,l}). \end{cases} \quad (12)$$

与将单个任务分配给每张测试图像相比，

此类混合适配器（MoA）结构可避免过拟合——任务 s 定义为 $R_r \rightarrow s(g) = \Pr(\mathbf{x}, y)_{D_r} [hg(\mathbf{z}_s(\mathbf{x})) \neq y]$ 。因此，CKAA 的鲁棒性目标可表述为：FIDENT聚合技术进一步增强了模型对误导性任务ID的鲁棒性，具体如下：

经聚合后的特征

L-th block $\mathbf{z}_i = \mathcal{F}(\mathbf{x}_i; \mathcal{P}_{sh}, \{A_{sp}^t\}_{t=1}^{t_c}, \{\alpha_i^t\}_{t=1}^{t_c})$ as the final prediction $\hat{k}_i$ is jointly determined by the global classification and the training classifier g :

$$\mathcal{L}_r^{\text{robust}} = R_{r \rightarrow r}(g) + \frac{1}{r-1} \sum_{s=1}^{r-1} [\lambda_1 R_{r \rightarrow s}(g) + \lambda_2 R_{s \rightarrow r}(g)], \quad (14)$$

$$\hat{k}_i = \arg \max_k \left[\frac{t_c}{t_c + 1} g^u(\hat{\mathbf{z}}_i)[k] + \frac{1}{t_c + 1} g(\hat{\mathbf{z}}_i)[k] \right]. \quad (13)$$

我们的 CKAA 框架包括：通过任务共享网络推导任务置信度分数，以及通过所提出的 TCMoA 方法聚合任务特定知识。整个推理流程如 Alg.2所示。

TC-MoA工作机理分析。我们在 10S-ImageNetA 数据集上，将具有正确与错误任务 ID 的样本的 Top-1 置信度得分分布可视化于图.4 中。

对于来自任务 t 的测试样本 $\mathbf{x}_{ti}$ ，其 top-1 置信度得分记作 α_{ti} ，其中 $\hat{t} = \arg \max_i \alpha_{ti}$ 。对于 $\mathbf{x}_{ti}$ ，若预测

当 top-1 任务匹配其真实任务 ID 时（即 $\hat{t} = t$ ），则认为该样本对任务 ID 的预测是正确的。

如图.4所示，具有正确任务ID的样本通常具有更高的 Top-1 分数。因此，对于被预测为具有正确任务ID的样本，TC-MoA 会为该样本分配一个置信度较高的、针对特定任务的模块，而忽略其他模块，从而充当一个**确定性模块选择器**；而对于被预测为具有错误任务ID的样本，TC-MoA 采用**一种类软标签方法**，通过来自潜在正确模块的中间特征对错误模块进行软性补偿。这种机制在减轻对错误任务ID的过度置信度的同时，仍能保持对正确任务ID的预测性能，凸显了 TC-MoA 在推理过程中处理错误任务ID的新颖性。

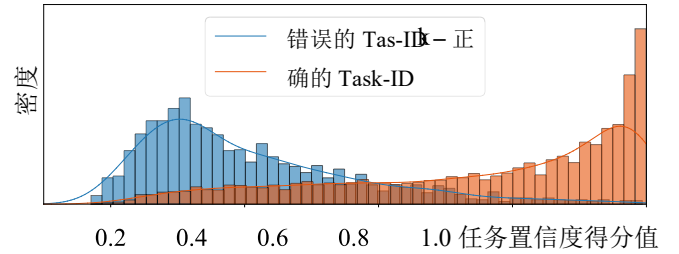


图 4: Top-1 任务置信度得分分布的可视化展示。

IV. 所提方法的理论分析：我们通过理论分析来阐明为何 CKAA 能增强对误导性任务ID预测的鲁棒性。

详细推导过程见附录A。考虑任务 r 的学习问题，其中 $(\mathbf{x}, y) \sim \mathcal{D}_r$ 表示来自任务分布 $\mathcal{D}_r$ 的一个输入-标签对。设 $F(\cdot)$ 为预训练主干网络， $\mathcal{P}_{sh}$ 为任务共享提示词， $Atsp$ 为任务 t 的任务特定适配器。通过将 $\mathbf{x}$ 路由至任务 t 而得到的特征记为 $\mathbf{z}_t(\mathbf{x}) = F(\mathbf{x}; \mathcal{P}_{sh}, Atsp)$ 。设 $g(\cdot)$ 为分类器， $g_k(\cdot)$ 为其对类别 k 的逻辑值， $h_g(\mathbf{z}) = \arg \max_k g_k(\mathbf{z})$ 为预测函数。风险

对来自任务 r 的样本进行基于子空间的分类

其中 λ_1 与 λ_2 为权衡超参数。第一项通过标准分类任务训练进行优化，而后两项分别对应新到旧及旧到新误路由风险；CKAA 通过特征层面对齐与分类器层面对齐分别降低这两种风险。

特征级对齐。对于点 $(\mathbf{x}, y)$ ，在正确子空间 r 与错误历史子空间 s 中定义分类间隔为 $m_r(\mathbf{x}, y) = g_y(\mathbf{z}_r(\mathbf{x})) - \max_{k \neq y} g_k(\mathbf{z}_r(\mathbf{x}))$ 与

and $m_s(\mathbf{x}, y) = g_y(\mathbf{z}_s(\mathbf{x})) - \max_{k \neq y} g_k(\mathbf{z}_s(\mathbf{x}))$, respectively. Assume that the logit difference $g_y(\mathbf{z}) - g_k(\mathbf{z})$ is L_g -Lipschitz with respect to $\mathbf{z}$, where L_g is the Lipschitz constant. Then, for any margin threshold $\gamma > 0$, we derive an upper bound:

$$R_{r \rightarrow s}(g) \leq \Pr[m_r(\mathbf{x}, y) < \gamma] + \Pr\left[\|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2 \geq \frac{\gamma}{L_g}\right]. \quad (15)$$

Eq. (15) shows that reducing the new-to-old misrouting risk requires both enlarging the correct-subspace margin and reducing the cross-subspace feature discrepancy. Specifically, our CSFA loss can be rewritten as follows:

$$\mathcal{L}_{\text{csfa}} = \frac{1}{B} \sum_{i=1}^B \log \left[\sum_{\mathbf{z}_j \in \mathcal{P}(\mathbf{z}_i) \cup \mathcal{N}(\mathbf{z}_i)} \exp\left(\frac{\text{sim}(\mathbf{z}_i, \mathbf{z}_j)}{\tau_f}\right) \right] - \frac{1}{B} \sum_{i=1}^B \log \left[\sum_{\mathbf{z}^+ \in \mathcal{P}(\mathbf{z}_i)} \exp\left(\frac{\text{sim}(\mathbf{z}_i, \mathbf{z}^+)}{\tau_f}\right) \right]. \quad (16)$$

Minimizing $\mathcal{L}_{\text{csfa}}$ suppresses the similarity between $\mathbf{z}_i$ and inter-class features $\mathbf{z}_j \in \mathcal{N}(\mathbf{z}_i)$, which encourages larger margins $m_r(\mathbf{x}, y)$ in the correct subspace. Meanwhile, it maximizes the similarity between $\mathbf{z}_i$ and intra-class positives $\mathbf{z}^+ \in \mathcal{P}(\mathbf{z}_i)$ from incorrect subspaces, reducing the cross-subspace discrepancy $\|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2$. Therefore, $\mathcal{L}_{\text{csfa}}$ jointly reduces the two terms in Eq. 15.

Classifier-level alignment. For an old task $s < r$, historical samples are unavailable when learning task r . Let $\mathbf{f} = \mathcal{F}(\mathbf{x}; \mathcal{P}_{sh})$ denote the task-shared feature of a historical sample, $\tilde{\mathbf{f}}_i$ denote a sampled historical feature from the class-wise Gaussian distribution, and $\tilde{\mathbf{z}}_i$ denote the simulated feature in the current subspace. Let g^r be the task-adaptive classifier after learning task r , and let L_{g^r} be the Lipschitz constant of its logit differences. Define the simulated margin as $\tilde{m}_{s \rightarrow r}(\tilde{\mathbf{z}}, y) = g_y^r(\tilde{\mathbf{z}}) - \max_{k \neq y} g_k^r(\tilde{\mathbf{z}})$. Then, for any $\gamma > 0$, the old-to-new misrouting risk is bounded by:

$$R_{s \rightarrow r}(g^r) \leq \Pr[\tilde{m}_{s \rightarrow r}(\tilde{\mathbf{z}}, y) < \gamma] + \frac{L_{g^r}}{\gamma} \left\{ (1 + L_\Delta) \mathbb{E}[\|\mathbf{f} - \tilde{\mathbf{f}}_i\|_2] + L_\Delta \mathbb{E} \left[\frac{\sum_{j=1}^B G_{ij} \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}} \right] \right\}. \quad (17)$$

The first term in Eq. 17 is explicitly reduced by training g^r with simulated features. The second term depends on the quality of feature simulation, which can be tightened via our Gaussian sampling and affinity-guided subspace transfer (the detailed analysis is provided in Appx.A).

TC-MoA. We further analyze why TC-MoA is more robust than deterministic maximum task selection. For a test sample $\mathbf{x}_i^t$ from task t , let α_i^t be the task confidence assigned to task t , and let $\hat{t} = \arg \max_{t'} \alpha_i^{t'}$ be the predicted task-ID. Define $p = \Pr(\hat{t} = t)$ as the task-ID accuracy, $\tau_1 = \mathbb{E}[\alpha_i^t | \hat{t} = t]$ as the expected confidence of the ground-truth task when the task-ID is correct, and $\tau_2 = \mathbb{E}[\alpha_i^t | \hat{t} \neq t]$ as the expected confidence of the ground-truth task when the top-1

task-ID is incorrect. Let $A_l^t(\cdot)$ be the adapter output of task t at Transformer block l , L be the number of blocks, $\mathbf{X}_{i,l}$ be the input tokens of $\mathbf{x}_i^t$ at block l , and ρ_l be the Lipschitz constant that measures how perturbations at block l affect the final prediction. The expected routing-induced error of TC-MoA is bounded by follows:

$$\mathbb{E}[\mathcal{E}_{\text{TC-MoA}}] \leq [p(1 - \tau_1) + (1 - p)(1 - \tau_2)] \cdot \max_{\mathbf{x}_i^t, t' \neq t} \sum_{l=1}^L \rho_l \|A_l^{t'}(\mathbf{X}_{i,l}) - A_l^t(\mathbf{X}_{i,l})\|_2, \quad (18)$$

$$\mathbb{E}[\mathcal{E}_{\text{TC-MoA}}] \leq \mathbb{E}[\mathcal{E}_{\text{Max}}] \quad \text{if } p < \frac{\tau_2}{1 - \tau_1 + \tau_2}.$$

Eq. (18) explains why TC-MoA tightens the routing-error bound under ambiguous task boundaries.

TABLE I: Empirical verification of the condition for TC-MoA.

Benchmark	Filtered	p	τ_1	τ_2	$\frac{\tau_2}{1 - \tau_1 + \tau_2}$	Satisfied
10S-ImageNetA	✗	0.541	0.843	0.247	0.611	✓ (0.611 > 0.541)
10S-ImageNetA	✓	0.096	0.546	0.253	0.358	✓ (0.358 > 0.096)
20S-ImageNetA	✗	0.484	0.815	0.224	0.548	✓ (0.548 > 0.484)
20S-ImageNetA	✓	0.042	0.476	0.208	0.284	✓ (0.284 > 0.042)
10S-ImageNetR	✗	0.847	0.953	0.313	0.869	✓ (0.869 > 0.847)
10S-ImageNetR	✓	0.030	0.541	0.310	0.403	✓ (0.403 > 0.030)
20S-ImageNetR	✗	0.830	0.916	0.292	0.777	✗ (0.777 < 0.830)
20S-ImageNetR	✓	0.067	0.680	0.275	0.462	✓ (0.462 > 0.067)

We empirically verify whether the theoretical condition $p < \tau_2/(1 - \tau_1 + \tau_2)$ holds across five CIL benchmarks. We report the results in Tab.XXI. “Filtered” denotes *excluding simple samples that can already be correctly classified using only the task-shared network*. Such samples do not require additional task-specific information and are therefore not the primary cases affected by adapter-routing errors. The results demonstrate that TC-MoA tightens the routing-error for ambiguous samples whose predictions genuinely depend on task-specific information. Further details are in Appx.A.C.

V. EXPERIMENTS

Dataset and Evaluation Metrics. We evaluate our proposed method on four public CIL benchmarks, namely ImagenetR [71], ImagenetA [72], CIFAR100 [73] and DomainNet [74]. ImagenetR comprises 30000 images of 200 classes, which share the same class names with Imagenet-21K [75] but belong to different domains. ImagenetA is a challenging dataset with 7500 images, which contains hard samples and exhibits significant class imbalance. Cifar100 is a commonly-used dataset in continual learning, which consists of 60000 32×32 images of 100 classes. DomainNet is a cross-domain dataset including images from 345 classes and 6 diverse domains. Following [3], [49], the 200 categories with the maximum number of images are selected for training and evaluation. We randomly split the datasets into 10 or 20 non-overlapping tasks. Note that for DomainNet, the 200 categories with the maximum number of images are selected for training and evaluation following [3], [49], [76].

Following existing PEFT-based CIL methods [2], [49], we evaluate the model performance through two commonly-used metrics: the accuracy of all classes after learning the last session (Last-acc), and the averaged accuracy of all incremental sessions (Avg-acc). We conduct all experiments under

$m_s(\mathbf{x}, y) = g_y(\mathbf{z}_s(\mathbf{x})) - \max_{k \neq y} g_k(\mathbf{z}_s(\mathbf{x}))$, 分别。假设对数几率差 $g_y(\mathbf{z}) - g_k(\mathbf{z})$ 关于 $\mathbf{z}$ 具有 L_g -Lipschitz 性质, 其中 L_g 为 Lipschitz 常数。那么, 对于任意边际阈值 $\gamma > 0$, 我们可推导出一个上界:

$$R_{r \rightarrow s}(g) \leq \Pr[m_r(\mathbf{x}, y) < \gamma] + \Pr\left[\|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2 \geq \frac{\gamma}{L_g}\right]. \quad (15)$$

式 (15) 表明, 降低新旧数据间误路由风险需要同时扩大正确子空间的边界并减小跨子空间特征差异。具体而言, 我们的 CSFA 损失函数可重写为以下形式:

$$\mathcal{L}_{\text{csfa}} = \frac{1}{B} \sum_{i=1}^B \log \left[\sum_{\mathbf{z}_j \in \mathcal{P}(\mathbf{z}_i) \cup \mathcal{N}(\mathbf{z}_i)} \exp\left(\frac{\text{sim}(\mathbf{z}_i, \mathbf{z}_j)}{\tau_f}\right) \right] - \frac{1}{B} \sum_{i=1}^B \log \left[\sum_{\mathbf{z}^+ \in \mathcal{P}(\mathbf{z}_i)} \exp\left(\frac{\text{sim}(\mathbf{z}_i, \mathbf{z}^+)}{\tau_f}\right) \right]. \quad (16)$$

最小化 $\mathcal{L}_{\text{csfa}}$ 可抑制 $\mathbf{z}_i$ 与类间特征 $\mathbf{z}_j \in \mathcal{N}(\mathbf{z}_i)$ 之间的相似性, 从而促使在正确子空间中获得更大的间隔 $m_r(\mathbf{x}, y)$; 同时, 它最大化了 $\mathbf{z}_i$ 与来自错误子空间的类内正样本 $\mathbf{z}^+ \in \mathcal{P}(\mathbf{z}_i)$ 之间的相似性, 从而减小跨子空间差异 $\|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2$ 。因此, $\mathcal{L}_{\text{csfa}}$ 可联合减小等式 15 中的两项。

分类器级对齐。对于旧任务 $s < r$, 在学习任务 r 时无法获取历史样本。令 $\mathbf{f} = F(\mathbf{x}; P_{sh})$ 表示历史样本的任务共享特征, $\tilde{\mathbf{f}}$ 表示从类别-wise 高斯分布中采样的历史特征, 而 $\tilde{\mathbf{z}}$ 表示当前子空间中的模拟特征。令 g^r 表示在学习任务 r 后得到的任务自适应分类器, 且令 L_{g^r} 表示其 logit 差值的 Lipschitz 常数。定义模拟边际为 $m_{s \rightarrow r}(\tilde{\mathbf{z}}, y) = g^r(y(\tilde{\mathbf{z}})) - \max_{k \neq y} g^r(k(\tilde{\mathbf{z}}))$ 。那么, 对于任意 $\gamma > 0$, 旧路由至新路由的误路由风险有界于:

$$R_{s \rightarrow r}(g^r) \leq \Pr[\tilde{m}_{s \rightarrow r}(\tilde{\mathbf{z}}, y) < \gamma] + \frac{L_{g^r}}{\gamma} \left\{ (1 + L_{\Delta}) \mathbb{E} \left[\|\mathbf{f} - \tilde{\mathbf{f}}\|_2 \right] + L_{\Delta} \mathbb{E} \left[\frac{\sum_{j=1}^B G_{ij} \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}} \right] \right\}. \quad (17)$$

等式 17 中的第一项通过使用模拟特征对 g^r 进行训练而显式简化; 第二项取决于特征模拟的质量, 这一项可通过我们的高斯采样与亲和力引导的子空间迁移方法加以收紧 (详细分析见附录 A)。

TC-MoA。我们进一步分析为何 TC-MoA 比确定性最大任务选择方法更具鲁棒性。对于来自任务 t 的测试样本 $\mathbf{x}_{ti}$, 令 $\hat{ati}$ 表示分配给任务 t 的任务置信度, 令 $\hat{t} = \arg \max_t \hat{ati}$ 表示预测的任务 ID。定义 $p = \Pr(\hat{t} = t)$ 为任务 ID 准确率, $\tau_1 = \mathbb{E}[\hat{ati} | \hat{t} = t]$ 表示当任务 ID 正确时真实任务的预期置信度, 而 $\tau_2 = \mathbb{E}[\hat{ati} | \hat{t} \neq t]$ 表示当 top-1 任务-ID 不正确时, 真实任务的

预期置信度。设 $Atl(\cdot)$ 为任务 t 在 Transformer 第 l 个块中的适配器输出, L 为块的数量, $\mathbf{X}_{i,l}$ 为第 l 个块中 $\mathbf{x}_{ti}$ 的输入标记, ρ_l 为衡量第 l 个块中的扰动对最终预测影响程度的 Lipschitz 常数。TCMoA 的预期路由诱导误差有如下界:

$$\mathbb{E}[\mathcal{E}_{\text{TC-MoA}}] \leq [p(1 - \tau_1) + (1 - p)(1 - \tau_2)] \cdot \max_{\mathbf{x}_i^t, t' \neq t} \sum_{l=1}^L \rho_l \|A_l^t(\mathbf{X}_{i,l}) - A_l^{t'}(\mathbf{X}_{i,l})\|_2, \quad (18)$$

$$\frac{\tau_2}{1 - \tau_1 + \tau_2}.$$

式 (18) 阐明了在任务边界不明确的情况下, TC-MoA 如何收紧路由误差界。

表 I: TC-MoA 条件的实证验证。

基准	已过滤	p	τ_1	τ_2	$\frac{\tau_2}{1 - \tau_1 + \tau_2}$	满意
10S-ImageNetA	✗	0.541	0.843	0.247	0.611	✓ (0.611 > 0.541)
10S-ImageNetA	✓	0.096	0.546	0.253	0.358	✓ (0.358 > 0.096)
20S-ImageNetA	✗	0.484	0.815	0.224	0.548	✓ (0.548 > 0.484)
20S-ImageNetA	✓	0.042	0.476	0.208	0.284	✓ (0.284 > 0.042)
10S-ImageNetR	✗	0.847	0.953	0.313	0.869	✓ (0.869 > 0.847)
10S-ImageNetR	✓	0.030	0.541	0.310	0.403	✓ (0.403 > 0.030)
20S-ImageNetR	✗	0.830	0.916	0.292	0.777	✗ (0.777 < 0.830)
20S-ImageNetR	✓	0.067	0.680	0.275	0.462	✓ (0.462 > 0.067)

我们通过实证检验了理论条件 $p < \tau_2 / (1 - \tau_1 + \tau_2)$ 在五个 CIL 基准测试集上是否成立。相关结果列于表 XXI 中。

“Filtered” 表示 **排除那些仅通过使用任务共享网络即可正确分类的简单样本**; 此类样本无需额外的任务特定信息, 因此并非受适配器路由错误影响的主要样本类型。实验结果表明, TC-MoA 可有效抑制对于预测结果确实依赖于任务特定信息的歧义样本所产生的路由错误。更多细节请参阅附录 A.C。

V. 实验

数据集与评估指标。我们在四个公开的持续学习基准数据集上对所提出的方法进行评估, 即 ImagenetR[71]、ImagenetA[72]、CIFAR100[73]和 DomainNet[74]。ImagenetR 包含来自 200 个类别的 30,000 张图像, 这些类别与 Imagenet-21 K[75]具有相同的类别名称, 但属于不同的领域。ImagenetA 是一个具有挑战性的数据集, 包含 7,500 张图像, 其中包含困难样本且存在显著的类别不平衡问题。CIFAR100 是持续学习领域中常用的数据集, 包含来自 100 个类别的 60,000 张 32x32 尺寸的图像。DomainNet 是一个跨领域数据集, 包含来自 345 个类别和 6 个不同领域的图像。根据[3], [49]选取图像数量最多的 200 个类别用于训练和评估。我们将数据集随机划分为 10 个或 20 个互不重叠的任务。请注意, 对于 DomainNet 数据集, 我们按照[3]的方法, 选取图像数量最多的 200 个类别用于训练和评估。[49], [76]。

基于现有基于 PEFT 的 CIL 方法[2], [49]我们通过两项常用指标来评估模型性能: 学习最后一个会话后所有类别的准确率 (Last-acc) 以及所有增量会话的平均准确率 (Avg-acc)。所有实验均在表 II “四个 CIL 基准测试集

TABLE II: Experimental results on four CIL benchmarks. We report the averaged results over 3 trials. VPS-NSP[†] denotes our reproduction of the results under the same settings and seeds in our method. The highest results are in **bold**, and the second best results are underlined.

Method	Venue	10s-ImageNetR		10s-ImageNetA		10s-CIFAR100		10s-DomainNet	
		Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$
L2P [20]	CVPR-22	72.34 $\pm$ 0.17	77.36 $\pm$ 0.64	44.04 $\pm$ 0.93	51.24 $\pm$ 2.26	84.06 $\pm$ 0.88	88.26 $\pm$ 1.34	81.17 $\pm$ 0.83	87.43 $\pm$ 0.95
DualPrompt [23]	ECCV-22	69.10 $\pm$ 0.62	74.28 $\pm$ 0.66	53.19 $\pm$ 0.74	64.59 $\pm$ 0.08	86.93 $\pm$ 0.24	91.13 $\pm$ 0.32	81.70 $\pm$ 0.78	87.80 $\pm$ 0.99
ADA [42]	NIPS-22	73.76 $\pm$ 0.27	79.57 $\pm$ 0.84	50.16 $\pm$ 0.20	59.43 $\pm$ 2.20	88.25 $\pm$ 0.26	91.85 $\pm$ 1.32	—	—
CODAPrompt [24]	CVPR-23	73.31 $\pm$ 0.50	78.47 $\pm$ 0.53	52.08 $\pm$ 0.12	63.92 $\pm$ 0.12	83.21 $\pm$ 3.39	87.71 $\pm$ 3.17	80.04 $\pm$ 0.79	86.27 $\pm$ 0.82
LAE [62]	ICCV-23	72.29 $\pm$ 0.14	77.99 $\pm$ 0.46	47.18 $\pm$ 1.17	58.15 $\pm$ 0.73	85.25 $\pm$ 0.43	89.80 $\pm$ 1.20	—	—
SLCA [48]	ICCV-23	79.35 $\pm$ 0.28	83.29 $\pm$ 0.46	61.05 $\pm$ 0.63	68.88 $\pm$ 2.31	91.26 $\pm$ 0.37	94.29 $\pm$ 0.92	—	—
Adam-adapter [63]	IJCV-24	65.79 $\pm$ 0.98	72.42 $\pm$ 1.41	48.81 $\pm$ 0.08	58.84 $\pm$ 1.37	87.29 $\pm$ 0.27	91.21 $\pm$ 1.33	—	—
RAPF [64]	ECCV-24	<u>79.62</u>	<u>86.28</u>	—	—	79.04	86.19	—	—
InfLoRA [21]	CVPR-24	75.65 $\pm$ 0.14	80.82 $\pm$ 0.24	47.04 $\pm$ 0.90	56.91 $\pm$ 1.27	86.51 $\pm$ 0.73	91.70 $\pm$ 0.32	81.45 $\pm$ 0.68	—
EASE [25]	CVPR-24	71.43 $\pm$ 0.18	78.04 $\pm$ 0.67	59.25 $\pm$ 0.88	68.92 $\pm$ 2.06	85.71 $\pm$ 0.76	90.96 $\pm$ 0.83	—	—
CPrompt [49]	CVPR-24	77.14 $\pm$ 0.11	82.92 $\pm$ 0.70	55.15 $\pm$ 0.85	65.49 $\pm$ 0.52	87.82 $\pm$ 0.21	92.53 $\pm$ 0.23	82.97 $\pm$ 0.34	88.54 $\pm$ 0.41
SSIAT [2]	CVPR-24	79.38 $\pm$ 0.59	83.63 $\pm$ 0.43	<u>62.43</u> $\pm$ 1.63	<u>70.83</u> $\pm$ 1.63	<u>91.35</u> $\pm$ 0.26	<u>94.35</u> $\pm$ 0.60	<u>85.11</u> $\pm$ 0.56	<u>89.80</u> $\pm$ 0.34
VQ-Prompt [28]	NeurIPS-24	78.71 $\pm$ 0.22	83.24 $\pm$ 0.68	—	—	88.73 $\pm$ 0.27	92.84 $\pm$ 0.73	—	—
VPT-NSP [†] [3]	NeurIPS-24	77.95 $\pm$ 0.22	83.44 $\pm$ 0.40	53.83 $\pm$ 0.37	63.93 $\pm$ 1.08	90.19 $\pm$ 0.43	94.12 $\pm$ 1.05	83.05 $\pm$ 1.06	87.83 $\pm$ 0.92
DIA [65]	CVPR-25	79.03	85.61	61.69	<u>71.58</u>	90.80	94.29	—	—
ACMap [66]	CVPR-25	70.49	77.31	56.19	65.19	87.81	92.04	—	—
SEMA [67]	CVPR-25	78.00	83.56	53.32	64.53	86.98	91.37	—	—
LoRA-DRS [68]	CVPR-25	74.74 $\pm$ 0.78	81.16 $\pm$ 0.59	—	—	89.14 $\pm$ 0.23	92.55 $\pm$ 0.25	—	—
CL-LoRA [15]	CVPR-25	—	—	60.54 $\pm$ 0.63	70.15 $\pm$ 2.23	85.32 $\pm$ 0.08	91.02 $\pm$ 0.12	—	—
CKAA (Ours)	—	82.41 $\pm$ 0.13	87.20 $\pm$ 0.35	63.53 $\pm$ 0.66	71.11 $\pm$ 1.03	92.26 $\pm$ 0.39	95.18 $\pm$ 0.86	86.72 $\pm$ 0.55	90.68 $\pm$ 0.78

TABLE III: Experimental results under the long-term setting (20 incremental tasks) on ImageNetR and ImageNetA.

Method	20s-ImageNetR		20s-ImageNetA	
	Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$
L2P [20]	69.64 $\pm$ 0.42	75.28 $\pm$ 0.57	40.48 $\pm$ 1.78	49.62 $\pm$ 1.46
DualPrompt [23]	66.61 $\pm$ 0.58	72.45 $\pm$ 0.37	42.28 $\pm$ 1.94	53.39 $\pm$ 1.64
CODAPrompt [24]	69.96 $\pm$ 0.50	75.34 $\pm$ 0.85	44.62 $\pm$ 1.92	54.86 $\pm$ 0.50
LAE [62]	69.86 $\pm$ 0.43	77.38 $\pm$ 0.61	39.52 $\pm$ 0.78	51.75 $\pm$ 2.15
SLCA [48]	74.63 $\pm$ 1.55	79.92 $\pm$ 1.29	36.69 $\pm$ 21.31	56.35 $\pm$ 7.09
Adam-adapter [63]	57.42 $\pm$ 0.84	64.75 $\pm$ 0.79	48.65 $\pm$ 0.12	59.55 $\pm$ 1.07
InfLoRA [21]	71.01 $\pm$ 0.45	77.28 $\pm$ 0.45	—	—
CPrompt [49]	74.79 $\pm$ 0.28	81.46 $\pm$ 0.93	—	—
SSIAT [2]	75.67 $\pm$ 0.14	82.30 $\pm$ 0.36	59.16 $\pm$ 1.03	68.45 $\pm$ 1.92
VPT-NSP [†] [3]	75.69 $\pm$ 0.61	81.87 $\pm$ 0.59	49.81 $\pm$ 1.29	61.41 $\pm$ 1.84
DIA [65]	76.32	83.51	—	—
CKAA (Ours)	80.98 $\pm$ 0.54	86.28 $\pm$ 0.18	60.54 $\pm$ 0.50	70.55 $\pm$ 0.64

TABLE IV: Results under the long-term setting (50 & 100 tasks) on ImageNetR and DomainNet. Last-acc is reported.

Method	S-ImageNetR		S-DomainNet	
	50s	100s	50s	100s
L2P [20]	51.38	41.51	63.13	54.83
OVOR-Deep [69]	63.25	43.02	68.29	52.09
ConvPrompt [70]	64.61	44.32	71.76	56.21
InfLoRA [21]	62.81	42.23	71.87	48.06
Cprompt [49]	70.75	59.90	70.74	<u>57.60</u>
EASE [25]	70.27	51.56	65.34	37.56
VPT-NSP [†] [3]	69.48	<u>62.23</u>	71.28	57.35
CKAA (Ours)	78.20 $\pm$ 0.65	73.41 $\pm$ 0.51	81.53 $\pm$ 2.60	77.28 $\pm$ 3.09

three different random seeds. The average performance and the maximum deviation are reported.

Implementation Details. Experiments are conducted on a single NVIDIA RTX 3090 GPU. We use ViT-B/16 [60] as our backbone, which is pre-trained on ImageNet-21K [75]. The Adam optimizer with an initial learning rate of 0.01 is employed for training. We train each task for 100 epochs and the mini-batch is 110. Following [2], [3], [21], the PEFT modules are inserted into all transformer blocks to avoid searching. The prompt length is 4 and the width of adapters are 64. In DKA, τ_f in Eq.4 is set to 0.05, τ_g and K_g in Eq.5 are 0.2 and 20. In TC-MoA, K_c in Eq.10 is set to 20 for ImageNetA and 10 for others, and τ in Eq.11 is set to 3.0.

TABLE V: Experimental results on 10S-CUB-200.

Method	Venue	Last-acc $\uparrow$	Avg-acc $\uparrow$
L2P [20]	CVPR-22	67.02 $\pm$ 1.90	79.62 $\pm$ 1.60
DualPrompt [23]	ECCV-22	68.48 $\pm$ 0.47	80.59 $\pm$ 1.50
CODAPrompt [24]	CVPR-23	77.23 $\pm$ 1.12	81.90 $\pm$ 0.85
LAE [62]	ICCV-23	80.97 $\pm$ 0.51	87.22 $\pm$ 1.21
SLCA [48]	ICCV-23	84.68 $\pm$ 0.09	90.77 $\pm$ 0.79
Adam-adapter [63]	IJCV-24	85.84 $\pm$ 0.08	91.33 $\pm$ 0.49
EASE [25]	CVPR-24	86.81	92.23
SSIAT [2]	CVPR-24	88.75 $\pm$ 0.38	93.00 $\pm$ 0.90
VPT-NSP [†] [3]	NIPS-24	83.08 $\pm$ 0.75	90.32 $\pm$ 0.62
CoFiMa [77]	ECCV-24	87.11 $\pm$ 0.56	91.87 $\pm$ 0.69
DIA [65]	CVPR-25	86.73	93.21
ACMap [66]	CVPR-25	86.66	91.56
CKAA (Ours)	—	89.48 $\pm$ 0.46	93.34 $\pm$ 0.26

A. Comparison with the State-of-the-Arts

We compare our CKAA with state-of-the-art PEFT-based CIL methods, as shown in Tab.II, Tab.III and Tab.IV. We compare our method with prompt-based methods [3], [20], [23], [24], adapter-based methods [2], [42], [63] and other PEFT-based methods [21], [64]. Tab.II shows the results of the standard 10-split incremental setting on four commonly-used benchmarks. Notably, our proposed method outperforms the state-of-the-art method SSAIT by 3.03%, 1.10%, 0.91%, and 1.61% in Last-acc on four datasets, respectively.

To further demonstrate the scalability of our method under long-term settings, we conduct experiments under the more challenging settings of 20, 50, and 100 tasks on ImageNetR, ImageNetA, and DomainNet. The results are shown in Tab.III and Tab.IV. Under the 20-split setting, our method outperforms SSAIT by 5.31% and 1.38% in Last-acc on two datasets. While under the 50-split and 100-split settings, our proposed CKAA exhibits superior performances, which surpasses VPT-NSP by 8.71% and 11.18% in Last-Acc on the ImageNetR dataset. For the 50-split and 100-split settings, the intermediate dimensions of adapters are set to 16 and 8 to avoid additional inference overhead. The results highlight its scalability and ability to address forgetting in long-term CIL problems.

上的实验结果”中进行；我们报告了3次独立实验的平均结果。VPS - NSP†表示我们在与原方法相同的实验设置和随机种子下复现的实验结果。最高性能结果以**粗体**标出，次优结果则以下划线标出。

方法	发生地	10s-ImageNetR		10s-ImageNetA		10s-CIFAR100		10s-DomainNet	
		最后到达↑	Avg-acc ↑	最后到达↑	Avg-acc ↑	最后到达↑	Avg-acc ↑	最后到达↑	Avg-acc ↑
L2P [20]	CVPR-22	72.34 $\pm$ 0.17	77.36 $\pm$ 0.64	44.04 $\pm$ 0.93	51.24 $\pm$ 2.26	84.06 $\pm$ 0.88	88.26 $\pm$ 1.34	81.17 $\pm$ 0.83	87.43 $\pm$ 0.95
DualPrompt [23]	ECCV-22	69.10 $\pm$ 0.62	74.28 $\pm$ 0.66	53.19 $\pm$ 0.74	64.59 $\pm$ 0.08	86.93 $\pm$ 0.24	91.13 $\pm$ 0.32	81.70 $\pm$ 0.78	87.80 $\pm$ 0.99
ADA [42]	NIPS-22	73.76 $\pm$ 0.27	79.57 $\pm$ 0.84	50.16 $\pm$ 0.20	59.43 $\pm$ 2.20	88.25 $\pm$ 0.26	91.85 $\pm$ 1.32	—	—
CODAPrompt [24]	CVPR-23	73.31 $\pm$ 0.50	78.47 $\pm$ 0.53	52.08 $\pm$ 0.12	63.92 $\pm$ 0.12	83.21 $\pm$ 3.39	87.71 $\pm$ 3.17	80.04 $\pm$ 0.79	86.27 $\pm$ 0.82
LAE [62]	ICCV-23	72.29 $\pm$ 0.14	77.99 $\pm$ 0.46	47.18 $\pm$ 1.17	58.15 $\pm$ 0.73	85.25 $\pm$ 0.43	89.80 $\pm$ 1.20	—	—
SLCA [48]	ICCV-23	79.35 $\pm$ 0.28	83.29 $\pm$ 0.46	61.05 $\pm$ 0.63	68.88 $\pm$ 2.31	91.26 $\pm$ 0.37	94.29 $\pm$ 0.92	—	—
Adam适配器[63]	IJCV-24	65.79 $\pm$ 0.98	72.42 $\pm$ 1.41	48.81 $\pm$ 0.08	58.84 $\pm$ 1.37	87.29 $\pm$ 0.27	91.21 $\pm$ 1.33	—	—
RAPF [64]	ECCV-24	79.62	86.28	—	—	79.04	86.19	—	—
InfLoRA [21]	CVPR-24	75.65 $\pm$ 0.14	80.82 $\pm$ 0.24	47.04 $\pm$ 0.90	56.91 $\pm$ 1.27	86.51 $\pm$ 0.73	91.70 $\pm$ 0.32	81.45 $\pm$ 0.68	—
轻松[25]	CVPR-24	71.43 $\pm$ 0.18	78.04 $\pm$ 0.67	59.25 $\pm$ 0.88	68.92 $\pm$ 2.06	85.71 $\pm$ 0.76	90.96 $\pm$ 0.83	—	—
CPrompt [49]	CVPR-24	77.14 $\pm$ 0.11	82.92 $\pm$ 0.70	55.15 $\pm$ 0.85	65.49 $\pm$ 0.52	87.82 $\pm$ 0.21	92.53 $\pm$ 0.23	82.97 $\pm$ 0.34	88.54 $\pm$ 0.41
SSIAT [2] VQ-Prompt [28]	CVPR-24	79.38 $\pm$ 0.59	83.63 $\pm$ 0.43	62.43 $\pm$ 1.63	70.83 $\pm$ 1.63	91.35 $\pm$ 0.26	94.35 $\pm$ 0.60	85.11 $\pm$ 0.56	89.80 $\pm$ 0.34
VPT-NSP† [3]	NeurIPS-24	78.71 $\pm$ 0.22	83.24 $\pm$ 0.68	—	—	88.73 $\pm$ 0.27	92.84 $\pm$ 0.73	—	—
DIA [65]	NeurIPS-24	77.95 $\pm$ 0.22	83.44 $\pm$ 0.40	53.83 $\pm$ 0.37	63.93 $\pm$ 1.08	90.19 $\pm$ 0.43	94.12 $\pm$ 1.05	83.05 $\pm$ 1.06	87.83 $\pm$ 0.92
ACMap [66]	CVPR-25	79.03	85.61	61.69	71.58	90.80	94.29	—	—
SEMA [67]	CVPR-25	70.49	77.31	56.19	65.19	87.81	92.04	—	—
LoRA-DRS [68]	CVPR-25	78.00	83.56	53.32	64.53	86.98	91.37	—	—
CL-LoRA [15]	CVPR-25	74.74 $\pm$ 0.78	81.16 $\pm$ 0.59	—	—	89.14 $\pm$ 0.23	92.55 $\pm$ 0.25	—	—
CKAA (我们)	—	—	—	60.54 $\pm$ 0.63	70.15 $\pm$ 2.23	85.32 $\pm$ 0.08	91.02 $\pm$ 0.12	—	—
CKAA (我们)	—	82.41 $\pm$ 0.13	87.20 $\pm$ 0.35	63.53 $\pm$ 0.66	71.11 $\pm$ 1.03	92.26 $\pm$ 0.39	95.18 $\pm$ 0.86	86.72 $\pm$ 0.55	90.68 $\pm$ 0.78

表 III：在 ImageNetR 和 ImageNetA 数据集上进行的长期场景（20 个增量任务）下的实验结果。

方法	20s-ImageNetR		20s-ImageNetA	
	最后到达↑	Avg-acc ↑	最后到达↑	Avg-acc ↑
L2P [20]	69.64 $\pm$ 0.42	75.28 $\pm$ 0.57	40.48 $\pm$ 1.78	49.62 $\pm$ 1.46
DualPrompt [23]	66.61 $\pm$ 0.58	72.45 $\pm$ 0.37	42.28 $\pm$ 1.94	53.39 $\pm$ 1.64
CODAPrompt [24]	69.96 $\pm$ 0.50	75.34 $\pm$ 0.85	44.62 $\pm$ 1.92	54.86 $\pm$ 0.50
LAE [62]	69.86 $\pm$ 0.43	77.38 $\pm$ 0.61	39.52 $\pm$ 0.78	51.75 $\pm$ 2.15
SLCA [48]	74.63 $\pm$ 1.55	79.92 $\pm$ 1.29	36.69 $\pm$ 21.31	56.35 $\pm$ 7.09
Adam适配器[63]	57.42 $\pm$ 0.84	64.75 $\pm$ 0.79	48.65 $\pm$ 0.12	59.55 $\pm$ 1.07
InfLoRA [21]	71.01 $\pm$ 0.45	77.28 $\pm$ 0.45	—	—
CPrompt [49]	74.79 $\pm$ 0.28	81.46 $\pm$ 0.93	—	—
SSIAT [2]	75.67 $\pm$ 0.14	82.30 $\pm$ 0.36	59.16 $\pm$ 1.03	68.45 $\pm$ 1.92
VPT-NSP† [3]	75.69 $\pm$ 0.61	81.87 $\pm$ 0.59	49.81 $\pm$ 1.29	61.41 $\pm$ 1.84
DIA [65]	76.32	83.51	—	—
CKAA (我们)	80.98 $\pm$ 0.54	86.28 $\pm$ 0.18	60.54 $\pm$ 0.50	70.55 $\pm$ 0.64

表 IV：在 ImageNetR 和 DomainNet 数据集上进行的长期场景（50 项与 100 项任务）下的实验结果。报告了最终准确率（Last-acc）。

方法	S-ImageNetR		S-DomainNet	
	50s	100s	50s	100s
L2P [20]	51.38	41.51	63.13	54.83
OVOR-Deep [69]	63.25	43.02	68.29	52.09
ConvPrompt [70]	64.61	44.32	71.76	56.21
InfLoRA [21]	62.81	42.23	71.87	48.06
CPrompt [49]	70.75	59.90	70.74	<u>57.60</u>
轻松[25]	70.27	51.56	65.34	37.56
VPT-NSP† [3]	69.48	<u>62.23</u>	71.28	57.35
CKAA (我们)	78.20 $\pm$ 0.65	73.41 $\pm$ 0.51	81.53 $\pm$ 2.60	77.28 $\pm$ 3.09

采用三种不同的随机种子；文中报告了平均性能与最大偏差。

实现细节。实验在单块 NVIDIA RTX 3090 GPU 上进行。我们采用 ViT-B/16[60]作为主干网络，该网络已在 ImageNet-21K [75]数据集上进行预训练。训练过程中使用 Adam 优化器，初始学习率为 0.01。每个任务训练 100 个 epoch，小批量大小为 110。遵循[2], [3], [21]将 PEFT 模块插入所有变压器块中以避免搜索操作。提示长度为 4，适配器宽度为 64。在 DKA 中，公式.4中的 τ_f 设为 0.05；公式.5中的 τ_g 与 K_g 分别为 0.2 和 20。在 TC-MoA 中，公式.10中的 K_c 对于 ImageNetA 设为 20，对于其他数据集设为 10；而公式.11中的 τ 设为 3.0。

表 V：10S-CUB-200 上的实验结果。

方法	发生地	最后到达↑	Avg-acc ↑
L2P [20]	CVPR-22	67.02 $\pm$ 1.90	79.62 $\pm$ 1.60
DualPrompt [23]	ECCV-22	68.48 $\pm$ 0.47	80.59 $\pm$ 1.50
CODAPrompt [24]	CVPR-23	77.23 $\pm$ 1.12	81.90 $\pm$ 0.85
LAE [62]	ICCV-23	80.97 $\pm$ 0.51	87.22 $\pm$ 1.21
SLCA [48]	ICCV-23	84.68 $\pm$ 0.09	90.77 $\pm$ 0.79
Adam适配器[63]	IJCV-24	85.84 $\pm$ 0.08	91.33 $\pm$ 0.49
轻松[25]	CVPR-24	86.81	92.23
SSIAT [2] VPT-NSP† [3]	CVPR-24	88.75 $\pm$ 0.38	93.00 $\pm$ 0.90
CoFiMa [77]	NIPS-24	83.08 $\pm$ 0.75	90.32 $\pm$ 0.62
DIA [65]	ECCV-24	87.11 $\pm$ 0.56	91.87 $\pm$ 0.69
ACMap [66]	CVPR-25	86.73	93.21
CKAA (我们)	CVPR-25	86.66	91.56
CKAA (我们)	—	89.48 $\pm$ 0.46	93.34 $\pm$ 0.26

A. 与当前技术水平的对比

我们将提出的 CKAA 方法与基于 PEFT 的先进 CIL 方法进行对比，如表II、表III及表IV所示。此外，我们还将本方法与基于提示的学习方法[3]进行对比。[20],[23], [24]基于适配器的方法[2], [42], [63] 以及其他基于 PEFT 的方法[21], [64]表II展示了标准的 10 分割增量设置在四个常用基准数据集上的实验结果。值得注意的是，本文提出的方法在 Last-acc 指标上分别比当前最先进的方法 SSAIT 高出 3.03%、1.10%、0.91% 和 1.61%。

为进一步验证本方法在长期场景下的可扩展性，我们针对 ImagenetR、ImageNetA 和 DomainNet 数据集，在更具挑战性的 20、50 和 100 个任务规模下开展了实验。实验结果如表III和表IV所示。在 20 个任务的划分场景下，本方法在两个数据集上的 Last-acc 指标均优于 SSIAT，分别高出 5.31% 和 1.38%；而在 50 个任务和 100 个任务的划分场景下，我们提出的 CKAA 方法展现出更优的性能——在 ImageNetR 数据集上的 Last-Acc 指标较 VPT - NSP 提升了 8.71% 和 11.18%。对于 50 个任务和 100 个任务的划分场景，适配器的中间维度分别设置为 16 和 8，以避免产生额外的推理开销。这些结果充分体现了该方法的可扩展性及其在解决长期 CIL 问题中对抗遗忘现象的能力。

TABLE VI: Experimental results on 10S-StanfordCars.

Method	Venue	Last-acc $\uparrow$	Avg-acc $\uparrow$
L2P [20]	CVPR-22	60.39 $\pm$ 1.99	71.92 $\pm$ 1.12
DualPrompt [23]	ECCV-22	57.27 $\pm$ 0.34	70.36 $\pm$ 2.33
ESN [76]	AAAI-23	56.91 $\pm$ 0.56	72.82 $\pm$ 0.79
CODAPrompt [24]	CVPR-23	62.24 $\pm$ 0.14	73.28 $\pm$ 0.93
SLCA [48]	ICCV-23	67.73 $\pm$ 0.85	76.93 $\pm$ 1.21
CPrompt [49]	CVPR-24	66.77 $\pm$ 0.37	76.81 $\pm$ 0.27
SSIAT [2]	CVPR-24	75.32 $\pm$ 0.45	82.87 $\pm$ 0.68
CoMA [77]	ECCV-24	73.35 $\pm$ 0.59	78.55 $\pm$ 0.42
CoFiMA [77]	ECCV-24	76.96 $\pm$ 0.64	82.65 $\pm$ 0.96
HiDe-PreT [78]	TPAMI-25	68.75 $\pm$ 1.01	69.62 $\pm$ 0.38
α_2 -SLDC [79]	AAAI-26	74.07 $\pm$ 0.78	83.32 $\pm$ 0.55
β_1 -SLDC+ADE [79]	AAAI-26	80.61 $\pm$ 0.36	85.51 $\pm$ 0.41
CKAA (Ours)	—	80.92$\pm$0.08	87.32$\pm$0.90

B. Comparison with Existing Methods on Fine-Grained Datasets

We evaluate our method on the fine-grained datasets CUB-200 [80] and StanfordCars [81] to demonstrate its effectiveness in fine-grained CIL classification. CUB-200 [80] contains 11,788 images of 200 bird species with fine-grained class labels. StanfordCars [81] is a fine-grained car dataset including 196 classes and 16185 car images. Following existing methods [2], [49], [63], we evaluate our method under the 10-split setting. The experiment results on CUB-200 and StanfordCars are outlined in Tab.V and Tab.VI, respectively. The results demonstrate the robustness of our proposed CKAA against the catastrophic forgetting of fine-grained knowledge.

C. Ablation Study

In this subsection, we perform ablation studies on the components of our framework on ImageNetR and ImageNetA datasets with 10- or 20-split tasks, as shown in Tab.VII. We also visualize the task-by-task accuracy changing curves of the ablation studies on four benchmarks with 10- or 100-split tasks, as shown in Fig.5. Our baseline is the frozen ViT backbone with a trainable classifier (Idx 1) and Fig.5.

The effectiveness of $\mathcal{P}_{sh}$. We conduct experiments on the learnable visual prompt-based task-shared network to demonstrate its effectiveness. Specifically, we directly replace the learnable task-shared network with the frozen pre-trained backbone. As indicated in Tab.VII, the learnable visual prompts $\mathcal{P}_{sh}$ improves the model’s Last-acc by 0.85% and Avg-acc by 0.57% on 10S-ImagenetR (Idx 5 *v.s.* Idx 10). This improvement is mainly due to the learnable task-shared network’s ability to capture common knowledge across downstream datasets, compared to a frozen pre-trained backbone. It results in more precise task identification during inference and more appropriate routing weights for aggregating task-specific knowledge in TC-MoA.

The effectiveness of DKA. The proposed DKA (FA & CA) yields an overall improvement of 3.27%~9.20% on Last-acc compared to the independent training of sub-modules. The Feature-level Alignment (FA) improves the performance by 0.60%~3.80% Last-acc (Idx 7 *v.s.* Idx 8). Our FA effectively aligns feature semantics within different subspaces, resulting in a performance gain. The Classifier-level Alignment (CA) further substantially improves the model performance of

2.67%~5.40% on Last-acc (Idx 8 *v.s.* Idx 10). Unlike existing methods that use features with the proper task-ids to train the classifier, our CA simulates the wrongly projected features to train a unified classifier, improving the classifier’s robustness.

Tab.VIII further shows the performance *without and with DKA under various task-specific knowledge aggregation strategies*. In Tab.VIII, “Query-key” matching designs a task-key for each task-specific sub-module and selects the adapter corresponding to the task-key with the highest similarity to the input for integrating task-specific knowledge, which is a widely-used technology in PEFT-based CIL methods [20], [23], [24], [49], [82]. “Maximum” selection directly selects the sub-module with the highest task-confidence score, representing a one-hot variant of our TC-MoA. The results (upper *vs.* lower in Tab.VIII) indicate that DKA gains model performance with various aggregation strategies, further demonstrating the effectiveness of our DKA.

To gain a more detailed understanding of our CA, we conduct ablation studies of *the feature simulation process* in CA on 10S-ImageNetA to indicate its efficacy, as shown in Tab.IX. “CA w/o subspace transfer” denotes training the classifier using features obtained directly from Gaussian Sampling without subspace transfer. “CA w/o affinities $\mathbf{G}$ ” denotes modeling the shift of previous features without considering their affinities $\mathbf{G}$ with current features, instead directly utilizing the average shift of current features. The results demonstrate the efficacy of each step in our affinity-guided feature simulation.

Additionally, we examine the effectiveness of our DKA through the task-by-task accuracy curves in Fig.5, highlighting DKA’s superiority under long-term settings.

TABLE X: Comparison of DKA with traditional alignment approaches.

Methods	10S-ImageNetA		10S-ImageNetR		Gain
	Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$	
w/o Alignment	54.33 $\pm$ 0.87	64.43 $\pm$ 1.22	79.14 $\pm$ 0.21	85.04 $\pm$ 0.33	—
+ L_2 -reg	50.82 $\pm$ 0.69	61.26 $\pm$ 1.32	79.16 $\pm$ 0.40	84.68 $\pm$ 0.28	-1.76
+ KL-reg	45.12 $\pm$ 1.12	56.33 $\pm$ 1.65	74.91 $\pm$ 0.53	81.99 $\pm$ 0.38	-6.15
+ Proto-reg	53.66 $\pm$ 0.83	63.22 $\pm$ 1.35	77.76 $\pm$ 0.15	84.05 $\pm$ 0.52	-1.06
+ FA (Ours)	58.13 $\pm$ 0.20	66.59 $\pm$ 0.84	79.74 $\pm$ 0.34	85.29 $\pm$ 0.19	+1.70
+ CA (Ours)	61.29 $\pm$ 0.46	69.31 $\pm$ 1.16	81.80 $\pm$ 0.38	86.59 $\pm$ 0.18	+4.01
+ DKA (Ours)	63.53$\pm$0.66	71.11$\pm$1.03	82.41$\pm$0.13	87.20$\pm$0.35	+5.33

Comparison of DKA (FA+CA) with traditional alignment approaches. We compare DKA with three traditional alignment objectives: (1) “ L_2 -reg” minimizes the sample-wise feature distance between the current and historical subspaces; (2) “KL-reg” minimizes the KL-divergence between their prediction distributions; (3) “Proto-reg” matches current class-wise mean features maintained by Exponential Moving Average (EMA) with historical prototypes. The results in Tab.X show that these approaches even underperform the model without alignment, whereas DKA achieves the best results. Since each task-specific module primarily captures distinctive task knowledge, such absolute consistency regularization suppresses the task-specific variations essential for learning new tasks. In contrast, DKA promotes *compatible discriminative subspaces* by structuring relative inter-class relationships across subspaces (FA) and adapting classifier boundaries to

表 VI: 在 10S-StanfordCars 数据集上的实验结果。

方法	发生地	最后到达↑	Avg-acc ↑
L2P [20]	CVPR-22	60.39±1.99	71.92±1.12
DualPrompt [23]	ECCV-22	57.27±0.34	70.36±2.33
ESN [76]	AAAI-23	56.91±0.56	72.82±0.79
CODAPrompt [24]	CVPR-23	62.24±0.14	73.28±0.93
SLCA [48]	ICCV-23	67.73±0.85	76.93±1.21
CPrompt [49]	CVPR-24	66.77±0.37	76.81±0.27
SSIAT [2]	CVPR-24	75.32±0.45	82.87±0.68
CoMA [77]	ECCV-24	73.35±0.59	78.55±0.42
CoFiMA [77]	ECCV-24	76.96±0.64	82.65±0.96
HiDe-PreT [78]	TPAMI-25	68.75±1.01	69.62±0.38
α_2 -SLDC [79]	AAAI-26	74.07±0.78	83.32±0.55
β_1 -SLDC+ADE [79]	AAAI-26	80.61±0.36	85.51±0.41
CKAA (我们)	—	80.92±0.08	87.32±0.90

B. 与现有方法在细粒度数据集上的对比实验

我们在细粒度数据集 CUB- 上对本文提出的方法进行了评估。200 [80]与 StanfordCars[81]用于验证其在细粒度 CIL 分类任务中的有效性。CUB-200[80]包含 200 种鸟类的 11,788 张带细粒度类别标签的图像；StanfordCars[81]是一个细粒度汽车数据集，包含 196 个类别和 16,185 张汽车图像。沿用现有方法[2], [49], [63]我们在“10折数据划分”设置下对本文提出的方法进行了评估。在 CUB-200 和 StanfordCars 数据集上的实验结果分别汇总于表.V和表.VI中。实验结果表明，我们提出的 CKAA 方法在应对细粒度知识的灾难性遗忘问题时具有较强的鲁棒性。

C. 消融术研究

在本小节中，我们针对本文提出的框架中的各个组件，在 ImageNetR 和 ImageNetA 数据集上开展了消融实验，实验采用 10 分割或 20 分割的任务设置，如表VII所示。此外，我们还对在四个基准数据集上、采用 10 分割或 100 分割任务设置的消融实验所得到的逐任务准确率变化曲线进行了可视化展示，如图.5所示。我们的基线模型为配备可训练分类器 (Idx 1) 的冻结 ViT 基干网络，相关结果见图.5。

P_{sh} 的有效性。我们通过可对学习的基于视觉提示的任务共享网络进行实验，以验证其有效性。具体而言，我们将可学习的任务共享网络直接替换为冻结的预训练主干网络。如表VII所示，在 10S-ImagenetR 数据集 (Idx 5对比Idx 10) 上，可学习的视觉提示 P_{sh} 将模型的 Last-acc 提升了 0.85%，Avg-acc 提升了 0.57%。这一提升主要归功于可学习任务共享网络相较于冻结预训练主干网络所具备的捕捉下游数据集间通用知识的能力；这使得在推理阶段能够实现更精确的任务识别，并为在 TC-MoA 中聚合任务特定知识时生成更合适的路由权重。

DKA 的有效性。所提出的 DKA (FA 与 CA) 相较于对子模块进行独立训练，在 Lastacc 指标上实现了 3.27%~9.20% 的整体性能提升；特征级对齐 (FA) 使 Last-acc 指标提升了 0.60%~3.80% (Idx 7对比Idx 8)。我们的 FA 有效对齐了不同子空间中的特征语义，从而带来了性能增益；分类器级对齐 (CA) 则进一步显著提升了模型性

能，Last-acc 指标提升了 2.67%~5.40% (Idx 8对比Idx 10)。与现有方法采用带有正确任务ID的特征来训练分类器不同，我们的 CA 通过模拟错误投影的特征来训练一个统一的分类器，从而增强了分类器的鲁棒性。表VIII进一步展示了在不同任务特定知识聚合策略下，采用与不采用 DKA 时的模型性能。在表VIII中，“Query-key” 匹配机制为每个任务特定子模块设计一个任务键，并选择与输入数据相似度最高的对应任务键所对应的适配器，以整合任务特定知识——这在 PEFT -ba 中是一种广泛应用的技术。sed CIL 方法[20]、[23], [24], [49], [82] “Maximum” 选项直接选择任务置信度得分最高的子模块，这代表了我们的 TC-MoA 的一种 One-Hot 变体。结果 (上图vs.) 表VIII中的较低数值表明，采用不同的聚合策略可提升 DKA 的模型性能，这进一步证明了我们所提出的 DKA 方法的有效性。

为更深入地理解我们的分类器 (CA)，我们在 10S-ImageNetA 数据集上对CA 中的特征模拟过程进行了消融实验以验证其有效性，如表IX所示。“CA w/o subspace transfer” 表示使用直接通过高斯采样获得的特征（不进行子空间迁移）对分类器进行训练；“CA w/o affinities G” 表示对先前特征的偏移量进行建模，而不考虑这些特征与当前特征之间的亲和性G，而是直接利用当前特征的平均偏移量。实验结果证明了我们在亲和性引导特征模拟方法中每个步骤的有效性。

此外，我们通过图.5中的逐任务准确率曲线来评估 DKA 的有效性，从而突显了 DKA 在长期应用场景中的优越性。

表 X: DKA 与传统对齐方法的对比。

方法	10S-ImageNetA		10S-ImageNetR		获得
	最后到达↑	Avg-acc ↑	最后到达↑	Avg-acc ↑	
无对齐	54.33±0.87	64.43±1.22	79.14±0.21	85.04±0.33	—
+ L_2 -reg	50.82±0.69	61.26±1.32	79.16±0.40	84.68±0.28	-1.76
+ KL-reg	45.12±1.12	56.33±1.65	74.91±0.53	81.99±0.38	-6.15
+ 原型-Reg	53.66±0.83	63.22±1.35	77.76±0.15	84.05±0.52	-1.06
+ FA (本人)	58.13±0.20	66.59±0.84	79.74±0.34	85.29±0.19	+1.70
+ CA (我)	61.29±0.46	69.31±1.16	81.80±0.38	86.59±0.18	+4.01
+ DKA (我们)	63.53±0.66	71.11±1.03	82.41±0.13	87.20±0.35	+5.33

DKA (FA+CA) 与传统对齐方法的对比。我们将 DKA 与三种传统对齐目标进行比较：(1) “ L_2 -reg” 旨在最小化当前子空间与历史子空间之间样本级特征距离；(2) “KL-reg” 旨在最小化两者预测分布之间的 KL 散度；(3) “Proto-reg” 将通过指数移动平均 (EMA) 维护的当前类别均值特征与历史原型进行匹配。表.X中的结果表明，这些方法的性能甚至不及未采用对齐机制的模型，而 DKA 则取得了最佳性能。由于每个任务特定模块主要捕获了特定任务的知识，因此这种绝对一致性正则化抑制了对学习新任务至关重要的任务特定变异性；相比之下，DKA 通过构建跨子空间的相对类间关系 (FA) 并使分类器边界适应表 VII：我们在 ImageNetR 和 ImageNetA 上对所提方法中每个组件进行的消融实验。

TABLE VII: Ablation Studies of each component in our proposed method on ImageNetR and ImageNetA.

Idx	$\mathcal{P}_{sh}$	TC-MoA	FA	CA	10S-ImageNetR		10S-ImageNetA		20S-ImageNetR	
					Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$
CKAA with frozen pre-trained task-shared network.										
1	—	—	—	—	65.65 $\pm$ 0.58	72.82 $\pm$ 0.67	46.01 $\pm$ 0.92	56.51 $\pm$ 0.70	63.67 $\pm$ 0.68	71.56 $\pm$ 1.14
2	—	✓	—	—	76.72 $\pm$ 0.61	83.14 $\pm$ 0.32	53.92 $\pm$ 0.85	64.16 $\pm$ 0.18	73.91 $\pm$ 0.35	81.15 $\pm$ 0.32
3	—	✓	✓	—	78.54 $\pm$ 0.39	84.36 $\pm$ 0.28	57.12 $\pm$ 1.75	65.68 $\pm$ 0.25	75.58 $\pm$ 0.19	82.43 $\pm$ 0.76
4	—	✓	—	✓	79.77 $\pm$ 0.12	85.09 $\pm$ 0.24	59.71 $\pm$ 0.20	68.55 $\pm$ 0.32	78.65 $\pm$ 0.27	84.52 $\pm$ 0.41
5	—	✓	✓	✓	81.56 $\pm$ 0.13	86.63 $\pm$ 0.13	62.49 $\pm$ 1.53	70.59 $\pm$ 0.56	79.86 $\pm$ 0.32	85.33 $\pm$ 0.52
CKAA with learnable visual prompt-based task-shared network.										
6	✓	—	—	—	77.95 $\pm$ 0.22	83.44 $\pm$ 0.40	53.83 $\pm$ 0.37	63.93 $\pm$ 1.08	75.69 $\pm$ 0.62	81.87 $\pm$ 0.56
7	✓	✓	—	—	79.14 $\pm$ 0.21	85.04 $\pm$ 0.33	54.33 $\pm$ 0.87	64.43 $\pm$ 1.22	76.88 $\pm$ 0.57	82.91 $\pm$ 0.52
8	✓	✓	✓	—	79.74 $\pm$ 0.34	85.29 $\pm$ 0.19	58.13 $\pm$ 0.20	66.59 $\pm$ 0.84	77.82 $\pm$ 0.55	83.71 $\pm$ 0.39
9	✓	✓	—	✓	81.80 $\pm$ 0.38	86.59 $\pm$ 0.18	61.29 $\pm$ 0.46	69.31 $\pm$ 1.16	80.18 $\pm$ 0.42	85.75 $\pm$ 0.24
10	✓	✓	✓	✓	82.41 $\pm$ 0.13	87.20 $\pm$ 0.35	63.53 $\pm$ 0.66	71.11 $\pm$ 1.03	80.98 $\pm$ 0.35	86.28 $\pm$ 0.18

TABLE VIII: Ablation studies of various task-specific knowledge aggregation strategies without and with DKA on two datasets.

Method	10S-ImageNetR		10S-ImageNetA	
	Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$
<i>Without Dual-level Knowledge Alignment (DKA)</i>				
Query-key	74.60 $\pm$ 0.58	80.75 $\pm$ 0.82	47.30 $\pm$ 0.88	58.21 $\pm$ 0.76
Maximum	77.03 $\pm$ 0.46	82.95 $\pm$ 0.98	49.18 $\pm$ 0.72	59.57 $\pm$ 1.43
TC-MoA	79.14 $\pm$ 0.21	85.04 $\pm$ 0.33	54.33 $\pm$ 0.87	64.43 $\pm$ 1.22
<i>With Dual-level Knowledge Alignment (DKA)</i>				
Query-key	76.02 $\pm$ 0.45	81.88 $\pm$ 0.67	56.39 $\pm$ 0.37	65.80 $\pm$ 1.28
Maximum	78.55 $\pm$ 0.25	84.20 $\pm$ 0.72	58.12 $\pm$ 0.57	67.53 $\pm$ 0.56
TC-MoA	82.41 $\pm$ 0.13	87.20 $\pm$ 0.35	63.53 $\pm$ 0.66	71.11 $\pm$ 1.03

TABLE IX: Ablation studies of feature simulation on 10S-ImageNetA.

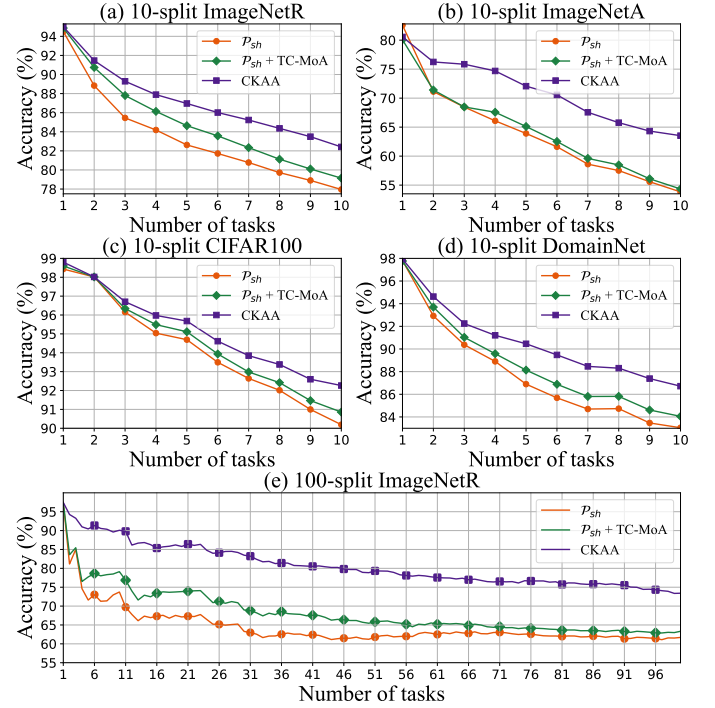
Method	Last-acc $\uparrow$	Avg-acc $\uparrow$
CKAA (Ours)	63.53 $\pm$ 0.66	71.11 $\pm$ 1.03
<i>The following methods are based on FA and TC-MoA.</i>		
CKAA w/o CA	58.13 $\pm$ 0.20	66.59 $\pm$ 0.84
+ CA w/o subspace transfer	62.38 $\pm$ 0.70	70.35 $\pm$ 1.28
+ CA w/o affinities $\mathbf{G}$	63.20 $\pm$ 0.63	70.83 $\pm$ 0.92

simulated cross-subspace variations (CA), without enforcing identical representations and preserving model plasticity.

The effectiveness of TC-MoA. TC-MoA achieves an improvement of 0.50%~1.19% Last-acc (Idx 6 *v.s.* Idx 7) compared to directly utilizing the task-shared network for classification. This observation indicates that our TC-MoA is a powerful tool for aggregating task-specific knowledge, while the task-shared network alone is insufficient for the final decision. We further compare *different aggregation strategies* in Tab.VIII, where existing “Query-Key” matching exhibits limited performance as a single task key is insufficient to represent the data distribution within an entire task. In contrast, our TC-MoA softly aggregates task-specific knowledge based on task-confidence scores, avoiding overconfidence in misleading task-ids and fully leveraging the fine-grained instance-class relationships.

TABLE XI: Ablation study of the dual classifier design.

Methods	10S-ImageNetA		10S-ImageNetR	
	Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$
Only g	58.13 $\pm$ 0.20	66.59 $\pm$ 0.84	79.74 $\pm$ 0.34	85.29 $\pm$ 0.19
Only g^u	63.00 $\pm$ 0.13	70.75 $\pm$ 0.90	82.10 $\pm$ 0.16	87.03 $\pm$ 0.08
$g^u + g$ ($w = 0.50$)	63.13 $\pm$ 0.78	70.94 $\pm$ 1.16	82.21 $\pm$ 0.24	87.15 $\pm$ 0.17
$g^u + g$ ($w = 0.80$)	63.25 $\pm$ 0.57	70.86 $\pm$ 1.23	82.18 $\pm$ 0.40	87.08 $\pm$ 0.12
$g^u + g$ ($w = \frac{t_c}{t_c+1}$)	63.53 $\pm$ 0.66	71.11 $\pm$ 1.03	82.41 $\pm$ 0.13	87.20 $\pm$ 0.35

Fig. 5: The task-by-task accuracy changing curves of our base-line ($\mathcal{P}_{sh}$), $\mathcal{P}_{sh}$ +TC-MoA and CKAA on four benchmarks.

Ablation studies of the global classifier g^u and the standard classifier g . The two classifiers provide complementary decision capabilities: (1) g learns stable category discrimination from shared and correct task-specific features, and (2) g^u aggregates task-adaptive classifiers trained with real and simulated incorrect-subspace features to improve robustness under misleading task-IDs. As shown in Tab. XI, their combination achieves the best performance, confirming their complementary roles. We also report performances under different fusion weights w in prediction, *i.e.*, $w \cdot g^u(\hat{\mathbf{z}}_i)[k] + (1-w) \cdot g(\hat{\mathbf{z}}_i)[k]$. The dynamic weight $w = \frac{t_c}{t_c+1}$ outperforms fixed weights by gradually increasing the contribution of g^u as more cross-subspace knowledge is accumulated.

D. Analysis of the Effectiveness of CKAA in handling Misleading Task-ids

We conduct ablation studies on 10S-ImageNetA to analyze the effectiveness of our proposed CKAA in enhancing the

Idx	$\mathcal{P}_{sh}$	TC-MoA	FA	CA	10S-ImageNetR		10S-ImageNetA		20S-ImageNetR	
					最后到达 $\uparrow$	Avg-acc $\uparrow$	最后到达 $\uparrow$	Avg-acc $\uparrow$	最后到达 $\uparrow$	Avg-acc $\uparrow$
CKAA 配备预训练的、任务共享的冻结网络。										
1	—	—	—	—	65.65 $\pm$ 0.58	72.82 $\pm$ 0.67	46.01 $\pm$ 0.92	56.51 $\pm$ 0.70	63.67 $\pm$ 0.68	71.56 $\pm$ 1.14
2	—	✓	—	—	76.72 $\pm$ 0.61	83.14 $\pm$ 0.32	53.92 $\pm$ 0.85	64.16 $\pm$ 0.18	73.91 $\pm$ 0.35	81.15 $\pm$ 0.32
3	—	✓	✓	—	78.54 $\pm$ 0.39	84.36 $\pm$ 0.28	57.12 $\pm$ 1.75	65.68 $\pm$ 0.25	75.58 $\pm$ 0.19	82.43 $\pm$ 0.76
4	—	✓	—	✓	79.77 $\pm$ 0.12	85.09 $\pm$ 0.24	59.71 $\pm$ 0.20	68.55 $\pm$ 0.32	78.65 $\pm$ 0.27	84.52 $\pm$ 0.41
5	—	✓	✓	✓	81.56$\pm$0.13	86.63$\pm$0.13	62.49$\pm$1.53	70.59$\pm$0.56	79.86$\pm$0.32	85.33$\pm$0.52
CKAA 基于可学习的视觉提示任务共享网络。										
6	✓	—	—	—	77.95 $\pm$ 0.22	83.44 $\pm$ 0.40	53.83 $\pm$ 0.37	63.93 $\pm$ 1.08	75.69 $\pm$ 0.62	81.87 $\pm$ 0.56
7	✓	✓	—	—	79.14 $\pm$ 0.21	85.04 $\pm$ 0.33	54.33 $\pm$ 0.87	64.43 $\pm$ 1.22	76.88 $\pm$ 0.57	82.91 $\pm$ 0.52
8	✓	✓	✓	—	79.74 $\pm$ 0.34	85.29 $\pm$ 0.19	58.13 $\pm$ 0.20	66.59 $\pm$ 0.84	77.82 $\pm$ 0.55	83.71 $\pm$ 0.39
9	✓	✓	—	✓	81.80 $\pm$ 0.38	86.59 $\pm$ 0.18	61.29 $\pm$ 0.46	69.31 $\pm$ 1.16	80.18 $\pm$ 0.42	85.75 $\pm$ 0.24
10	✓	✓	✓	✓	82.41$\pm$0.13	87.20$\pm$0.35	63.53$\pm$0.66	71.11$\pm$1.03	80.98$\pm$0.35	86.28$\pm$0.18

表 VIII: 在两个数据集上对不同任务特定知识聚合策略

(无论是否结合 DKA) 进行的消融实验。

方法	10S-ImageNetR		10S-ImageNetA	
	最后到达 $\uparrow$	Avg-acc $\uparrow$	最后到达 $\uparrow$	Avg-acc $\uparrow$
不使用双层知识对齐 (DKA)				
查询键	74.60 $\pm$ 0.58	80.75 $\pm$ 0.82	47.30 $\pm$ 0.88	58.21 $\pm$ 0.76
最大值	77.03 $\pm$ 0.46	82.95 $\pm$ 0.98	49.18 $\pm$ 0.72	59.57 $\pm$ 1.43
TC-MoA	79.14$\pm$0.21	85.04$\pm$0.33	54.33$\pm$0.87	64.43$\pm$1.22
支持双层知识对齐 (DKA)				
查询键	76.02 $\pm$ 0.45	81.88 $\pm$ 0.67	56.39 $\pm$ 0.37	65.80 $\pm$ 1.28
最大值	78.55 $\pm$ 0.25	84.20 $\pm$ 0.72	58.12 $\pm$ 0.57	67.53 $\pm$ 0.56
TC-MoA	82.41$\pm$0.13	87.20$\pm$0.35	63.53$\pm$0.66	71.11$\pm$1.03

表 IX: 基于 10SImageNetA 数据集的特征模拟消融实验。

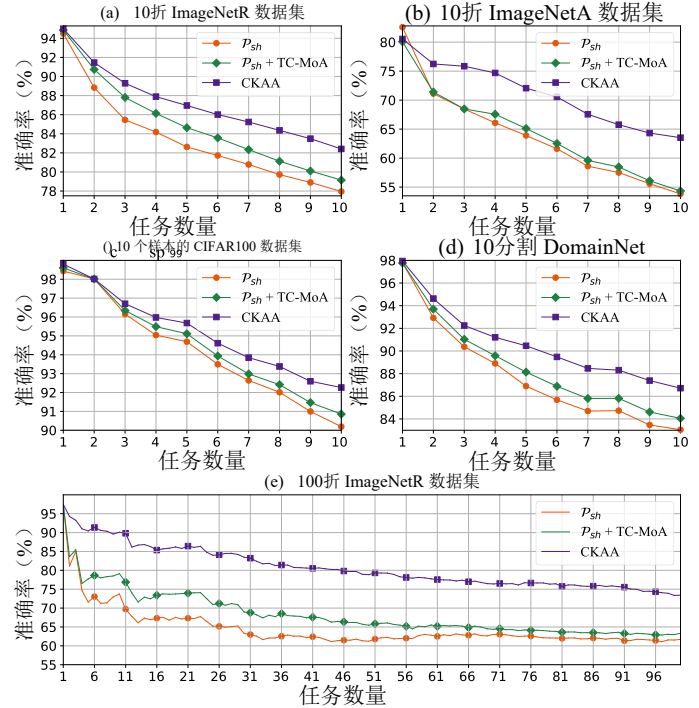
方法	最后到达 $\uparrow$	Avg-acc $\uparrow$
CKAA (我们)	63.53$\pm$0.66	71.11$\pm$1.03
以下方法为: 基于 FA 与 TC-MoA 的分析。		
CKAA w/o CA	58.13 $\pm$ 0.20	66.59 $\pm$ 0.84
+ 无子空间传递的 CA + 无亲和力的 CA G	62.38 $\pm$ 0.70	70.35 $\pm$ 1.28
	63.20 $\pm$ 0.63	70.83 $\pm$ 0.92

模拟交叉子空间变化 (CA), 无需强制要求表示形式完全一致, 同时保留模型的可塑性。

TC-MoA 的有效性。与直接使用任务共享网络进行分类相比, TC-MoA 在 Last-acc (Idx 6 对比 Idx 7) 指标上实现了 0.50%~1.19% 的提升。这一现象表明, 我们的 TC-MoA 是一个强大的聚合任务特定知识的工具, 而仅使用任务共享网络不足以做出最终决策。我们进一步在表 VIII 中对比了 **不同的聚合策略**: 其中, 现有的“查询-键”匹配方法性能有限, 因为单个任务键不足以表征整个任务中的数据分布。相比之下, 我们的 TC-MoA 基于任务置信度分数对任务特定知识进行软聚合, 从而避免因过度置信而导致对任务 ID 的误判, 并充分挖掘细粒度的实例-类别关系。

表 XI: 双分类器设计的消融研究。

方法	10S-ImageNetA		10S-ImageNetR	
	最后到达 $\uparrow$	Avg-acc $\uparrow$	最后到达 $\uparrow$	Avg-acc $\uparrow$
仅 g	58.13 $\pm$ 0.20	66.59 $\pm$ 0.84	79.74 $\pm$ 0.34	85.29 $\pm$ 0.19
仅 g^u	63.00 $\pm$ 0.13	70.75 $\pm$ 0.90	82.10 $\pm$ 0.16	87.03 $\pm$ 0.08
$g^u + g$ ($w = 0.50$)	63.13 $\pm$ 0.78	70.94 $\pm$ 1.16	82.21 $\pm$ 0.24	87.15 $\pm$ 0.17
$g^u + g$ ($w = 0.80$)	63.25 $\pm$ 0.57	70.86 $\pm$ 1.23	82.18 $\pm$ 0.40	87.08 $\pm$ 0.12
$g^u + g$ ($w = \frac{t_c}{t_c+1}$)	63.53$\pm$0.66	71.11$\pm$1.03	82.41$\pm$0.13	87.20$\pm$0.35

图 5: 我们的基线模型 ($\mathcal{P}_{sh}$), $\mathcal{P}_{sh}$ 、 $\mathcal{P}_{sh}$ + TC-MoA 以及 CKAA 在四个基准数据集上的逐任务准确率变化曲线。

全局分类器 g^u 与标准分类器 g 的消融实验。这两个分类器提供了互补的决策能力: (1) g 通过共享且正确的任务特定特征学习稳定的类别判别能力; (2) g^u 聚合了基于真实及模拟的错误子空间特征训练的任务自适应分类器, 以提升在误导性任务 ID 条件下的鲁棒性。如表 XI 所示, 二者结合可实现最佳性能, 这证实了它们的互补作用。我们还报告了在不同预测融合权重 w 下的性能表现, $\mathbb{E}[w \cdot g^u(\mathbf{z}_i) + (1-w) \cdot g(\mathbf{z}_i)]$ 。动态权重 $w = \frac{t_c}{t_c+1}$ 通过随着更多跨子空间知识的积累而逐步增加 g^u 的贡献度, 从而优于固定权重方法。

D. 分析 CKAA 在处理误导性任务 ID 方面的有效性

我们在 10S-ImageNetA 数据集上开展消融实验, 以分析

TABLE XII: Task-by-task accuracy of the *challenging samples* on 10S-ImageNetA.

Methods	Ses.2	Ses.3	Ses.4	Ses.5	Ses.6	Ses.7	Ses.8	Ses.9	Ses.10
<i>CKAA with frozen pre-trained task-shared network.</i>									
pre-trained backbone	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
+ TC-MoA	43.26 $\pm$ 5.96	44.84 $\pm$ 5.08	43.67 $\pm$ 1.45	39.47 $\pm$ 2.20	36.09 $\pm$ 2.26	35.01 $\pm$ 2.87	32.29 $\pm$ 1.01	31.00 $\pm$ 1.75	30.20 $\pm$ 1.46
+ TC-MoA + CA	52.20 $\pm$ 2.80	51.74 $\pm$ 4.33	48.49 $\pm$ 1.28	46.61 $\pm$ 1.56	43.99 $\pm$ 1.64	41.93 $\pm$ 2.53	40.51 $\pm$ 0.51	39.19 $\pm$ 0.41	37.88 $\pm$ 0.47
+ TC-MoA + FA	43.72 $\pm$ 7.96	48.07 $\pm$ 7.06	47.08 $\pm$ 2.65	43.17 $\pm$ 2.16	41.81 $\pm$ 2.52	37.80 $\pm$ 3.44	36.68 $\pm$ 3.04	34.73 $\pm$ 1.53	33.76 $\pm$ 1.24
+ TC-MoA + DKA	53.99 $\pm$ 6.41	53.06 $\pm$ 2.95	52.60 $\pm$ 1.86	49.65 $\pm$ 0.27	47.33 $\pm$ 2.52	45.11 $\pm$ 2.82	44.54 $\pm$ 0.87	42.87 $\pm$ 0.89	42.01 $\pm$ 0.66
<i>CKAA with learnable visual prompt-based task-shared network.</i>									
$\mathcal{P}_{sh}$	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
+ TC-MoA	23.90 $\pm$ 5.54	19.97 $\pm$ 1.67	17.62 $\pm$ 1.23	14.82 $\pm$ 2.42	13.29 $\pm$ 1.13	12.31 $\pm$ 0.33	12.48 $\pm$ 1.05	11.41 $\pm$ 0.5	10.48 $\pm$ 0.22
+ TC-MoA + CA	30.48 $\pm$ 5.96	29.59 $\pm$ 0.88	30.80 $\pm$ 6.53	28.56 $\pm$ 5.17	27.54 $\pm$ 2.62	27.36 $\pm$ 3.25	25.97 $\pm$ 1.60	25.50 $\pm$ 1.08	25.68 $\pm$ 0.39
+ TC-MoA + FA	20.96 $\pm$ 4.42	21.41 $\pm$ 4.48	21.95 $\pm$ 3.96	19.60 $\pm$ 2.12	17.94 $\pm$ 1.73	17.59 $\pm$ 3.21	17.25 $\pm$ 2.05	16.81 $\pm$ 1.57	16.43 $\pm$ 1.52
+ TC-MoA + DKA	35.76 $\pm$ 0.93	33.75 $\pm$ 8.83	36.02 $\pm$ 2.03	31.84 $\pm$ 2.85	31.28 $\pm$ 2.78	29.91 $\pm$ 2.85	29.25 $\pm$ 1.42	28.31 $\pm$ 1.97	27.71 $\pm$ 0.53

TABLE XIII: Task-by-task accuracy of the *easy samples* on 10S-ImageNetA.

Methods	Ses.2	Ses.3	Ses.4	Ses.5	Ses.6	Ses.7	Ses.8	Ses.9	Ses.10
<i>CKAA with frozen pre-trained task-shared network.</i>									
pre-trained backbone	67.05 $\pm$ 3.05	67.15 $\pm$ 0.96	69.63 $\pm$ 1.68	71.60 $\pm$ 2.15	72.61 $\pm$ 3.42	73.43 $\pm$ 2.46	74.12 $\pm$ 1.52	72.56 $\pm$ 1.93	72.57 $\pm$ 1.84
+ TC-MoA	79.50 $\pm$ 3.98	82.11 $\pm$ 3.68	82.65 $\pm$ 1.49	83.54 $\pm$ 1.81	82.73 $\pm$ 0.99	82.48 $\pm$ 0.69	82.91 $\pm$ 2.30	82.53 $\pm$ 1.06	82.41 $\pm$ 1.36
+ TC-MoA + CA	81.92 $\pm$ 2.65	85.46 $\pm$ 1.61	86.08 $\pm$ 1.03	87.17 $\pm$ 1.66	87.25 $\pm$ 1.13	86.24 $\pm$ 0.63	86.09 $\pm$ 1.34	85.92 $\pm$ 1.70	86.36 $\pm$ 1.68
+ TC-MoA + FA	81.07 $\pm$ 3.75	82.75 $\pm$ 3.30	82.68 $\pm$ 1.20	84.89 $\pm$ 0.40	83.85 $\pm$ 1.63	84.49 $\pm$ 1.52	84.97 $\pm$ 0.86	84.76 $\pm$ 2.38	84.35 $\pm$ 2.39
+ TC-MoA + DKA	83.12 $\pm$ 1.31	86.40 $\pm$ 1.86	86.19 $\pm$ 0.99	88.81 $\pm$ 1.34	87.79 $\pm$ 1.34	86.78 $\pm$ 0.75	86.89 $\pm$ 2.12	86.87 $\pm$ 2.89	86.83 $\pm$ 3.30
<i>CKAA with learnable visual prompt-based task-shared network.</i>									
$\mathcal{P}_{sh}$	81.92 $\pm$ 2.49	83.42 $\pm$ 2.79	85.05 $\pm$ 3.00	85.80 $\pm$ 2.99	85.86 $\pm$ 2.48	85.14 $\pm$ 1.72	85.79 $\pm$ 1.49	85.45 $\pm$ 1.79	85.98 $\pm$ 1.17
+ TC-MoA	79.01 $\pm$ 3.99	80.84 $\pm$ 2.24	83.49 $\pm$ 4.47	84.31 $\pm$ 1.51	84.60 $\pm$ 1.27	83.62 $\pm$ 0.48	83.07 $\pm$ 0.64	82.33 $\pm$ 1.43	82.85 $\pm$ 1.97
+ TC-MoA + CA	80.10 $\pm$ 9.33	84.19 $\pm$ 3.20	86.26 $\pm$ 3.78	85.54 $\pm$ 1.66	86.09 $\pm$ 1.42	86.64 $\pm$ 3.07	86.84 $\pm$ 0.39	85.71 $\pm$ 0.49	85.76 $\pm$ 0.50
+ TC-MoA + FA	80.22 $\pm$ 1.81	82.35 $\pm$ 2.26	84.26 $\pm$ 0.55	85.22 $\pm$ 1.74	85.30 $\pm$ 1.09	85.02 $\pm$ 2.25	83.91 $\pm$ 1.05	84.34 $\pm$ 1.93	83.49 $\pm$ 2.56
+ TC-MoA + DKA	82.81 $\pm$ 4.11	85.13 $\pm$ 1.71	86.95 $\pm$ 2.04	87.76 $\pm$ 2.66	88.07 $\pm$ 1.11	87.39 $\pm$ 1.38	87.23 $\pm$ 0.85	86.08 $\pm$ 0.81	86.51 $\pm$ 0.52

model’s robustness against misleading task-ids. Specifically, we divide the testing samples into two subsets: (1) *Challenging samples*: These are misclassified by the task recognizer and assigned with incorrect task-ids; (2) *Easy samples*: These are classified correct task-ids. We report the task-by-task accuracy of these two types of samples, respectively. The results are shown in Tab.XII and Tab.XIII.

As shown in Tab.XII, the proposed TC-MoA and DKA effectively improve the prediction accuracy for the challenging samples, delivering more reliable predictions for features under incorrect task-specific sub-modules. Notably, the accuracy in the half bottom of Tab.XII is lower compared to the top half part. It is mainly due to our learnable task-shared network (the half bottom) encapsulates sufficient downstream information, the samples it fails to distinguish are inherently more challenging. Overall, our CKAA achieves significant improvement in the classification accuracy of these challenging samples.

TABLE XIV: Ablation studies under different task-ID error rate groups on 10S-ImageNetA.

Idx	TC-MoA	FA	CA	Task-ID Error Rate (%)				
				0-20	20-40	40-60	60-80	80-100
1	—	—	—	83.87	66.96	55.18	32.18	19.25
2	✓	—	—	79.57	68.76	55.70	40.65	19.91
3	✓	—	✓	79.05	74.51	64.22	49.80	33.64
4	✓	✓	—	79.72	70.99	61.72	50.63	23.11
5	✓	✓	✓	79.05	76.02	64.85	58.74	34.50

We further evaluate CKAA across different task-ID error-rate groups. For each class c from task t^c , we define its class-wise task-ID error rate as $\mathcal{E}_{TID}(c) = \frac{1}{|\mathcal{D}_c|} \sum_{\mathbf{x}^i \in \mathcal{D}_c} \mathbb{I}[\arg \max_t \alpha_t^i \neq t^c] \times 100\%$, where $\mathcal{D}_c$ is the test set of class c and α_t^i is the task-confidence score of sample i for task t . We divide all classes into five groups according to $\mathcal{E}_{TID}(c)$, i.e., 0–20%, 20–40%, 40–60%, 60–80%, and 80–100%, and report the accuracy of samples belonging to each

TABLE XV: Ablation studies under different task-ID error rate groups on 10S-CUB.

Idx	TC-MoA	FA	CA	Task-ID Error Rate (%)				
				0-20	20-40	40-60	60-80	80-100
1	—	—	—	94.04	71.78	59.64	47.73	17.69
2	✓	—	—	93.54	73.18	60.30	47.97	18.82
3	✓	—	✓	94.57	78.85	69.75	54.43	45.77
4	✓	✓	—	92.92	75.62	65.40	51.12	33.95
5	✓	✓	✓	95.37	82.61	72.43	57.67	56.98

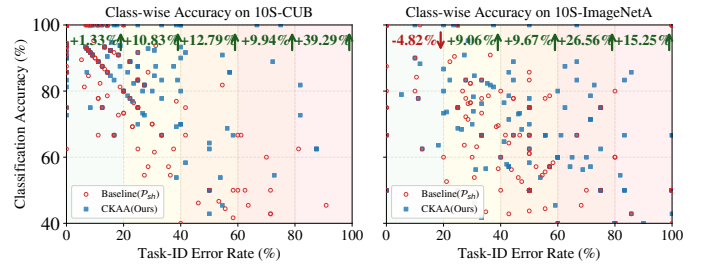


Fig. 6: Class-wise accuracy versus task-ID error rate on 10S-ImageNetA and 10S-CUB.

group in Tabs. XIV and XV. The baseline performance drops substantially as the task-ID error rate increases, confirming that misleading task-IDs severely affect prediction reliability. In contrast, CKAA consistently improves the accuracy of high-error classes: On 10S-ImageNetA, the accuracy increases from 19.25% to 34.50% in the 80–100% group, and On 10S-CUB, the accuracy in the 80–100% group increases from 17.69% to 56.98%. These results indicate that CKAA is particularly effective for samples that are likely to be routed to incorrect task-specific subspaces.

Fig.6 further visualizes the class-wise relationship between task-ID errors and classification performance on 10S-ImageNetA and 10S-CUB. The visualization compares CKAA

表 XII: 具有挑战性的样本在 10S-ImageNetA 上的逐任务准确率。

方法	Ses.2	Ses.3	Ses.4	Ses.5	Ses.6	Ses.7	Ses.8	Ses.9	Ses.10
CKAA 配备预训练的、任务共享的冻结网络。									
预训练主干网络	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00
+ TC-MoA	43.26±5.96	44.84±5.08	43.67±1.45	39.47±2.20	36.09±2.26	35.01±2.87	32.29±1.01	31.00±1.75	30.20±1.46
+ TC-MoA + CA	52.20±2.80	51.74±4.33	48.49±1.28	46.61±1.56	43.99±1.64	41.93±2.53	40.51±0.51	39.19±0.41	37.88±0.47
+ TC-MoA + FA	43.72±7.96	48.07±7.06	47.08±2.65	43.17±2.16	41.81±2.52	37.80±3.44	36.68±3.04	34.73±1.53	33.76±1.24
+ TC-MoA + DKA	53.99 ±6.41	53.06 ±2.95	52.60 ±1.86	49.65 ±0.27	47.33 ±2.52	45.11 ±2.82	44.54 ±0.87	42.87 ±0.89	42.01 ±0.66
CKAA 基于可学习的视觉提示任务共享网络。									
$\mathcal{P}_{sh}$	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00
+ TC-MoA	23.90±5.54	19.97±1.67	17.62±1.23	14.82±2.42	13.29±1.13	12.31±0.33	12.48±1.05	11.41±0.5	10.48±0.22
+ TC-MoA + CA	30.48±5.96	29.59±0.88	30.80±6.53	28.56±5.17	27.54±2.62	27.36±3.25	25.97±1.60	25.50±1.08	25.68±0.39
+ TC-MoA + FA	20.96±4.42	21.41±4.48	21.95±3.96	19.60±2.12	17.94±1.73	17.59±3.21	17.25±2.05	16.81±1.57	16.43±1.52
+ TC-MoA + DKA	35.76 ±0.93	33.75 ±8.83	36.02 ±2.03	31.84 ±2.85	31.28 ±2.78	29.91 ±2.85	29.25 ±1.42	28.31 ±1.97	27.71 ±0.53

表 XIII: 在 10S-ImageNetA 数据集上简单样本的逐任务准确率。

方法	Ses.2	Ses.3	Ses.4	Ses.5	Ses.6	Ses.7	Ses.8	Ses.9	Ses.10
CKAA 配备预训练的、任务共享的冻结网络。									
预训练主干网络	67.05±3.05	67.15±0.96	69.63±1.68	71.60±2.15	72.61±3.42	73.43±2.46	74.12±1.52	72.56±1.93	72.57±1.84
+ TC-MoA	79.50±3.98	82.11±3.68	82.65±1.49	83.54±1.81	82.73±0.99	82.48±0.69	82.91±2.30	82.53±1.06	82.41±1.36
+ TC-MoA + CA	81.92±2.65	85.46±1.61	86.08±1.03	87.17±1.66	87.25±1.13	86.24±0.63	86.09±1.34	85.92±1.70	86.36±1.68
+ TC-MoA + FA	81.07±3.75	82.75±3.30	82.68±1.20	84.89±0.40	83.85±1.63	84.49±1.52	84.97±0.86	84.76±2.38	84.35±2.39
+ TC-MoA + DKA	83.12 ±1.31	86.40 ±1.86	86.19 ±0.99	88.81 ±1.34	87.79 ±1.34	86.78 ±0.75	86.89 ±2.12	86.87 ±2.89	86.83 ±3.30
CKAA 基于可学习的视觉提示任务共享网络。									
$\mathcal{P}_{sh}$	81.92±2.49	83.42±2.79	85.05±3.00	85.80±2.99	85.86±2.48	85.14±1.72	85.79±1.49	85.45±1.79	85.98±1.17
+ TC-MoA	79.01±3.99	80.84±2.24	83.49±4.47	84.31±1.51	84.60±1.27	83.62±0.48	83.07±0.64	82.33±1.43	82.85±1.97
+ TC-MoA + CA	80.10±9.33	84.19±3.20	86.26±3.78	85.54±1.66	86.09±1.42	86.64±3.07	86.84±0.39	85.71±0.49	85.76±0.50
+ TC-MoA + FA	80.22±1.81	82.35±2.26	84.26±0.55	85.22±1.74	85.30±1.09	85.02±2.25	83.91±1.05	84.34±1.93	83.49±2.56
+ TC-MoA + DKA	82.81 ±4.11	85.13 ±1.71	86.95 ±2.04	87.76 ±2.66	88.07 ±1.11	87.39 ±1.38	87.23 ±0.85	86.08 ±0.81	86.51 ±0.52

所提出的 CKAA 方法在增强模型对误导性任务 ID 的鲁棒性方面的有效性。具体而言，我们将测试样本划分为两个子集：(1) **挑战性样本**：这些样本被任务识别器错误分类，并被分配了错误的任务 ID；(2) **简单样本**：这些样本被正确分类为对应的任务 ID。我们分别报告了这两类样本在各任务下的准确率。实验结果如表 XII 和表 XIII 所示。

如表. XII 所示，所提出的 TC-MoA 和 DKA 方法有效提升了对具有挑战性的样本的预测精度，为处于错误任务特定子模块下的特征提供了更可靠的预测结果。值得注意的是，表. XII 下半部分的准确率低于上半部分；这主要是因为我们的可学习任务共享网络（下半部分）包含了充分的下游信息，而该网络无法有效区分的样本本身更具挑战性。总体而言，我们的 CKAA 方法在这些具有挑战性的样本的分类准确率上取得了显著提升。

表 XIV: 在 10S-ImageNetA 数据集上针对不同任务-ID 错误率组开展的消融实验。

Idx	TC-MoA	FA	CA	任务ID错误率 (%)				
				0-20	20-40	40-60	60-80	80-100
1	—	—	—	83.87	66.96	55.18	32.18	19.25
2	✓	—	—	79.57	68.76	55.70	40.65	19.91
3	✓	—	✓	79.05	74.51	64.22	49.80	33.64
4	✓	✓	—	79.72	70.99	61.72	50.63	23.11
5	✓	✓	✓	79.05	76.02	64.85	58.74	34.50

我们进一步对不同任务-ID 错误率组别的 CKAA 进行评估。对于任务 t 中的每个类别 c ，我们将其类别级任务-ID 错误率定义为 $\mathcal{E}_{TID}(c) = \frac{1}{|\mathcal{D}_c|} \sum_{\mathbf{x}^i \in \mathcal{D}_c} \mathbb{I}[\arg \max_t \alpha_t^i \neq tc] \times 100\%$ ，其中 $\mathcal{D}_c$ 表示类别 c 的测试集，而 α_t 表示样本 i 对于任务 t 的任务置信度得分。我们根据 $\mathcal{E}_{TID}(c)$ 将所有类别划分为五个区间：0-20%、20-40%、40-60%、60-80% 以及 80-100%，并报告属于每个区间的样本的准确率（见表 XV：在

10S-CUB 数据集上针对不同任务-ID 错误率组进行的消融实验）。

Idx	TC-MoA	FA	CA	任务ID错误率 (%)				
				0-20	20-40	40-60	60-80	80-100
1	—	—	—	94.04	71.78	59.64	47.73	17.69
2	✓	—	—	93.54	73.18	60.30	47.97	18.82
3	✓	—	✓	94.57	78.85	69.75	54.43	45.77
4	✓	✓	—	92.92	75.62	65.40	51.12	33.95
5	✓	✓	✓	95.37	82.61	72.43	57.67	56.98

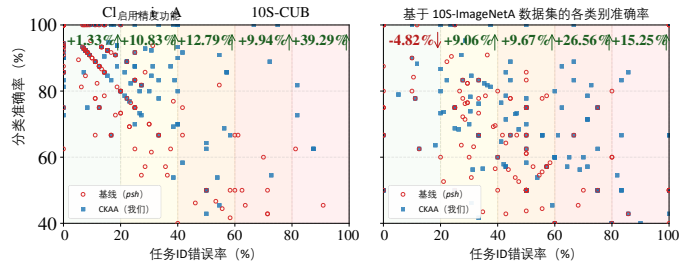


图 6: 在 10S-ImageNetA 和 10S-CUB 数据集上，按类别划分的准确率与任务 ID 误判率的关系曲线。

见第14章与第15章中的“Tab”部分。随着任务ID错误率的升高，基线性能显著下降，这证实了误导性任务ID会严重影响预测的可靠性。相比之下，CKAA consistently 提高了高错误率类别的准确率：在 10S-ImageNetA 数据集上，80-100% 组的准确率从 19.25% 提高至 34.50%；在 10S-CUB 数据集上，80-100% 组的准确率从 17.69% 提高至 56.98%。这些结果表明，CKAA 对于那些很可能被路由至错误的任务特定子空间的样本尤为有效。

图.6 进一步直观展示了在 10S-ImageNetA 和 10S-CUB 数据集上，任务ID错误率与分类性能之间的类别间关系。该可视化结果对比了 CKAA 与基于共享提示的基线模型

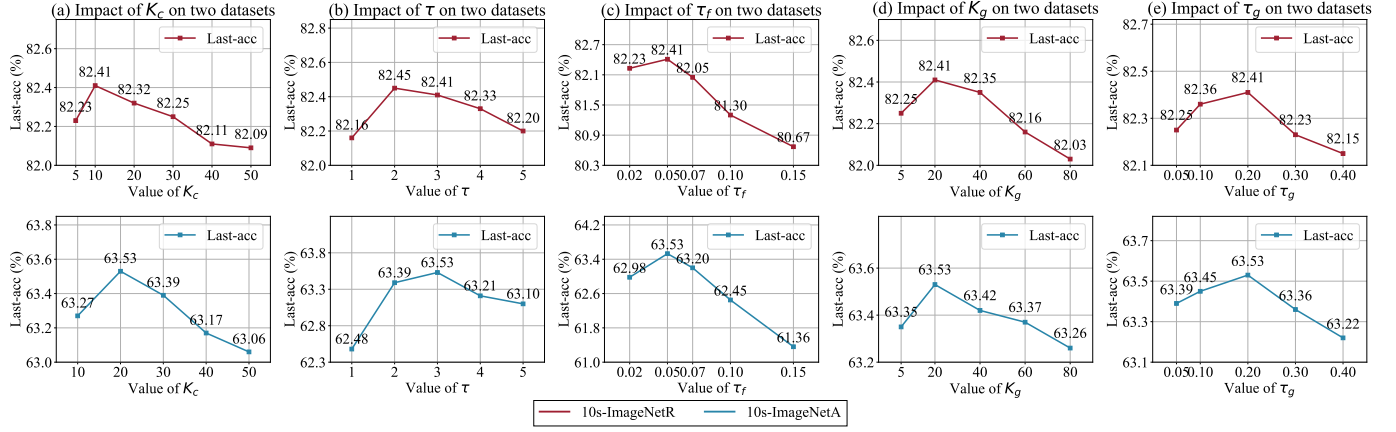


Fig. 7: Parameter analysis of K_c and τ in TC-MoA, and τ_f , K_g and τ_g in DKA on 10S-ImageNetR and 10S-ImageNetA.

and the shared-prompt-based baseline (P_{sh}). The horizontal axis represents class-wise task-ID error rate $\mathcal{E}_{TID}(c)$, while the vertical axis represents the classification accuracy of that class. These visualizations further confirm that CKAA effectively improves robustness for classes affected by incorrect task-ID estimation.

E. Hyper-Parameter Analysis

Key hyper-parameters. We conduct experiments to analyze the impact of the hyper-parameters in our proposed method on 10S-ImageNetR and 10S-ImageNetA benchmarks, including τ_f in Eq.4, K_g and τ_g in Eq.5, K_c in Eq.10 and τ in Eq.11. We first analyze the hyper-parameters in TC-MoA, including K_c and τ . The results are shown in Fig.7(a) and Fig.7(b). K_c is designed to select the Top- K_c classes with the highest logits to compute the task-confidence scores. A too small K_c makes task-confidence estimation vulnerable to a few spuriously activated classes, whereas a too large K_c includes responses from irrelevant tasks and weakens the dominance of the correct adapter. A small τ makes the aggregation weights overly sharp, while an excessively large τ over-smooths the weights and introduces interference from irrelevant adapters. We set $K_c = 10$ for ImageNetR and $K_c = 20$ for ImageNetA, $\tau = 2.0$ for all datasets based on experimental results. Then we analyze the hyper-parameters in DKA, including τ_f , K_g , and τ_g . The results are shown in Fig.7(c), Fig.7(d), and Fig.7(e). τ_f is a temperature factor of the Cross Subspace Feature Alignment (CSFA) loss in Eq.4. K_g and τ_g are two parameters for modeling the affinities between current and previous data. Based on the experimental results, we set $\tau_f = 0.05$, $K_g = 20$ and $\tau_g = 0.2$ for all datasets.

The loss weights λ_{fa} and λ_{ca} in Eq. 9. We vary $\lambda_{fa}, \lambda_{ca} \in \{0.2, 0.5, 1.0, 2.0, 3.0\}$ while fixing the coefficient of $\mathcal{L}_{ce}$ as 1.0. As shown in Fig. 8, CKAA remains stable across a broad range of weight combinations: the default setting $(\lambda_{fa}, \lambda_{ca}) = (1.0, 1.0)$ achieves results close to the best ones on both 10S-ImageNetA and 10S-ImageNetR. Since the optimal trade-off slightly varies across datasets, we adopt $\lambda_{fa} = \lambda_{ca} = 1.0$ by default to avoid dataset-specific tuning.

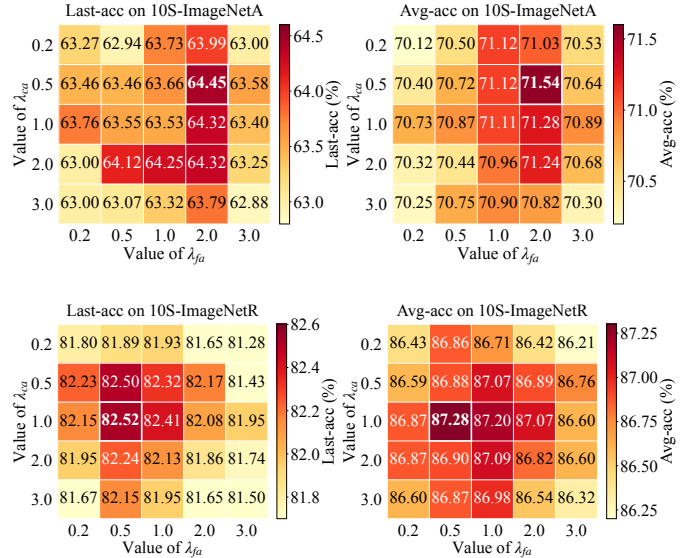


Fig. 8: Model performances under varying values of λ_{ca} and λ_{fa} on 10S-ImageNetA and 10S-ImageNetR.

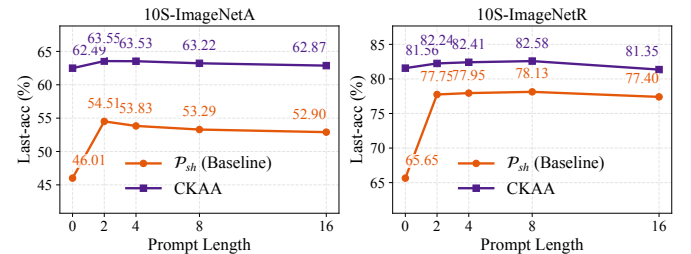


Fig. 9: Performances under different prompt lengths.

Prompt length and adapter width. We added a sensitivity analysis by varying the visual shared prompt length N_p over $\{0, 2, 4, 8, 16\}$ on 10S-ImageNetA and 10S-ImageNetR. The results are shown in Fig.9. The results demonstrate the effectiveness of the shared prompt for task-ID recognition.

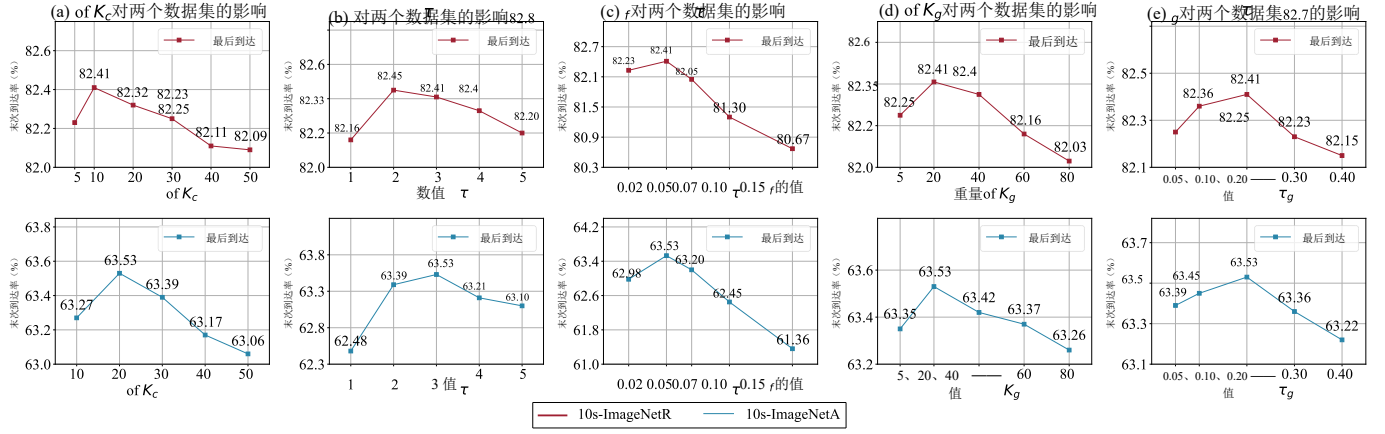


图 7: 在 10S-ImageNetR 和 10S-ImageNetA 数据集上, TC-MoA 中 K_c 与 τ 以及 DKA 中 τ_f , K_g 与 τ_g 的参数分析。

($\mathcal{P}_{sh}$)。横轴表示按类别划分的任务 ID 错误率 $\mathcal{E}_{TID}$ (c), 而纵轴表示该类别的分类准确率。这些可视化结果进一步证实, CKAA 能有效提升受错误任务 ID 估计影响的类别的鲁棒性。

E. 超参数分析

关键超参数。我们通过实验分析所提方法中各超参数对 10S-ImageNetR 和 10S-ImageNetA 基准集的影响, 包括等式 4 中的 τ_f 、等式 5 中的 K_g 与 τ_g 、等式 10 中的 K_c 以及等式 11 中的 τ 。我们首先分析 TC-MoA 中的超参数, 包括 K_c 与 τ 。结果如图 7(a) 和图 7(b) 所示。 K_c 用于筛选具有最高 logits 值的 Top- K_c 个类别以计算任务置信度分数; 若 K_c 过小, 任务置信度估计易受少数虚假激活类别的影响; 而若 K_c 过大, 则会包含来自无关任务的响应, 削弱正确适配器的主导性。较小的 τ 会使聚合权重过于尖锐, 而过大的 τ 则会使权重过度平滑, 并引入来自无关适配器的干扰。根据实验结果, 我们为 ImageNetR 设置 $K_c=10$, 为 ImageNetA 设置 $K_c=20$, 且对所有数据集均设 $\tau=2.0$ 。随后, 我们分析了 DKA 中的超参数, 包括 τ_f , K_g 与 τ_g 。结果如图 7(c)、图 7(d) 和图 7(e) 所示。 τ_f 是公式 4 中 Cross Subspace Feature Alignment (CSFA) 损失函数的温度因子; K_g 和 τ_g 是用于建模当前数据与先前数据间亲和力的两个参数。基于实验结果, 我们为所有数据集设置 $\tau_f=0.05$ 、 $K_g=20$ 以及 $\tau_g=0.2$ 。

损失权重 λ_{fa} 与 λ_{ca} 见等式 9。我们对 $\lambda_{fa}, \lambda_{ca} \in \{0.2, 0.5, 1.0, 2.0, 3.0\}$ 进行参数调优, 同时将 $\mathcal{L}_{ce}$ 的系数固定为 1.0。如图 8 所示, CKAA 在广泛的权重组合范围内均保持稳定: 默认设置 $(\lambda_{fa}, \lambda_{ca})=(1.0, 1.0)$ 在 10S-ImageNetA 和 10S-ImageNetR 两个数据集上均能获得接近最优的结果。由于最优均衡点在不同数据集间略有差异, 我们默认采用 $\lambda_{fa} = \lambda_{ca} = 1.0$, 以避免针对特定数据集进行参数调优。

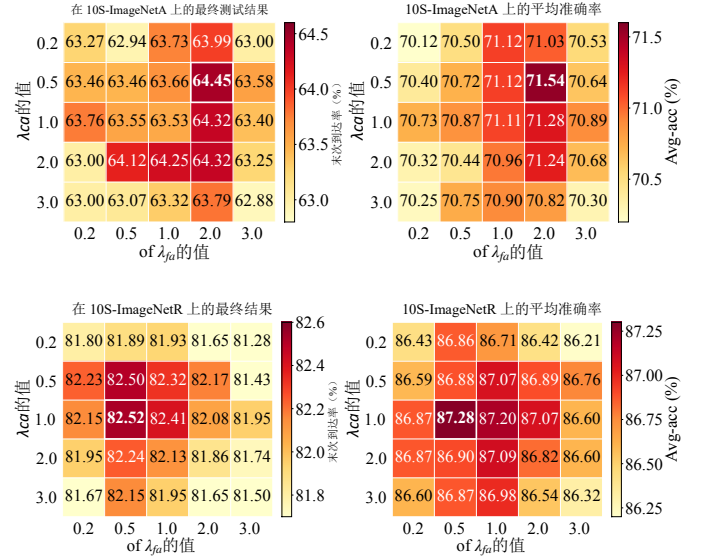


图 8: 在 10S-ImageNetA 和 10S-ImageNetR 数据集上, 模型在不同 λ_{ca} 与 λ_{fa} 值下的性能表现。

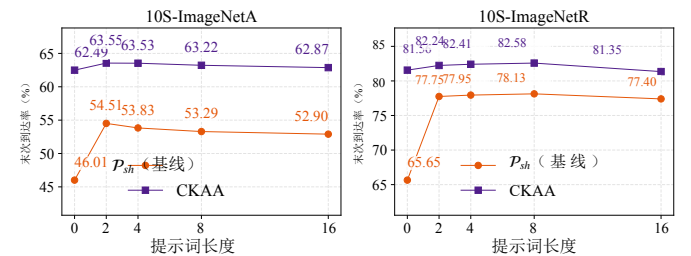


图 9: 不同提示长度下的性能表现。

提示长度与适配器宽度。我们通过在 10S-ImageNetA 和 10S-ImageNetR 数据集上将视觉共享提示长度 N_p 取值为 $\{0, 2, 4, 8, 16\}$ 来开展灵敏度分析。结果如图 9 所示。这些结果证明了共享提示在任务 ID 识别任务中的有效性。表 XVI: 不

TABLE XVI: Performances under different adapter widths.

Width	Params	FLOPs	10S-ImageNetA	10S-ImageNetR
2	0.42M (0.48%)	17.09 GMac	63.40 $\pm$ 0.66	81.63 $\pm$ 0.70
4	0.83M (0.96%)	17.16 GMac	63.84 $\pm$ 0.47	82.04 $\pm$ 0.20
8	1.66M (1.93%)	17.31 GMac	64.03 $\pm$ 1.06	81.77 $\pm$ 0.75
16	3.32M (3.86%)	17.60 GMac	63.56 $\pm$ 0.32	82.06 $\pm$ 1.09
32	6.64M (7.72%)	18.18 GMac	63.27 $\pm$ 0.74	82.15 $\pm$ 0.47
64	13.28M (15.44%)	19.34 GMac	63.53 $\pm$ 0.66	82.41 $\pm$ 0.13

Compared with the setting without the shared prompt, *i.e.*, $N_p = 0$, introducing a short shared prompt with $N_p = 2$ clearly improves the performance of our CKAA. The results also indicate that excessively long prompts may introduce redundant task-specific information and lead to overfitting. Based on the results, we set $N_p = 4$. We also conducted experiments to analyze the impact of adapter widths on 10S-ImageNetA and 10S-ImageNetR datasets, the results are shown in Tab.???. Our method performs well even with a width of 2, achieving a favorable trade-off between efficiency and performances, referred to as ‘CKAA-’.

TABLE XVII: Sensitivity analysis of the sample size in DKA on 10S-ImageNetA.

Sampling strategy	Sample size	Last-acc $\uparrow$	Avg-acc $\uparrow$
Class-balanced	1 per class	63.16 $\pm$ 0.47	70.57 $\pm$ 1.10
	2 per class	63.16 $\pm$ 0.27	70.83 $\pm$ 1.16
	5 per class	63.53 $\pm$ 0.66	71.11 $\pm$ 1.03
	10 per class	63.39 $\pm$ 0.24	71.12 $\pm$ 0.98
Fixed-budget	128 in total	62.88 $\pm$ 0.76	70.32 $\pm$ 1.08
	256 in total	63.26 $\pm$ 0.53	70.95 $\pm$ 1.23

The number of sampled pseudo features B_s . We consider two sampling strategies: (1) the class-balanced strategy, where “ M per class” means that M Gaussian features are generated from the estimated distribution of every class, and (2) the fixed-budget strategy, where the total number of sampled features is limited to a predefined budget, *i.e.*, 128 or 256. The results are in Tab.XVII. The performance remains relatively stable across different sample sizes and strategies, while the class-balanced strategy achieves slightly better performances. Based on the experimental results, we adopt 5 sampled features per class in all experiments. The additional cost of feature simulation mainly comes from affinity construction and subspace-shift transfer, resulting in $\Delta C_{\text{sim}}^{\text{train}} = \mathcal{O}(B_s d(B + K_g))$. Even with a large B_s when $B_s = 1000$, the cost introduced by feature simulation is $\Delta C_{\text{sim}}^{\text{train}} = 21,626,880 + 3,932,160 = 25,559,040 \approx 0.01\text{GMac}$ per batch ($\approx 0.0009\text{GMac}$ per sample). This overhead is negligible compared with the backbone computation ($\approx 17.23\text{GMac}$ per sample).

F. Analysis of additional computational costs

TABLE XVIII: Analysis of the complexity of components.

Component	Training Complexity	Inference Complexity
Task-specific adapters	$\mathcal{O}(BLNdw)$	$\mathcal{O}(TLNdw)$
FA ($\mathcal{L}_{csfa}$)	$\mathcal{O}(B^2d)$	–
CA (feature simulation)	$\mathcal{O}(B_s d(B + K_g))$	–
CA (global classifier g^u)	$\mathcal{O}(dK_T)$	$\mathcal{O}(dK_T)$
TC-MoA	–	$\mathcal{O}(TLNd)$

TABLE XIX: Practical additional computational costs under ViT-B/16, where $L = 12$, $N = 197$, $d = 768$.

Setting	Extra Params	Train Cost/Sample	Inference Cost/Image
CKAA ⁻ , $T = 10$, $w = 2$	+0.52M	+0.01 GMac	+0.091 GMac
CKAA, $T = 10$, $w = 64$	+11.95M	+0.23 GMac	+2.342 GMac
CKAA, $T = 100$, $w = 8$	+14.90M	+0.03 GMac	+3.086 GMac

We analyze the additional computational costs introduced by CKAA. Let T denote the number of learned tasks, L the number of Transformer blocks, N the number of tokens, d the feature dimension, w the adapter width. For CA, let B denote the mini-batch size, B_s the number of sampled pseudo features, K_g the number of nearest neighbors, and K_T the total number of seen classes. As shown in Tab. XVIII, the additional training cost mainly comes from the task-specific adapter and the feature simulation in CA, while the inference cost comes mainly from the aggregation of task-specific adapters in TC-MoA. The cost of the classifier increases with K_T , but remains negligible compared to the ViT backbone.

Under the standard 10-task setting with $K_T = 200$, our standard version CKAA with $w = 64$ introduces 11.95M extra parameters, while our efficient version CKAA- with $w = 2$ only introduces 0.52M. For computation, the additional adapter cost is $2LNd w$ per sample during training and $LTNd(2w+1)$ per image during inference with TC-MoA. The CA feature simulation cost is $\mathcal{O}(B_s d(B + K_g))$ per batch. With $B = 110$, $d = 768$, and $K_g = 20$, even $B_s = 1000$ only introduces about 0.10GMac per batch (0.0009 GMac per sample), which is negligible compared to the ViT-B/16 backbone computation ($\approx 17.00\text{GMac}$).

We provide a comparison of computational overhead and inference latency with existing PEFT-based CL methods. For fair, comparisons, all latency results are measured using a single NVIDIA RTX 4090 GPU under our standard setting with a batch size of 110, and the inference time per sample is computed by averaging the total inference time over all batch samples. As shown in Tab.XX, CKAA- exhibits an inference latency comparable to SSIAT and VPT-NSP, while achieving better performances.

VI. CONCLUSION

In this paper, we propose the Cross-subspace Knowledge Alignment and Aggregation (CKAA) framework to resolve feature subspace misalignment and improve model robustness against misleading task-ids. During training, CKAA introduces Dual-level Knowledge Alignment (DKA), which aligns feature semantics and unifies decision boundaries across task-specific subspaces, addressing the misalignment issue caused by independent sub-module training. During inference, CKAA designs Task-Confidence-guided Mixture of Adapters (TC-MoA), a dynamic aggregation scheme that softly combines task-specific knowledge based on confidence scores, mitigating overconfidence in erroneous task-id predictions.

Limitations. CKAA retains task-specific adapters for inference. Although the adapter width can be reduced to $w = 2$ for efficiency, the storage and inference costs still grow with the number of tasks. Future work will investigate more compact adapter compression strategies for long task sequences.

同适配器宽度下的性能表现。

宽度	参数	FLOPs	10S-ImageNetA	10S-ImageNetR
2	0.42M (0.48%)	17.09 GMac	63.40 $\pm$ 0.66	81.63 $\pm$ 0.70
4	0.83M (0.96%)	17.16 GMac	63.84 $\pm$ 0.47	82.04 $\pm$ 0.20
8	1.66M (1.93%)	17.31 GMac	64.03 $\pm$ 1.06	81.77 $\pm$ 0.75
16	3.32M (3.86%)	17.60 GMac	63.56 $\pm$ 0.32	82.06 $\pm$ 1.09
32	6.64M (7.72%)	18.18 GMac	63.27 $\pm$ 0.74	82.15 $\pm$ 0.47
64	13.28M (15.44%)	19.34 GMac	63.53 $\pm$ 0.66	82.41 $\pm$ 0.13

与不使用共享提示词的设置（即 $N_p=0$ ）相比，引入一个长度为 $N_p=2$ 的短共享提示词可显著提升我们所提出的CKAA的性能。实验结果还表明，过长的提示词可能会引入冗余的任务特定信息，从而导致过拟合。基于上述结果，我们设定 $N_p=4$ 。此外，我们还开展了实验以分析适配器宽度对10S-ImageNetA和10S-ImageNetR数据集的影响，结果如表？所示。即使采用适配器宽度为2的设置，我们的方法仍表现优异，在效率与性能之间取得了良好的平衡，该版本称为“CKAA-”。

表 XVII: 在 DKA 上对 10S-ImageNetA 数据集进行的样本量敏感性分析。

采样策略	样本量	最后到达个	Avg-acc $\uparrow$
平衡类	每班 1 人	63.16 $\pm$ 0.47	70.57 $\pm$ 1.10
	每班 2 人	63.16 $\pm$ 0.27	70.83 $\pm$ 1.16
	每班 5 人	63.53 $\pm$ 0.66	71.11 $\pm$ 1.03
	每班 10 人	63.39 $\pm$ 0.24	71.12 $\pm$ 0.98
固定预算	共计 128 个	62.88 $\pm$ 0.76	70.32 $\pm$ 1.08
	共计 256 个	63.26 $\pm$ 0.53	70.95 $\pm$ 1.23

采样伪特征的数量 B_s 。我们考虑两种采样策略：(1) 类别平衡策略，其中“ M per class”意味着从每个类别的估计分布中生成 M 个高斯特征；(2) 固定预算策略，其中采样特征的总数受限于预定义的预算（即 128 或 256）。结果见表. XVII。在不同样本量和策略下，模型性能保持相对稳定，而类别平衡策略的性能略优。基于实验结果，我们在所有实验中均采用每个类别采样 5 个特征。特征模拟带来的额外成本主要源于亲和度构建与子空间偏移迁移环节，由此得出 $\Delta Ctsraim^n = \mathcal{O}(B_s d(B + K_g))$ 。即使在 B_s 值较大（如 $B_s = 1000$ ）的情况下，特征模拟带来的计算成本为 $\Delta Ctsraim^n = 21,626,880 + 3,932,160 = 25,559,040 \approx 0.01$ GMac/批次（ ≈ 0.0009 GMac/样本）。与主干网络计算量（ ≈ 17.23 GMac/样本）相比，该开销可忽略不计。

F. 额外计算成本分析

表 XVIII: 组件复杂度分析。

组件	训练复杂度	推理复杂度
任务专用适配器	$\mathcal{O}(BLNdw)$	$\mathcal{O}(TLNdw)$
FA ($\mathcal{L}_{csfa}$)	$\mathcal{O}(B^2d)$	-
CA (特征仿真)	$\mathcal{O}(B_s d(B + K_g))$	-
CA (全局分类器 g_u)	$\mathcal{O}(dK_T)$	$\mathcal{O}(dK_T)$
TC-MoA	-	$\mathcal{O}(TLN)$

表 XIX: ViT-B/16 模型下的实际额外计算成本（其中 $L=12$, $N=197$, $d=768$ ）。

设置	额外参数	列车票价/示例	推理成本/图像
CKAA ⁻ , $T=10, w=2$	+0.52M	+0.01 GMac	+0.091 GMac
CKAA, $T=10, w=64$	+11.95M	+0.23 GMac	+2.342 GMac
CKAA, $T=100, w=8$	+14.90M	+0.03 GMac	+3.086 GMac

我们分析了CKAA带来的额外计算成本。设 T 表示学习到的任务数量， L 表示Transformer模块数量， N 表示Token数量， d 表示特征维度， w 表示适配器宽度。对于CA，设 B 表示小批量大小， B_s 表示采样伪特征数量， K_g 表示最近邻数量， K_T 表示已见类别的总数。如表xviii所示，额外的训练成本主要源于CA中的任务特定适配器和特征模拟，而推理成本则主要源于TCMoA中任务特定适配器的聚合操作。分类器的计算成本随 K_T 增大而增加，但与ViT基础模型相比仍可忽略不计。

在标准的10个任务设置（ $K_T=200$ ）下，我们的标准版本CKAA（ $w=64$ ）引入了11.95M个额外参数，而我们的版本CKAA-（ $w=2$ ）仅引入了0.52M个参数。就计算成本而言，在训练期间每个样本的额外适配器成本为 $2LNdw$ ，而在使用TC-MoA进行推理时，每个图像的额外成本为 $LTNd(2w+1)$ 。CA特征模拟成本为每批 $\mathcal{O}(B_s d(B + K_g))$ 。当 $B=110$ 、 $d=768$ 且 $K_g=20$ 时，即使 $B_s=1000$ ，每个批次也仅引入约0.10GMac（每个样本0.0009 GMac），与ViT-B/16主干网络的计算量（ ≈ 17.00 GMac）相比可忽略不计。

我们对计算开销与推理延迟与现有基于PEFT的CL方法进行了对比分析。为确保比较的公平性，所有延迟结果均在单块NVIDIA RTX 4090 GPU上、采用标准配置（批量大小为110）下测得；每个样本的推理时间是通过对所有批次样本的总推理时间取平均值计算得出的。如表. XX所示，CKAA的推理延迟与SSIAT和VPT-NSP相当，但其性能更优。

VI. 结论

本文提出了一种名为“跨子空间知识对齐与聚合”（CKAA）的框架，用于解决特征子空间错配问题并增强模型对误导性任务ID的鲁棒性。在训练阶段，CKAA引入了“双层知识对齐”（DKA）机制，该机制可对齐特征语义并统一跨任务特定子空间的决策边界，从而解决由独立子模块训练所导致的子空间错配问题；在推理阶段，CKAA设计了“任务置信度引导的适配器混合机制”（TCMoA）——这是一种基于置信度分数对任务特定知识进行软性融合的动态聚合方案，可缓解对错误任务ID预测所产生的过度置信问题。**局限性：**CKAA在推理时仍保留任务特定的适配器。尽管为提高效率可将适配器宽度缩减至 $w=2$ ，但其存储和推理成本仍随任务数量的增加而增长。未来的工作将探索适用于长任务序列的更紧凑的适配器压缩策略。

TABLE XX: Comparison with representative existing approaches on 10S-ImageNetR in terms of computational overhead, inference time per image, and CL performance. The latency is measured on a single NVIDIA RTX 4090 GPU. “–” indicates that the corresponding task recognition process is not involved in the framework

Method	Venue	Task Recognition		Inference		10S-ImageNetR	
		Overhead	Time/Image	Overhead	Time/Image	Last-acc $\uparrow$	Avg-acc $\uparrow$
L2P [20]	CVPR-22	17.0GMac	0.515ms	19.9GMac	0.658ms	72.34 $\pm$ 0.17	77.36 $\pm$ 0.64
DualPrompt [23]	ECCV-22	17.0GMac	0.515ms	17.6GMac	0.607ms	69.10 $\pm$ 0.62	74.28 $\pm$ 0.66
CODAPrompt [24]	CVPR-23	17.0GMac	0.515ms	18.2GMac	0.720ms	73.31 $\pm$ 0.50	78.47 $\pm$ 0.53
Adam-adapter [63]	IJCV-24	–	–	17.8GMac	0.599ms	65.79 $\pm$ 0.98	72.42 $\pm$ 1.41
SSIAT [2]	CVPR-24	–	–	17.8GMac	0.599ms	79.38 $\pm$ 0.59	83.63 $\pm$ 0.43
InfLoRA [21]	CVPR-24	–	–	20.5GMac	0.650ms	75.65 $\pm$ 0.14	80.82 $\pm$ 0.24
VQ-Prompt [28]	NeurIPS-24	17.0GMac	0.515ms	18.2GMac	0.622ms	78.71 $\pm$ 0.22	83.24 $\pm$ 0.68
VPT-NSP [3]	NeurIPS-24	–	–	17.0GMac	0.536ms	77.95 $\pm$ 0.22	83.44 $\pm$ 0.40
LoRA P&M [83]	NeurIPS-25	–	–	20.5GMac	0.650ms	79.95 $\pm$ 0.18	85.29 $\pm$ 0.93
CKAA- (Ours)	–	17.0GMac	0.536ms	17.1GMac	0.566ms	81.63$\pm$0.70	86.93$\pm$0.04
CKAA (Ours)	–	17.0GMac	0.536ms	19.3GMac	0.629ms	82.41$\pm$0.13	87.20$\pm$0.35

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表 XX: 在 10S-ImageNetR 数据集上与现有代表性方法在计算开销、单张图像推理时间及 CL 性能方面的对比。延迟值是在单块 NVIDIA RTX 4090 GPU 上测得的。“-”表示相应的任务识别流程未包含在该框架中。

方法	发生地	任务识别		推理		10S-ImageNetR	
		吊装	时间/图像	吊装	时间/图像	最后到达↑	Avg-acc ↑
L2P [20]	CVPR-22	17.0GMac	0.515ms	19.9GMac	0.658ms	72.34±0.17	77.36±0.64
DualPrompt [23]	ECCV-22	17.0GMac	0.515ms	17.6GMac	0.607ms	69.10±0.62	74.28±0.66
CODAPrompt [24]	CVPR-23	17.0GMac	0.515ms	18.2GMac	0.720ms	73.31±0.50	78.47±0.53
Adam适配器[63]	IJCV-24	-	-	17.8GMac	0.599ms	65.79±0.98	72.42±1.41
SSIAT [2]	CVPR-24	-	-	17.8GMac	0.599ms	79.38±0.59	83.63±0.43
InfLoRA [21]	CVPR-24	-	-	20.5GMac	0.650ms	75.65±0.14	80.82±0.24
VQ-Prompt [28]	NeurIPS-24	17.0GMac	0.515ms	18.2GMac	0.622ms	78.71±0.22	83.24±0.68
VPT-NSP [3]	NeurIPS-24	-	-	17.0GMac	0.536ms	77.95±0.22	83.44±0.40
LoRA P&M [83]	NeurIPS-25	-	-	20.5GMac	0.650ms	79.95±0.18	85.29±0.93
CKAA - (我们)	-	17.0GMac	0.536ms	17.1GMac	0.566ms	81.63±0.70	86.93±0.04
CKAA (我们)	-	17.0GMac	0.536ms	19.3GMac	0.629ms	82.41±0.13	87.20±0.35

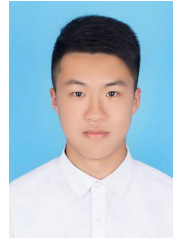
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APPENDIX A THEORETICAL ANALYSIS

In this section, we would like to provide detailed theoretical analysis of the working mechanism of the proposed Dual-level Knowledge Alignment (DKA) and Task-Confidence-guided Mixture-of-Adapters (TC-MoA) modules.

Optimization Objective. Consider a sample $(\mathbf{x}, y) \sim \mathcal{D}_r$ from task r . Let the feature generated by the correct task-specific module be $\mathbf{z}_r(\mathbf{x}) = F(\mathbf{x}; \mathcal{P}_{sh}, \mathbf{A}_{sp}^r)$, and let the feature generated by an incorrectly selected historical module $s < r$ be $\mathbf{z}_s(\mathbf{x}) = F(\mathbf{x}; \mathcal{P}_{sh}, \mathbf{A}_{sp}^s)$. Let $g(\cdot)$ denote the classifier and $h_g(\mathbf{z}) = \arg \max_k g_k(\mathbf{z})$ denote its prediction function. The risks under correct and incorrect routing are respectively defined as $R_{r \rightarrow r}(g) = \Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} [h_g(\mathbf{z}_r(\mathbf{x})) \neq y]$ and $R_{r \rightarrow s}(g) = \Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} [h_g(\mathbf{z}_s(\mathbf{x})) \neq y]$. Our robustness objective is to minimize the classification risk over both the correct subspace and possible incorrect historical subspaces:

$$\min \mathcal{L}_r^{\text{robust}}(g) = R_{r \rightarrow r}(g) + \frac{1}{r-1} \sum_{s=1}^{r-1} [\lambda_1 R_{r \rightarrow s}(g) + \lambda_2 R_{s \rightarrow r}(g)], \quad (19)$$

where λ_1 and λ_2 are trade-off hyper-parameters. The first term can be optimized by the standard cross-entropy loss. Therefore, this work mainly focuses on reducing the two cross-subspace misrouting terms. Specifically, the proposed CSFA loss targets the incorrect new-to-old routing risk $R_{r \rightarrow s}(g)$, and the proposed CA targets the incorrect old-to-new routing risk $R_{s \rightarrow r}(g)$. In this way, CSFA and CA complementarily improve model robustness against bidirectional misleading task-ID selection.

A. Theoretical analysis of FA

In this section, we would like to explain how $\mathcal{L}_{csfa}$ minimizes $R_{r \rightarrow s}(g)$. For $(\mathbf{x}, y)$, we define its classification margin in the correct subspace as $m_r(\mathbf{x}, y) = g_y(\mathbf{z}_r(\mathbf{x})) - \max_{k \neq y} g_k(\mathbf{z}_r(\mathbf{x}))$. If $m_r(\mathbf{x}, y) > 0$, the sample is correctly classified when routed to its correct task-specific subspace. We further define its classification margin in the incorrectly subspace of the s -th task as $m_s(\mathbf{x}, y) = g_y(\mathbf{z}_s(\mathbf{x})) - \max_{k \neq y} g_k(\mathbf{z}_s(\mathbf{x}))$. We assume that the logit difference between any two classes is L_g -Lipschitz with respect to the input feature representation. Specifically, for any classes y and k , and any feature representations $\mathbf{z}_r$ and $\mathbf{z}_s$, we have:

$$|(g_y(\mathbf{z}_r(\mathbf{x})) - g_k(\mathbf{z}_r(\mathbf{x}))) - (g_y(\mathbf{z}_s(\mathbf{x})) - g_k(\mathbf{z}_s(\mathbf{x})))| \leq L_g \|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2. \quad (20)$$

Therefore, the margin in the incorrect subspace s is lower bounded by:

$$g_y(\mathbf{z}_s(\mathbf{x})) - g_k(\mathbf{z}_s(\mathbf{x})) \geq g_y(\mathbf{z}_r(\mathbf{x})) - g_k(\mathbf{z}_r(\mathbf{x})) - L_g \|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2. \quad (21)$$

Since the above inequality holds for every competing class $k \neq y$ for an input $\mathbf{x}$, we take the minimum over all $k \neq y$ on both sides. By the definitions of $m_r(\mathbf{x}, y)$ and $m_s(\mathbf{x}, y)$, we obtain the following inequality:

$$m_s(\mathbf{x}, y) \geq m_r(\mathbf{x}, y) - L_g \|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2. \quad (22)$$

Consider an arbitrary margin threshold $\gamma > 0$, we can derive an upper bound on this risk:

$$R_{r \rightarrow s}(g) \leq \Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} [m_r(\mathbf{x}, y) < \gamma] + \Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} \left[\|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2 \geq \frac{\gamma}{L_g} \right]. \quad (23)$$

The above upper bound indicates that this risk $R_{r \rightarrow s}(g)$ is associated with two factors: (1) the probability that a sample has an insufficient classification margin in its correct subspace, and (2) the probability that its features induce a large discrepancy when projected into an incorrect historical subspace.

Then we would like to explain why the proposed CSFA loss minimize this upper bound. Specifically, our CSFA loss can be rewritten as follows:

$$\mathcal{L}_{csfa} = \frac{1}{B} \sum_{i=1}^B \log \left[\sum_{\mathbf{z}_j \in \mathcal{P}(\mathbf{z}_i) \cup \mathcal{N}(\mathbf{z}_i)} \exp \left(\frac{\text{sim}(\mathbf{z}_i, \mathbf{z}_j)}{\tau_f} \right) \right] - \frac{1}{B} \sum_{i=1}^B \log \left[\sum_{\mathbf{z}^+ \in \mathcal{P}(\mathbf{z}_i)} \exp \left(\frac{\text{sim}(\mathbf{z}_i, \mathbf{z}^+)}{\tau_f} \right) \right]. \quad (24)$$

The first term in $\mathcal{L}_{csfa}$ contains features with different labels $\mathbf{z}_j \in \mathcal{N}(\mathbf{z}_i)$. Minimizing $\mathcal{L}_{csfa}$ suppresses their similarity to $\mathbf{z}_i$, thereby enlarging the separation between different classes in the current subspace. Such enhanced inter-class discriminability encourages larger classification margins $m_r(\mathbf{x}, y)$ and thus helps reduce the first term $\Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} [m_r(\mathbf{x}, y) < \gamma]$ in Eq.23.

The second term maximizes the similarity between each feature $\mathbf{z}_i$ and its same-class positive features $\mathbf{z}^+ \in \mathcal{P}(\mathbf{z}_i)$ extracted from incorrect subspaces. This cross-subspace feature alignment reduces the L_2 -distance between representations across subspaces, thereby helping reduce the second term in $\Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} [\|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2 \geq \frac{\gamma}{L_g}]$ in Eq.23.

B. Theoretical analysis of CA

In this section, we would like to analyze why our CA minimizes the old-to-new misrouting term $R_{s \rightarrow r}(g)$. For a historical sample $(\mathbf{x}, y) \sim \mathcal{D}_s$ with $s < r$, its feature from the task-shared network is denoted as $\mathbf{f} = F(\mathbf{x}; \mathcal{P}_{sh})$, and its feature generated after being incorrectly routed into the current task-specific subspace r is denoted as $\mathbf{z}_r(\mathbf{x}) = F(\mathbf{x}; \mathcal{P}_{sh}, \mathbf{A}_{sp}^r)$. Accordingly, the incorrect old-to-new routing risk is defined as

$$R_{s \rightarrow r}(g^r) = \Pr_{(\mathbf{x}, y) \sim \mathcal{D}_s} [h_{g^r}(\mathbf{z}_r(\mathbf{x})) \neq y]. \quad (25)$$

However, historical training samples are unavailable during the learning of task r . Therefore, CA constructs a simulated feature $\tilde{\mathbf{z}}$ to approximate the representation that a historical sample would produce in the current subspace. We define the classification margin of the real incorrectly routed feature as $m_{s \rightarrow r}(\mathbf{z}_r(\mathbf{x}), y) = g_y^r(\mathbf{z}_r(\mathbf{x})) - \max_{k \neq y} g_k^r(\mathbf{z}_r(\mathbf{x}))$, and define the classification margin of its simulated feature as $\tilde{m}_{s \rightarrow r}(\tilde{\mathbf{z}}, y) = g_y^r(\tilde{\mathbf{z}}) - \max_{k \neq y} g_k^r(\tilde{\mathbf{z}})$.

附录A 理论分析

本节将对所提出的双层知识对齐 (DKA) 模块与任务置信度引导的适配器混合 (TC-MoA) 模块的工作机制进行详细的理论分析。

优化目标。考虑来自任务 r 的样本 $(\mathbf{x}, y) \sim \mathcal{D}_r$ 。令由正确任务特定模块生成的特征为 $\mathbf{z}_r(\mathbf{x}) = F(\mathbf{x}; \mathcal{P}_{sh}, \mathbf{Arsp})$ ；令由错误选择的历史模块 s_r 生成的特征为 $\mathbf{z}_s(\mathbf{x}) = F(\mathbf{x}; \mathcal{P}_{sh}, \mathbf{Assp})$ 。设 $g(\cdot)$ 表示分类器， $h_g(\mathbf{z}) = \arg \max_k g_k(\mathbf{z})$ 表示其预测函数。正确路由与错误路由下的风险分别定义为 $R_{r \rightarrow r}(g) = \Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} [h_g(\mathbf{z}_r(\mathbf{x})) \neq y]$ 且 $R_{r \rightarrow s}(g) = \Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} [h_g(\mathbf{z}_s(\mathbf{x})) \neq y]$ 。我们的鲁棒性目标是 minimized 在正确子空间以及可能存在的错误历史子空间上的分类风险：

$$\min \mathcal{L}_r^{\text{robust}}(g) = R_{r \rightarrow r}(g) + \frac{1}{r-1} \sum_{s=1}^{r-1} [\lambda_1 R_{r \rightarrow s}(g) + \lambda_2 R_{s \rightarrow r}(g)], \quad (19)$$

其中 λ_1 与 λ_2 为权衡超参数。第一项可通过标准交叉熵损失函数进行优化。因此，本研究主要致力于降低两项交叉子空间误路由项。具体而言，所提出的 CSFA 损失项针对错误的“新到旧”路由风险 $R_{r \rightarrow s}(g)$ ，而所提出的 CA 损失项则针对错误的“旧到新”路由风险 $R_{s \rightarrow r}(g)$ 。通过这种方式，CSFA 与 CA 相互补充，增强了模型对双向误导性任务-ID选择的鲁棒性。

A. FA的理论分析

在本节中，我们将解释 $\mathcal{L}_{csfa}$ 如何最小化 $R_{r \rightarrow s}(g)$ 。对于点 $(\mathbf{x}, y)$ ，我们定义其在正确子空间中的分类间隔为 $m_r(\mathbf{x}, y) = g_y(\mathbf{z}_r(\mathbf{x})) - \max_{k \neq y} g_k(\mathbf{z}_r(\mathbf{x}))$ 。若 $m_r(\mathbf{x}, y) > 0$ ，则当样本被路由至其正确的任务特定子空间时，该样本被正确分类。我们进一步定义其在第 s 个任务的错误子空间中的分类间隔为 $m_s(\mathbf{x}, y) = g_y(\mathbf{z}_s(\mathbf{x})) - \max_{k \neq y} g_k(\mathbf{z}_s(\mathbf{x}))$ 。我们假设任意两个类别之间的 Logit 差值相对于输入特征表示具有 L_g -Lipschitz 连续性。具体而言，对于任意类别 y 与 k 以及任意特征表示 $\mathbf{z}_r$ 与 $\mathbf{z}_s$ ，有：

$$|(g_y(\mathbf{z}_r(\mathbf{x})) - g_k(\mathbf{z}_r(\mathbf{x}))) - (g_y(\mathbf{z}_s(\mathbf{x})) - g_k(\mathbf{z}_s(\mathbf{x})))| \leq L_g \|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2. \quad (20)$$

因此，错误子空间 s 中的边际值下界为：

$$g_y(\mathbf{z}_s(\mathbf{x})) - g_k(\mathbf{z}_s(\mathbf{x})) \geq g_y(\mathbf{z}_r(\mathbf{x})) - g_k(\mathbf{z}_r(\mathbf{x})) - L_g \|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2. \quad (21)$$

由于上述不等式对每个竞争类 k 均成立。 $\neq y$ 对于输入 $\mathbf{x}$ ，我们对所有 k 取最小值。 $\neq y$ 位于两侧。根据 $m_r(\mathbf{x}, y)$ 与 $m_s(\mathbf{x}, y)$ 的定义，可得如下不等式：

$$m_s(\mathbf{x}, y) \geq m_r(\mathbf{x}, y) - L_g \|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2. \quad (22)$$

对于任意满足 $\gamma > 0$ 的边际阈值，我们可以推导出该风险的一个上界：

$$R_{r \rightarrow s}(g) \leq \Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} [m_r(\mathbf{x}, y) < \gamma] + \Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} \left[\|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2 \geq \frac{\gamma}{L_g} \right]. \quad (23)$$

上述上界表明，该风险 $R_{r \rightarrow s}(g)$ 与两个因素相关：(1) 样本在正确子空间中具有不足分类间隔的概率；(2) 其特征在投影到错误的历史子空间时引发较大差异的概率。

接下来，我们想解释为何所提出的 CSFA 损失函数能够最小化这一上界。具体而言，我们的 CSFA 损失函数可重写为如下形式：

$$\mathcal{L}_{csfa} = \frac{1}{B} \sum_{i=1}^B \log \left[\sum_{\mathbf{z}_j \in \mathcal{P}(\mathbf{z}_i) \cup \mathcal{N}(\mathbf{z}_i)} \exp \left(\frac{\text{sim}(\mathbf{z}_i, \mathbf{z}_j)}{\tau_f} \right) \right] - \frac{1}{B} \sum_{i=1}^B \log \left[\sum_{\mathbf{z}^+ \in \mathcal{P}(\mathbf{z}_i)} \exp \left(\frac{\text{sim}(\mathbf{z}_i, \mathbf{z}^+)}{\tau_f} \right) \right]. \quad (24)$$

$\mathcal{L}_{csfa}$ 中的第一项包含具有不同标签 $\mathbf{z}_j \in \mathcal{N}(\mathbf{z}_i)$ 的特征。最小化 $\mathcal{L}_{csfa}$ 可抑制这些特征与 $\mathbf{z}_i$ 之间的相似性，从而扩大当前子空间中不同类别的分隔度。这种增强的类间判别能力有助于获得更大的分类间隔 $m_r(\mathbf{x}, y)$ ，进而有助于减小公式23中第一项 $\Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} [m_r(\mathbf{x}, y) < \gamma]$ 。

第二项最大化了每个特征 $\mathbf{z}_i$ 与其从错误子空间中提取出的同类别正样本特征 $\mathbf{z}^+ \in \mathcal{P}(\mathbf{z}_i)$ 之间的相似度。这种跨子空间特征对齐方法降低了不同子空间间表示之间的 L_2 距离，从而有助于减小 $\Pr_{(\mathbf{x}, y) \sim \mathcal{D}_r} [\|\mathbf{z}_r(\mathbf{x}) - \mathbf{z}_s(\mathbf{x})\|_2 \geq \frac{\gamma}{L_g}]$ 中的第二项。见公式23。

B. CA的理论分析

在本节中，我们旨在分析为何我们的CA能最小化旧到新路径错误项 $R_{s \rightarrow r}(g)$ 。对于历史样本 $(\mathbf{x}, y) \sim \mathcal{D}_s$ (其中 $s < r$)，其来自任务共享网络的特征记为 $\mathbf{f} = F(\mathbf{x}; \mathcal{P}_{sh})$ ；而该特征在错误路由至当前任务特定子空间 r 后生成的特征记为 $\mathbf{z}_r(\mathbf{x}) = F(\mathbf{x}; \mathcal{P}_{sh}, \mathbf{Arsp})$ 。据此，错误的旧到新路由风险定义为：

$$R_{s \rightarrow r}(g^r) = \Pr_{(\mathbf{x}, y) \sim \mathcal{D}_s} [h_{g^r}(\mathbf{z}_r(\mathbf{x})) \neq y]. \quad (25)$$

然而，在任务 r 的学习过程中，历史训练样本不可用。因此，CA 构建了一个模拟特征 $\tilde{\mathbf{z}}$ 以近似历史样本在当前子空间中所产生的表征。我们将真实错误路由特征的分类间隔定义为 $m_{s \rightarrow r}(\mathbf{z}_r(\mathbf{x}), y) = g^r(\mathbf{z}_r(\mathbf{x})) - \max_{k \neq y} g^r_k(\mathbf{z}_r(\mathbf{x}))$ ，并将其模拟特征的分类间隔定义为 $m_{s \rightarrow r}(\tilde{\mathbf{z}}, y) = g^r(\tilde{\mathbf{z}}) - \max_{k \neq y} g^r_k(\tilde{\mathbf{z}})$ 。

Similar to the analysis of FA, we assume that the logit difference between any two classes under g^r is L_{g^r} -Lipschitz with respect to the input feature representation. the logit difference between the ground-truth class and the competing class on the real incorrectly routed feature $\mathbf{z}_r(\mathbf{x})$ is lower bounded by the corresponding logit difference on the simulated feature $\tilde{\mathbf{z}}$, minus their feature discrepancy scaled by L_{g^r} . Since this relation holds for every competing class $k \neq y$, taking the minimum over all competing classes yields the following inequality:

$$m_{s \rightarrow r}(\mathbf{z}_r(\mathbf{x}), y) \geq \tilde{m}_{s \rightarrow r}(\tilde{\mathbf{z}}, y) - L_{g^r} \|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}\|_2. \quad (26)$$

Given an arbitrary margin threshold $\gamma > 0$, we can derive an upper bound on this risk:

$$R_{s \rightarrow r}(g^r) \leq \Pr[\tilde{m}_{s \rightarrow r}(\tilde{\mathbf{z}}, y) < \gamma] + \Pr\left[\|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}\|_2 \geq \frac{\gamma}{L_{g^r}}\right]. \quad (27)$$

The above upper bound indicates that the incorrect old-to-new routing risk $R_{s \rightarrow r}(g^r)$ is determined by two factors: (1) the classification capability $\Pr[\tilde{m}_{s \rightarrow r}(\tilde{\mathbf{z}}, y) < \gamma]$ on the simulated historical features, and (2) the approximation error $\Pr\left[\|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}\|_2 \geq \frac{\gamma}{L_{g^r}}\right]$ between the simulated features and the real incorrectly routed features. The first term is explicitly optimized by training the task-adaptive classifier g^r with the simulated features in $\mathcal{L}_{ca}$. Therefore, we mainly focus on analyzing the second term, which characterizes the quality of feature simulation.

Since $\|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}\|_2$ is non-negative, applying Markov's inequality yields:

$$\Pr\left[\|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}\|_2 \geq \frac{\gamma}{L_{g^r}}\right] \leq \frac{L_{g^r}}{\gamma} \mathbb{E}[\|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}\|_2]. \quad (28)$$

CA samples a same-class pseudo feature $\tilde{\mathbf{f}}_i$ and constructs its simulated feature as follows:

$$\tilde{\mathbf{z}}_i = \tilde{\mathbf{f}}_i + \frac{\sum_{j=1}^B G_{ij}(\mathbf{z}_j - \mathbf{f}_j)}{\sum_{j=1}^B G_{ij}}. \quad (29)$$

Accordingly, the feature simulation error can be decomposed as follows:

$$\begin{aligned} & \|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}_i\|_2 \\ &= \left\| \mathbf{f} + (\mathbf{z}_r(\mathbf{x}) - \mathbf{f}) - \tilde{\mathbf{f}}_i - \frac{\sum_{j=1}^B G_{ij}(\mathbf{z}_j - \mathbf{f}_j)}{\sum_{j=1}^B G_{ij}} \right\|_2 \\ &\leq \left\| \mathbf{f} - \tilde{\mathbf{f}}_i \right\|_2 + \left\| (\mathbf{z}_r(\mathbf{x}) - \mathbf{f}) - \frac{\sum_{j=1}^B G_{ij}(\mathbf{z}_j - \mathbf{f}_j)}{\sum_{j=1}^B G_{ij}} \right\|_2. \end{aligned} \quad (30)$$

The first term measures the approximation error introduced by Gaussian sampling, and the second term measures the error of transferring the subspace shift from current samples to the sampled historical feature through the affinity matrix.

To further bound the affinity-guided subspace shift transfer error, we first note that the affinity matrix only transfers shifts from features in the neighborhood. Specifically, according to Eq. 5, $G_{ij} > 0$ only when $\mathbf{f}_j \in \mathcal{K}(\tilde{\mathbf{f}}_i, K_g)$. We assume that the subspace shift from the shared space to the current

task-specific subspace is smooth within the local neighborhood selected by G , where $\|(\mathbf{z}_r(\mathbf{x}) - \mathbf{f}) - (\mathbf{z}_j - \mathbf{f}_j)\|_2 \leq L_\Delta \|\mathbf{f} - \mathbf{f}_j\|_2$ and L_Δ characterizes the smoothness of the subspace shift. Since $G_{ij} \geq 0$ and $\sum_{j=1}^B G_{ij} > 0$, given the vector triangle inequality to the weighted combination and the local smoothness assumption, we have:

$$\begin{aligned} & \left\| (\mathbf{z}_r(\mathbf{x}) - \mathbf{f}) - \frac{\sum_{j=1}^B G_{ij}(\mathbf{z}_j - \mathbf{f}_j)}{\sum_{j=1}^B G_{ij}} \right\|_2 \\ &\leq \frac{\sum_{j=1}^B G_{ij} \|(\mathbf{z}_r(\mathbf{x}) - \mathbf{f}) - (\mathbf{z}_j - \mathbf{f}_j)\|_2}{\sum_{j=1}^B G_{ij}} \\ &\leq L_\Delta \frac{\sum_{j=1}^B G_{ij} \|\mathbf{f} - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}}. \end{aligned} \quad (31)$$

Furthermore, by the inequality $\|\mathbf{f} - \mathbf{f}_j\|_2 \leq \|\mathbf{f} - \tilde{\mathbf{f}}_i\|_2 + \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2$, the feature simulation error is bounded by:

$$\begin{aligned} & \|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}_i\|_2 \leq \\ & (1 + L_\Delta) \|\mathbf{f} - \tilde{\mathbf{f}}_i\|_2 + L_\Delta \frac{\sum_{j=1}^B G_{ij} \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}}. \end{aligned} \quad (32)$$

Then we obtain the upper bound of the risk as follows:

$$\begin{aligned} R_{s \rightarrow r}(g^r) &\leq \Pr[\tilde{m}_{s \rightarrow r}(\tilde{\mathbf{z}}, y) < \gamma] \\ &+ \frac{L_{g^r}}{\gamma} \left\{ (1 + L_\Delta) \mathbb{E}[\|\mathbf{f} - \tilde{\mathbf{f}}_i\|_2] \right. \\ &\left. + L_\Delta \mathbb{E} \left[\frac{\sum_{j=1}^B G_{ij} \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}} \right] \right\}. \end{aligned} \quad (33)$$

Specifically, the first term is explicitly optimized through $\mathcal{L}_{ca}$. The term $\mathbb{E}[\|\mathbf{f} - \tilde{\mathbf{f}}_i\|_2]$ corresponds to the approximation error from Gaussian sampling. The term $\mathbb{E} \left[\frac{\sum_{j=1}^B G_{ij} \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}} \right]$ measures the affinity-weighted neighborhood radius between the sampled historical feature and the current-task features in the shared feature space. The second term becomes small when historical and current tasks exhibit locally related semantics.

The Gaussian sampling error term $\mathbb{E}[\|\mathbf{f} - \tilde{\mathbf{f}}_i\|_2]$ is small when the shared features of historical classes can be well approximated by Gaussian distributions. Although this assumption is supported by existing works [84], [85], we would like to provide further evidence to analyze why it is reasonable. For the k -th class in task t , let its shared feature distribution be approximated by $G_{k,t} = \mathcal{N}(\boldsymbol{\mu}_{k,t}, \boldsymbol{\Sigma}_{k,t})$. A fundamental property of the multivariate Gaussian distribution is that, if $\mathbf{f} \sim \mathcal{N}(\boldsymbol{\mu}_{k,t}, \boldsymbol{\Sigma}_{k,t})$, then its squared Mahalanobis distance to the class center follows a chi-square distribution:

$$m(\mathbf{f}, \boldsymbol{\mu}_{k,t}) = (\mathbf{f} - \boldsymbol{\mu}_{k,t})^\top \boldsymbol{\Sigma}_{k,t}^{-1} (\mathbf{f} - \boldsymbol{\mu}_{k,t}) \sim \chi_d^2, \quad (34)$$

where d is the feature dimension. This property provides a direct diagnostic for examining whether class-wise shared features follow the assumed Gaussian distribution. Specifically, We randomly select 16 historical classes from 10S-ImageNetR

与FA的分析类似，我们假设在 g^r 下任意两个类别之间的logit差值相对于输入特征表示满足 L_{gr} -Lipschitz条件。在真实错误路由特征 $\mathbf{z}_r(\mathbf{x})$ 上，真实类别与竞争类别之间的logit差值下界等于在模拟特征 $\tilde{\mathbf{z}}$ 上对应的logit差值减去其特征差异乘以 L_{gr} 。由于这一关系对每个竞争类别 k 均成立。 $\neq y$ ，对所有竞争类别取最小值可得到以下不等式：

$$m_{s \rightarrow r}(\mathbf{z}_r(\mathbf{x}), y) \geq \tilde{m}_{s \rightarrow r}(\tilde{\mathbf{z}}, y) - L_{gr} \|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}\|_2. \quad (26)$$

给定任意边际阈值 $\gamma > 0$ ，我们可以推导出该风险的一个上界：

$$R_{s \rightarrow r}(g^r) \leq \Pr[\tilde{m}_{s \rightarrow r}(\tilde{\mathbf{z}}, y) < \gamma] + \Pr\left[\|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}\|_2 \geq \frac{\gamma}{L_{gr}}\right]. \quad (27)$$

上述上界表明，错误的“旧到新”路由选择风险 $R_{s \rightarrow r}(g^r)$ 由两个因素决定：

(1) 基于模拟历史特征的分类能力 $\Pr[m_{s \rightarrow r}(\tilde{\mathbf{z}}, y) < \gamma]$ ，以及(2)近似误差

$\Pr\left[\|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}\|_2 \geq \frac{\gamma}{L_{gr}}\right]$ between the simulated features and 实际特征路由错误问题。第一项通过使用 $\mathcal{L}_{ca}$ 中的模拟特征对任务自适应分类器 g^r 进行训练而实现显式优化。因此，我们主要关注对第二项进行分析，该项用于表征特征模拟的质量。

由于 $\|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}\|_2$ 为非负数，应用马尔可夫不等式可得：

$$\Pr\left[\|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}\|_2 \geq \frac{\gamma}{L_{gr}}\right] \leq \frac{L_{gr}}{\gamma} \mathbb{E}[\|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}\|_2]. \quad (28)$$

CA 对同一类别的伪特征 $\tilde{\mathbf{f}}_i$ 进行采样，并按以下方式构建其模拟特征：

$$\tilde{\mathbf{z}}_i = \tilde{\mathbf{f}}_i + \frac{\sum_{j=1}^B G_{ij} (\mathbf{z}_j - \mathbf{f}_j)}{\sum_{j=1}^B G_{ij}}. \quad (29)$$

因此，特征仿真误差可分解为以下形式：

$$\begin{aligned} \|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}_i\|_2 &= \left\| \mathbf{f} + (\mathbf{z}_r(\mathbf{x}) - \mathbf{f}) - \tilde{\mathbf{f}}_i - \frac{\sum_{j=1}^B G_{ij} (\mathbf{z}_j - \mathbf{f}_j)}{\sum_{j=1}^B G_{ij}} \right\|_2 \\ &\leq \left\| \mathbf{f} - \tilde{\mathbf{f}}_i \right\|_2 + \left\| (\mathbf{z}_r(\mathbf{x}) - \mathbf{f}) - \frac{\sum_{j=1}^B G_{ij} (\mathbf{z}_j - \mathbf{f}_j)}{\sum_{j=1}^B G_{ij}} \right\|_2. \end{aligned} \quad (30)$$

第一项衡量由高斯采样引入的近似误差，第二项衡量通过亲和力矩阵将子空间偏移从当前样本传递至采样历史特征时所产生的误差。

为进一步约束亲和力引导的子空间位移转移误差，我们首先注意到，亲和力矩阵仅将邻域内特征的位移进行转移。具体而言，根据等式5， $G_{ij} > 0$ 仅当 $\mathbf{f}_j \in K(\tilde{\mathbf{f}}_i, K_g)$ 时成立。我们假设从共享空间到当前任务特定子空间的子

空间位移在由 G 选定的局部邻域内是平滑的，其中 $\|\mathbf{z}_r(\mathbf{x}) - \mathbf{f}\|_2 - \|\mathbf{z}_j - \mathbf{f}_j\|_2 \leq L_\Delta \|\mathbf{f} - \mathbf{f}_j\|_2$ 且 L_Δ 表征了该子空间位移的平滑度。由于 $G_{ij} \geq 0$ 且 $\sum_{j=1}^B G_{ij} > 0$ ；基于加权组合的向量三角不等式及局部平滑性假设，可得：

$$\begin{aligned} &\left\| (\mathbf{z}_r(\mathbf{x}) - \mathbf{f}) - \frac{\sum_{j=1}^B G_{ij} (\mathbf{z}_j - \mathbf{f}_j)}{\sum_{j=1}^B G_{ij}} \right\|_2 \\ &\leq \frac{\sum_{j=1}^B G_{ij} \|\mathbf{z}_r(\mathbf{x}) - \mathbf{f}\|_2 - \sum_{j=1}^B G_{ij} \|\mathbf{z}_j - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}} \\ &\leq L_\Delta \frac{\sum_{j=1}^B G_{ij} \|\mathbf{f} - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}}. \end{aligned} \quad (31)$$

此外，根据不等式 $\|\mathbf{f} - \mathbf{f}_j\|_2 \leq \|\mathbf{f} - \tilde{\mathbf{f}}_i\|_2 + \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2$ ，特征仿真误差有界于：

$$\begin{aligned} \|\mathbf{z}_r(\mathbf{x}) - \tilde{\mathbf{z}}_i\|_2 &\leq \\ (1 + L_\Delta) \|\mathbf{f} - \tilde{\mathbf{f}}_i\|_2 + L_\Delta \frac{\sum_{j=1}^B G_{ij} \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}}. \end{aligned} \quad (32)$$

随后，我们可得出风险的上界如下：

$$\begin{aligned} R_{s \rightarrow r}(g^r) &\leq \Pr[\tilde{m}_{s \rightarrow r}(\tilde{\mathbf{z}}, y) < \gamma] \\ &\quad + \frac{L_{gr}}{\gamma} \left\{ (1 + L_\Delta) \mathbb{E}[\|\mathbf{f} - \tilde{\mathbf{f}}_i\|_2] \right. \\ &\quad \left. + L_\Delta \mathbb{E} \left[\frac{\sum_{j=1}^B G_{ij} \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}} \right] \right\}. \end{aligned} \quad (33)$$

具体而言，第一项通过 $\mathcal{L}_{ca}$ 进行显式优化。项 $\mathbb{E}[\|\mathbf{f} - \tilde{\mathbf{f}}_i\|_2]$ 对应于高斯采样产生的近似误差。该项为 $\mathbb{E} \left[\frac{\sum_{j=1}^B G_{ij} \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}} \right]$ 该项衡量了在共享特征空间中，采样历史特征与当前任务特征之间的亲和力加权邻域半径；当历史任务与当前任务在局部语义上相关时，该第二项的值较小。高斯采样误差项 $\mathbb{E}[\|\mathbf{f} - \tilde{\mathbf{f}}_i\|_2]$ 当历史类别的共享特征能够很好地用高斯分布近似时，该值较小。尽管这一假设得到了现有研究[84]的支持，[85]我们希望提供进一步的证据，以分析这一做法为何合理。

对于任务 t 中的第 k 类，其共享特征分布可近似为 $G_{k,t} = \mathcal{N}(\boldsymbol{\mu}_{k,t}, \boldsymbol{\Sigma}_{k,t})$ 。多元高斯分布的一个基本性质是：若 $\mathbf{f} \sim \mathcal{N}(\boldsymbol{\mu}_{k,t}, \boldsymbol{\Sigma}_{k,t})$ ，则其到类中心的平方马氏距离服从卡方分布：

$$m(\mathbf{f}, \boldsymbol{\mu}_{k,t}) = (\mathbf{f} - \boldsymbol{\mu}_{k,t})^\top \boldsymbol{\Sigma}_{k,t}^{-1} (\mathbf{f} - \boldsymbol{\mu}_{k,t}) \sim \chi_d^2, \quad (34)$$

其中 d 表示特征维度。该性质为检验类别间共享特征是否符合假设的高斯分布提供了直接的诊断方法。具体而言，我们从10S-ImageNetR数据集中随机选取16个历史类

and construct chi-square Q-Q plots for their squared Mahalanobis distances. It compares the ordered empirical values of $m(\mathbf{f})$ with the corresponding theoretical quantiles of χ_r^2 . As shown in Fig.10, the points for the randomly selected classes are well aligned with the diagonal line, supporting that the Gaussian sampling error term is small.

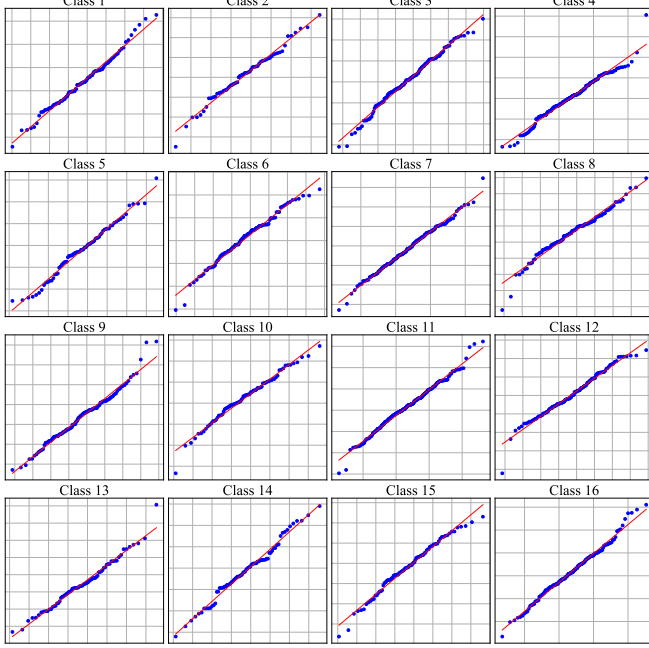


Fig. 10: Chi-square Q-Q plots of the squared Mahalanobis distances for 16 randomly selected classes on 10S-ImageNetA.

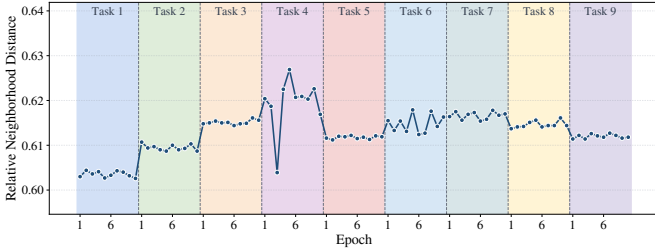


Fig. 11: Training dynamics of the relative neighborhood distance $\mathcal{R}_{\text{rel}}$.

Then, we further investigate the weighted neighborhood distance term $\mathbb{E} \left[\frac{\sum_{j=1}^B G_{ij} \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}} \right]$. Since the absolute magnitude of this term depends on the scale of features, its value alone cannot directly indicate whether affinity-guided transfer is effective. Therefore, we evaluate its relative magnitude with respect to the average distance obtained without affinity-guided weighting. Specifically, we define the relative neighborhood distance as follows:

$$\mathcal{R}_{\text{rel}} = \mathbb{E} \left[\frac{\sum_{j=1}^B G_{ij} \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}} \right] / \mathbb{E} \left[\frac{1}{B} \sum_{j=1}^B \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2 \right]. \quad (35)$$

$\mathcal{R}_{\text{rel}} < 1$ indicates that, compared with feature simulation without affinity guidance, the proposed affinity-guided transfer

reduces this term. We visualize the training dynamics of $\mathcal{R}_{\text{rel}}$ in Fig.11 on 10S-ImageNetA. The figure shows that $\mathcal{R}_{\text{rel}}$ remains consistently within 0.60–0.63 throughout continual training across different tasks. This result indicates that affinity-guided transfer reduces the neighborhood distance term to about 60% of that obtained without affinity guidance.

Therefore, global classifier training explicitly reduces the classification-related term, while Gaussian sampling and affinity-guided subspace shift transfer reduce the simulation-error term, thereby tightening the upper bound of $R_{s \rightarrow r}(g^r)$.

C. Theoretical analysis of TC-MoA

We further provide an theoretical explanation for why our proposed TC-MoA is more robust than deterministic maximum selection under confusing task boundaries. For a test sample $\mathbf{x}_i^t$ from task t , let $\hat{t} = \arg \max_{t'} \alpha_i^{t'}$ denote the predicted top-1 task-id. Given the test dataset, we define $\mathbb{P}(\hat{t} = t) = p$ and $\mathbb{P}(\hat{t} \neq t) = 1 - p$. For easy samples with correct task-IDs, we define the expected confidence score assigned to the ground-truth task as $\mathbb{E}[\alpha_i^t | \hat{t} = t] = \tau_1$. For challenging samples with misleading task-ids, although the predicted top-1 task is incorrect, the ground-truth task may still receive a relative high confidence. We define $\mathbb{E}[\alpha_i^t | \hat{t} \neq t] = \tau_2$. This formulation is consistent with the observation in Fig.12 on 10S-ImageNetA. For easy samples with correct task-IDs, the top-1 task-confidence scores are highly concentrated near 1. This observation directly indicates that $\tau_1 = \mathbb{E}[\alpha_i^t | \hat{t} = t]$ is close to 1. In contrast, for challenging samples with misleading task-IDs, the top-1 task-confidence scores are substantially lower. This indicates that the incorrectly selected adapter is generally not assigned large confidence.

To analyze how the perturbation introduced by adapter routing propagates to the final prediction, we make a standard Lipschitz smoothness assumption. Specifically, for the l -th Transformer block, we assume that the subsequent mapping from its adapter residual output to the final classification scores defined in Eq.13 is ρ_l -Lipschitz:

$$\begin{aligned} \mathcal{E}(\mathbf{x}_i^t) &= \left\| \frac{t_c}{t_c + 1} g^u(\hat{\mathbf{z}}_i) + \frac{1}{t_c + 1} g(\hat{\mathbf{z}}_i) \right. \\ &\quad - \frac{t_c}{t_c + 1} g^u(\mathcal{F}(\mathbf{x}_i^t; \mathcal{P}_{sh}, \mathcal{A}_{sp}^t)) \\ &\quad \left. - \frac{1}{t_c + 1} g(\mathcal{F}(\mathbf{x}_i^t; \mathcal{P}_{sh}, \mathcal{A}_{sp}^t)) \right\|_2 \\ &\leq (1 - \alpha_i^t) \max_{\mathbf{x}_i^t, t' \neq t} \sum_{l=1}^L \rho_l \|A_l^{t'}(\hat{\mathbf{X}}_{i,l}) - A_l^t(\hat{\mathbf{X}}_{i,l})\|_2. \end{aligned} \quad (36)$$

Therefore, the expected error upper bound of TC-MoA is

别，并为其平方马氏距离构建卡方 Q-Q 图。该方法将 $m(\mathbf{f})$ 的排序经验值与 χ^2 的相应理论分位数进行比较。如图.10所示，随机选取类别的数据点与对角线高度吻合，这表明高斯抽样误差项较小。

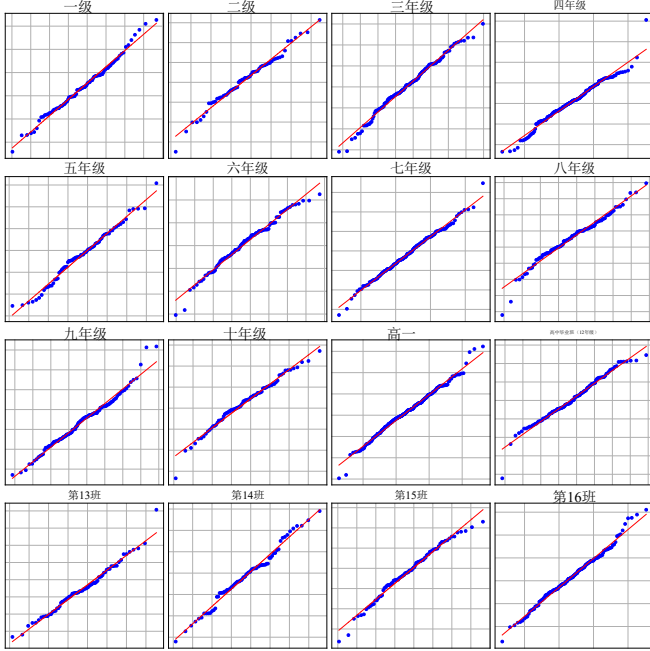


图 10: 在 10S-ImageNetR 数据集上对 16 个随机选取的类别计算的平方 Mahalanobis 距离的卡方 Q-Q 图。

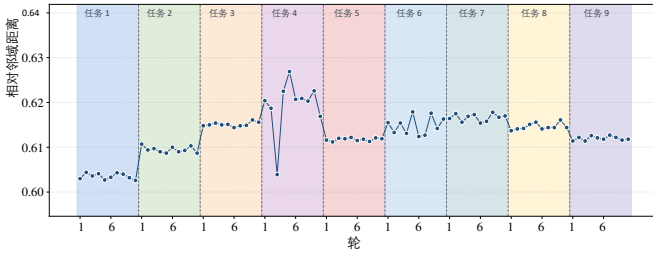


图 11: 相对邻域距离 R_{rel} 的训练动态变化。

随后，我们进一步研究加权邻域距离项 $\mathcal{S}\mathcal{E}\mathcal{S}$ 。
$$\left[\frac{\sum_{j=1}^B G_{ij} \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}} \right]$$
。由于该项的绝对值取决于特征的尺度，因此仅凭其数值无法直接判断亲和力引导迁移是否有效。因此，我们通过将其与未采用亲和力引导加权时获得的平均距离进行比较，来评估其相对大小。具体而言，我们定义相对邻域距离如下：

$$\mathcal{R}_{rel} = \mathbb{E} \left[\frac{\sum_{j=1}^B G_{ij} \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2}{\sum_{j=1}^B G_{ij}} \right] / \mathbb{E} \left[\frac{1}{B} \sum_{j=1}^B \|\tilde{\mathbf{f}}_i - \mathbf{f}_j\|_2 \right]. \quad (35)$$

$\mathcal{R}_{rel} < 1$ 表示，与未采用亲和力引导的特征模拟方法相比，所提

出的亲和力引导迁移方法降低了该项的值。我们将在 10S-ImageNetA 数据集上对 $\mathcal{R}_{rel}$ 的训练动态过程进行可视化展示，如图.11所示。该图显示，在跨任务的持续训练过程中， $\mathcal{R}_{rel}$ 始终保持在 0.60–0.63 的范围内。这一结果表明，亲和力引导迁移方法将邻域距离项降低至未采用亲和力引导时所得值的约 60%。

因此，全局分类器训练可显式地减小与分类相关的项，而高斯采样与亲和力引导的子空间偏移迁移则可减小仿真误差项，从而收紧 $R_s \rightarrow r(gr)$ 的上界。

C. TC-MoA 的理论分析

我们进一步从理论上解释了为何所提出的 TC-MoA 在存在混淆任务边界的情况下比确定性最大选择方法更具鲁棒性。对于来自任务 t 的测试样本 $\mathbf{x}t_i$ ，令 $\hat{t} = \arg \max_t \alpha t_i$ 表示预测结果 top-1 任务 ID。给定测试数据集，我们定义 $\mathcal{P}(\hat{t}=t) = p$ 且 $\mathcal{P}(t \neq \hat{t}) = 1 - p$ 。对于任务ID正确的简单样本，我们将分配给真实任务的期望置信度得分定义为 $\mathbb{E}[\alpha t_i | \hat{t}=t] = \tau_1$ 。对于任务ID存在误导性的复杂样本，尽管预测的 top-1 任务不正确，但真实任务仍可能获得相对较高的置信度。我们将 $\mathbb{E}[\alpha t_i | \hat{t} \neq t] = \tau_2$ 。这一表述与图.12中在 10S-

ImageNetA 数据集上的观测结果一致。对于任务ID正确的简单样本，top-1 任务置信度得分高度集中于 1 附近；这一现象直接表明 $\tau_1 = \mathbb{E}[\alpha t_i | \hat{t}=t]$ 接近于 1。相比之下，对于任务ID存在误导性的复杂样本，top-1 任务置信度得分则显著较低；这表明被错误选择的适配器通常未被赋予较高的置信度。

为分析适配器路由引入的扰动如何传播至最终预测结果，我们采用标准的Lipschitz平滑性假设。具体而言，对于第 l 个Transformer块，我们假设其从适配器残差输出到由等式 13 定义的最终分类分数之间的后续映射满足 ρ_l -Lipschitz条件：

$$\begin{aligned} \mathcal{E}(\mathbf{x}_i^t) &= \left\| \frac{t_c}{t_c + 1} g^u(\hat{\mathbf{z}}_i) + \frac{1}{t_c + 1} g(\hat{\mathbf{z}}_i) \right. \\ &\quad - \frac{t_c}{t_c + 1} g^u(\mathcal{F}(\mathbf{x}_i^t; \mathcal{P}_{sh}, \mathcal{A}_{sp}^t)) \\ &\quad \left. - \frac{1}{t_c + 1} g(\mathcal{F}(\mathbf{x}_i^t; \mathcal{P}_{sh}, \mathcal{A}_{sp}^t)) \right\|_2 \\ &\leq (1 - \alpha_i^t) \max_{\mathbf{x}_i^t, t' \neq t} \sum_{l=1}^L \rho_l \|A_l^t(\hat{\mathbf{X}}_{i,l}) - A_l^{t'}(\hat{\mathbf{X}}_{i,l})\|_2. \end{aligned} \quad (36)$$

因此，TC-MoA 的预期误差上界推导如下：

derived as follows:

$$\begin{aligned}
& \mathbb{E}[\mathcal{E}_{\text{TC-MoA}}(\mathbf{x}_i^t)] \\
&= p \mathbb{E}[\mathcal{E}_{\text{TC-MoA}}(\mathbf{x}_i^t) \mid \hat{t} = t] + (1-p) \mathbb{E}[\mathcal{E}_{\text{TC-MoA}}(\mathbf{x}_i^t) \mid \hat{t} \neq t] \\
&\leq [p \mathbb{E}[1 - \alpha_i^t \mid \hat{t} = t] + (1-p) \mathbb{E}[1 - \alpha_i^t \mid \hat{t} \neq t]] \\
&\quad \cdot \max_{\mathbf{x}_i^t, t' \neq t} \sum_{l=1}^L \rho_l \left\| A_l^{t'}(\hat{\mathbf{X}}_{i,l}) - A_l^t(\hat{\mathbf{X}}_{i,l}) \right\|_2 \\
&= [p(1 - \tau_1) + (1-p)(1 - \tau_2)] \\
&\quad \cdot \max_{\mathbf{x}_i^t, t' \neq t} \sum_{l=1}^L \rho_l \left\| A_l^{t'}(\hat{\mathbf{X}}_{i,l}) - A_l^t(\hat{\mathbf{X}}_{i,l}) \right\|_2.
\end{aligned} \tag{37}$$

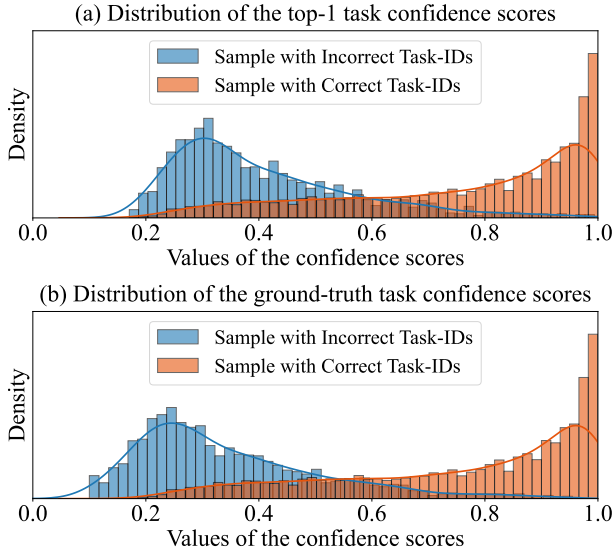


Fig. 12: Visualization of the distributions of the top-1 task-confidence scores (the upper figure) and the task-confidence scores of the ground-truth task (the lower figure).

In contrast, deterministic maximum selection introduces no routing-induced perturbation when $\hat{t} = t$, but fully commits to an incorrect adapter when $\hat{t} \neq t$. The expected error upper bound is formulated as follows:

$$\mathbb{E}[\mathcal{E}_{\text{Max}}] \leq (1-p) \max_{\mathbf{x}_i^t, t' \neq t} \sum_{l=1}^L \rho_l \left\| A_l^{t'}(\hat{\mathbf{X}}_{i,l}) - A_l^t(\hat{\mathbf{X}}_{i,l}) \right\|_2. \tag{38}$$

Therefore, under the following condition, TC-MoA achieves a tighter expected routing-induced error upper bound than deterministic maximum selection, *i.e.*, $\mathbb{E}[\mathcal{E}_{\text{TC-MoA}}(\mathbf{x}_i^t)] \leq \mathbb{E}[\mathcal{E}_{\text{Max}}(\mathbf{x}_i^t)]$:

$$p < \frac{\tau_2}{1 - \tau_1 + \tau_2}. \tag{39}$$

This condition is especially likely to hold when task boundaries are ambiguous. As shown in Fig. 12, on 10S-ImageNetA with ambiguous task boundaries, $1 - \tau_1$ is small, indicating that TC-MoA introduces negligible additional perturbation on samples with correctly predicted task-IDs. Moreover, the task-ID accuracy is relatively low ($p < 0.5$ on 10S-ImageNetA) but τ_2 remains at a moderate level, indicating that TC-MoA

TABLE XXI: Validation of the theoretical condition for TC-MoA.

Benchmark	Filtered	p	τ_1	τ_2	$\frac{\tau_2}{1 - \tau_1 + \tau_2}$	Satisfied
10S-ImageNetA	✗	0.541	0.843	0.247	0.611	✓ (0.611 > 0.541)
10S-ImageNetA	✓	0.096	0.546	0.253	0.358	✓ (0.358 > 0.096)
20S-ImageNetA	✗	0.484	0.815	0.224	0.548	✓ (0.548 > 0.484)
20S-ImageNetA	✓	0.042	0.476	0.208	0.284	✓ (0.284 > 0.042)
10S-ImageNetR	✗	0.847	0.953	0.313	0.869	✓ (0.869 > 0.847)
10S-ImageNetR	✓	0.030	0.541	0.310	0.403	✓ (0.403 > 0.030)
20S-ImageNetR	✗	0.830	0.916	0.292	0.777	✗ (0.777 < 0.830)
20S-ImageNetR	✓	0.067	0.680	0.275	0.462	✓ (0.462 > 0.067)

still preserves a relatively large contribution from the ground-truth adapter for these challenging samples. In this case, the condition is easily to be satisfied.

Empirical Validation. We empirically verify whether the theoretical condition $p < \tau_2/(1 - \tau_1 + \tau_2)$ holds across five representative benchmarks, including 10S-ImageNetA, 20S-ImageNetA, 10S-ImageNetR, 20S-ImageNetR, and 10S-CIFAR100. We report the results in Tab.XXI. “Filtered” in Tab.XXI denotes *excluding simple samples that can already be correctly classified using only the task-shared network*.

As shown in Tab.XXI, the condition is easily to be satisfied on ImageNetA, where its ambiguous task boundaries lead to low task-ID accuracy. In contrast, task-ID prediction on ImageNet-R is more accurate ($p > 0.8$), making the condition more stringent and causing it to fail on 20S-ImageNetR. However, our theoretical error quantifies score discrepancy rather than actual misclassification. For easy samples already correctly classified by the task-shared network, this discrepancy rarely changes the predicted class because they rely little on task-specific information and generally have sufficient classification margins.

Therefore, to remove this confounding effect and provide a cleaner comparison, we exclude these routing-insensitive samples, denoted as “Filtered”. After filtering, the condition is satisfied by a substantial margin across all benchmarks, further demonstrating that TC-MoA provides a clear advantage over deterministic selection for ambiguous samples whose predictions genuinely depend on task-specific information.

APPENDIX B ADDITIONAL EXPERIMENTAL RESULTS

A. Comparison with Traditional CL Methods

We compare CKAA with representative traditional CL methods under the ImageNet-21K supervised pre-trained ViT-B/16 backbone, including rehearsal-free regularization-based methods, *i.e.*, EWC [4] and LwF [5], and replay-based methods, *i.e.*, GDumb [86], DER++ [87], iCaRL [88], and BiC [89]. For replay-based methods, we use a memory buffer of 1,000 images, while the other methods do not store historical images. As shown in Tab. XXIII, CKAA consistently outperforms these non-PEFT CL baselines without exemplar replay. Compared with the strongest traditional baseline EWC in terms of Last-acc, CKAA improves the performance by 2.96% on 10S-CIFAR100 and 12.14% on 10S-ImageNetR. These results show that CKAA benefits from preserving the generalization ability of the pre-trained backbone via lightweight task-specific adapters.

$$\begin{aligned}
& \mathbb{E}[\mathcal{E}_{\text{TC-MoA}}(\mathbf{x}_i^t)] \\
&= p \mathbb{E}[\mathcal{E}_{\text{TC-MoA}}(\mathbf{x}_i^t) \mid \hat{t} = t] + (1-p) \mathbb{E}[\mathcal{E}_{\text{TC-MoA}}(\mathbf{x}_i^t) \mid \hat{t} \neq t] \\
&\leq [p \mathbb{E}[1 - \alpha_i^t \mid \hat{t} = t] + (1-p) \mathbb{E}[1 - \alpha_i^t \mid \hat{t} \neq t]] \\
&\quad \cdot \max_{\mathbf{x}_i^t, t' \neq t} \sum_{l=1}^L \rho_l \left\| A_l^{t'}(\hat{\mathbf{X}}_{i,l}) - A_l^t(\hat{\mathbf{X}}_{i,l}) \right\|_2 \\
&= [p(1 - \tau_1) + (1-p)(1 - \tau_2)] \\
&\quad \cdot \max_{\mathbf{x}_i^t, t' \neq t} \sum_{l=1}^L \rho_l \left\| A_l^{t'}(\hat{\mathbf{X}}_{i,l}) - A_l^t(\hat{\mathbf{X}}_{i,l}) \right\|_2.
\end{aligned} \tag{37}$$

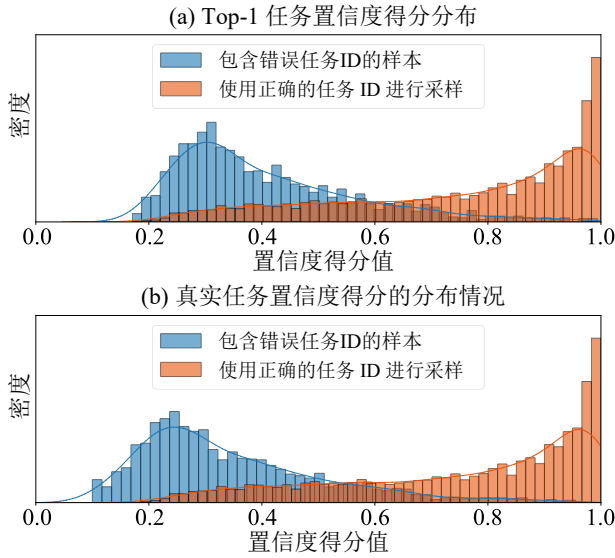


图 12: Top-1 任务置信度得分分布 (上图) 与真实任务置信度得分分布 (下图) 的可视化展示。

相比之下, 确定性最大选择并未引入任何 $\hat{t}$ 当 $t=t$ 时, 出现路由诱导的扰动, 但随后完全采用 $\hat{t}$ 时使用了错误的适配器 $\neq t$ 。期望误差上界表述如下:

$$\mathbb{E}[\mathcal{E}_{\text{Max}}] \leq (1-p) \max_{\mathbf{x}_i^t, t' \neq t} \sum_{l=1}^L \rho_l \left\| A_l^{t'}(\hat{\mathbf{X}}_{i,l}) - A_l^t(\hat{\mathbf{X}}_{i,l}) \right\|_2. \tag{38}$$

因此, 在以下条件下, TC-MoA 所达到的预期路由诱导误差上界比确定性最大选择法更小, $\mathbb{E}[\mathcal{E}_{\text{TC-MoA}}(\mathbf{x}_i^t)] \leq \mathbb{E}[\mathcal{E}_{\text{Max}}(\mathbf{x}_i^t)]$:

$$\tau_2 p < \frac{1}{1 - \tau_1 + \tau_2}. \tag{39}$$

当任务边界不明确时, 这一条件尤为容易成立。如图12所示, 在任务边界不明确的 10S-ImageNetA 数据集上, $1 - \tau_1$ 值较小, 表明对于任务ID被正确预测的样本, TC-MoA 引入的额外扰动可忽略不计。此外, 任务ID的预测准确率相对较低 (在 10S-ImageNetA 上 $p < 0.5$), 但 τ_2 仍保持在中等水平, 这表明 TC-MoA 表 XXI: TCMoA 理论条件的验

证。

基准	已过滤	p	τ_1	τ_2	$\frac{\tau_2}{1 - \tau_1 + \tau_2}$	满意
10S-ImageNetA	✗	0.541	0.843	0.247	0.611	✓ (0.611 > 0.541)
10S-ImageNetA	✓	0.096	0.546	0.253	0.358	✓ (0.358 > 0.096)
20S-ImageNetA	✗	0.484	0.815	0.224	0.548	✓ (0.548 > 0.484)
20S-ImageNetA	✓	0.042	0.476	0.208	0.284	✓ (0.284 > 0.042)
10S-ImageNetR	✗	0.847	0.953	0.313	0.869	✓ (0.869 > 0.847)
10S-ImageNetR	✓	0.030	0.541	0.310	0.403	✓ (0.403 > 0.030)
20S-ImageNetR	✗	0.830	0.916	0.292	0.777	✗ (0.777 < 0.830)
20S-ImageNetR	✓	0.067	0.680	0.275	0.462	✓ (0.462 > 0.067)

对于这些具有挑战性的样本, 该方法仍保留了来自真实适配器的相对较大的贡献度; 在这种情况下, 该条件很容易满足。

实证验证。我们通过实证检验理论条件 $p < \tau_2 / (1 - \tau_1 + \tau_2)$ 在五个代表性基准数据集 (包括 10S-ImageNetA、20S-ImageNetA、10S-ImageNetR、20S-ImageNetR 和 10SCIFAR 100) 上是否成立。相关结果列于表 XXI 中。表 XXI 中的 “Filtered” 表示 **排除那些仅使用任务共享网络即可正确分类的简单样本**。如表 XXI 所示, 在 ImageNetA 上该条件很容易满足——该数据集的模糊任务边界导致了较低的任务 ID 准确率; 相比之下, 在 ImageNet-R 上的任务 ID 预测精度更高 ($p > 0.8$), 这使得该条件更为严苛, 导致其在 20S-ImageNetR 上不成立。

然而, 我们的理论误差度量的是评分差异, 而非实际的误分类情况。对于那些已由任务共享网络正确分类的简单样本而言, 这种差异通常不会改变其预测类别, 因为这些样本对任务特定信息的依赖程度较低, 且通常具有足够的分类置信度。

因此, 为消除这一混杂效应并获得更清晰的对比结果, 我们剔除了这些对路由路径不敏感的样本 (标记为 “Filtered”)。经过筛选后, 该条件在所有基准测试中均以显著优势得到满足; 这进一步证明: 对于那些预测结果确实依赖于任务特定信息的模糊样本而言, TC-MoA 相较于确定性选择方法具有明显优势。

附录B: 额外实验结果

A. 与传统 CL 方法的对比

我们将 CKAA 与具有代表性的传统 CL 方法进行对比, 这些方法均基于 ImageNet-21K 监督预训练的 ViTB/16 骨干网络构建, 包括基于无 rehearsal 正则化的方法 (即 EWC[4] 和 LwF[5]) 以及基于 replay 的方法 (即 GDumb[86]、DER++[87]、iCaRL[88] 和 BiC[89])。对于基于 replay 的方法, 我们使用包含 1,000 张图像的记忆缓冲区, 而其他方法则不存储历史图像。如表 XXIII 所示, CKAA 在所有不采用示例 replay 的非 PEFT CL 基线方法中均表现最优。与最强的传统基线 EWC 在 Last-acc 指标上的表现相比, CKAA 在 10S-CIFAR100 数据集上提升了 2.96%, 在 10S-ImageNetR 数据集上提升了 12.14%。这些结果表明, 通过轻量级任务特定适配器来保留预训练骨干网络的泛化能力, 使 CKAA 获得了显著优势。

TABLE XXII: Experiments with ViT models with different sizes.

Models	Params	Methods	10s-ImageNetA		10s-ImageNetR	
			Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$
ViT-Tiny	5.5M	$\mathcal{P}_{sh}$ (Baseline)	21.02 $\pm$ 1.27	31.20 $\pm$ 1.66	47.82 $\pm$ 0.68	56.76 $\pm$ 0.77
		CKAA (Ours)	27.48 $\pm$ 0.42	39.54 $\pm$ 0.54	58.49 $\pm$ 0.14	67.65 $\pm$ 1.08
ViT-Small	22M	$\mathcal{P}_{sh}$ (Baseline)	37.85 $\pm$ 0.60	48.37 $\pm$ 0.99	63.61 $\pm$ 0.34	71.08 $\pm$ 0.20
		CKAA (Ours)	47.07 $\pm$ 0.40	57.60 $\pm$ 1.21	73.77 $\pm$ 0.15	81.05 $\pm$ 1.09
ViT-Base	86M	$\mathcal{P}_{sh}$ (Baseline)	53.83 $\pm$ 0.66	63.93 $\pm$ 0.66	77.95 $\pm$ 0.66	83.44 $\pm$ 0.66
		CKAA (Ours)	63.53 $\pm$ 0.66	71.11 $\pm$ 1.03	82.15 $\pm$ 0.47	87.21 $\pm$ 0.13
ViT-Base (CLIP)	86M	$\mathcal{P}_{sh}$ (Baseline)	51.07 $\pm$ 0.69	63.08 $\pm$ 0.88	80.55 $\pm$ 0.62	86.78 $\pm$ 0.32
		CKAA (Ours)	59.25 $\pm$ 0.33	69.63 $\pm$ 1.36	83.94 $\pm$ 0.46	89.42 $\pm$ 0.40

TABLE XXIII: Comparison with representative traditional CL methods on 10S-CIFAR100 and 10S-ImageNetR. For fair comparisons, we report baseline results under ImageNet-21K supervised pre-trained ViT-B/16 backbone.

Method	Buffer	10S-CIFAR100		10S-ImageNetR	
		Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$
GDumb [86]	1,000	81.92 $\pm$ 0.15	89.46 $\pm$ 0.94	24.23 $\pm$ 0.35	43.48 $\pm$ 0.49
DER++ [87]	1,000	84.50 $\pm$ 1.67	91.49 $\pm$ 0.61	67.75 $\pm$ 0.93	78.13 $\pm$ 1.14
iCARL [88]	1,000	85.68 $\pm$ 1.33	92.45 $\pm$ 0.98	63.75 $\pm$ 1.58	71.93 $\pm$ 0.45
BiC [89]	1,000	88.45 $\pm$ 0.57	93.37 $\pm$ 0.32	64.89 $\pm$ 0.80	73.66 $\pm$ 1.61
EWC [4]	0	89.30 $\pm$ 0.23	92.31 $\pm$ 1.66	70.27 $\pm$ 1.99	76.27 $\pm$ 2.13
LwF [5]	0	87.99 $\pm$ 0.05	92.13 $\pm$ 1.16	67.29 $\pm$ 1.67	74.47 $\pm$ 1.48
CKAA	0	92.26 $\pm$ 0.39	95.18 $\pm$ 0.86	82.41 $\pm$ 0.13	87.20 $\pm$ 0.35

B. Generality under Various Models and Settings

TABLE XXIV: Results of existing methods under various pre-trained models. Here, DINO-1k and iBOT-1k denote that the frozen backbone is pre-trained on ImageNet-1k through DINO and iBOT pre-training algorithms, respectively.

PTM	Method	Last-acc $\uparrow$	Avg-acc $\uparrow$
DINO-1k	L2P [20]	56.71 $\pm$ 0.12	63.59 $\pm$ 0.21
	DualPrompt [23]	60.23 $\pm$ 0.42	66.57 $\pm$ 0.25
	CODAPrompt [24]	64.02 $\pm$ 0.68	71.50 $\pm$ 0.42
	C-LoRA [90]	63.07 $\pm$ 0.36	68.09 $\pm$ 0.41
	LAE [62]	61.03 $\pm$ 0.27	69.89 $\pm$ 0.15
	HiDe-Prompt [84]	68.11 $\pm$ 0.18	71.70 $\pm$ 0.01
	InfLoRA [21]	68.31 $\pm$ 0.28	76.15 $\pm$ 0.05
	VPS-NSP [†] [3]	68.96 $\pm$ 0.94	76.22 $\pm$ 0.56
	CKAA (Ours)	75.19 $\pm$ 0.29	81.17 $\pm$ 0.21
iBOT-1k	L2P [20]	60.80 $\pm$ 0.35	66.58 $\pm$ 0.28
	DualPrompt [23]	63.78 $\pm$ 0.38	68.88 $\pm$ 0.16
	CODAPrompt [24]	68.02 $\pm$ 0.48	74.28 $\pm$ 0.47
	C-LoRA [90]	68.60 $\pm$ 0.07	73.47 $\pm$ 0.28
	LAE [62]	64.14 $\pm$ 0.29	72.59 $\pm$ 0.22
	HiDe-Prompt [84]	71.33 $\pm$ 0.21	73.62 $\pm$ 0.13
	InfLoRA [21]	71.84 $\pm$ 0.09	78.29 $\pm$ 0.09
	VPS-NSP [†] [3]	73.25 $\pm$ 0.78	79.65 $\pm$ 0.63
	CKAA (Ours)	77.96 $\pm$ 0.22	83.26 $\pm$ 0.15

TABLE XXV: Results under the class-imbalanced setting.

Methods	10S-ImageNetA-TI		10S-ImageNetR-TI	
	Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$
Only $\mathcal{P}_{sh}$ (Baseline)	46.57 $\pm$ 0.74	56.99 $\pm$ 0.51	70.54 $\pm$ 0.12	76.03 $\pm$ 0.74
$\mathcal{P}_{sh}$ + TC-MoA	48.45 $\pm$ 0.99	58.34 $\pm$ 0.05	72.17 $\pm$ 0.29	77.50 $\pm$ 0.95
$\mathcal{P}_{sh}$ + TC-MoA+FA	52.60 $\pm$ 0.99	60.44 $\pm$ 0.23	73.22 $\pm$ 0.52	78.03 $\pm$ 0.73
CKAA	58.18 $\pm$ 0.38	64.93 $\pm$ 0.15	75.24 $\pm$ 0.21	79.60 $\pm$ 0.80

Performances under various pre-trained models. We conduct experiments on 10s-ImageNetR with ViT-B/16 under various self-supervised pre-training methods, including DINO [98] and iBOT [99], following [21], [84]. The results are

TABLE XXVI: Results under the task-imbalance setting.

Methods	10S-ImageNetA-TI		10S-ImageNetR-TI	
	Last-acc $\uparrow$	Avg-acc $\uparrow$	Last-acc $\uparrow$	Avg-acc $\uparrow$
Only $\mathcal{P}_{sh}$ (Baseline)	52.86 $\pm$ 0.92	62.91 $\pm$ 1.07	77.16 $\pm$ 0.24	81.99 $\pm$ 0.44
$\mathcal{P}_{sh}$ + TC-MoA	55.69 $\pm$ 0.69	64.39 $\pm$ 0.45	79.37 $\pm$ 0.21	83.97 $\pm$ 0.24
$\mathcal{P}_{sh}$ + TC-MoA+FA	57.71 $\pm$ 0.83	66.29 $\pm$ 0.58	80.39 $\pm$ 0.37	84.39 $\pm$ 0.46
CKAA	62.58 $\pm$ 0.42	70.77 $\pm$ 0.79	82.15 $\pm$ 0.17	86.34 $\pm$ 0.31

TABLE XXVII: Comparison with existing Class-Incremental Semantic Segmentation (CISS) methods on Pascal VOC under the disjoint setting of 15-5 and 15-1 scenarios.

Method	15-5 (2 steps)			15-1 (6 steps)		
	1-15	16-20	All	1-15	16-20	All
CNN-based Methods						
EWC [4]	26.7	37.7	29.4	0.3	4.3	1.3
ILT [91]	63.2	39.5	57.3	3.7	5.7	4.2
MiB [92]	71.8	43.3	64.7	46.2	12.9	37.9
RECALL [93]	66.3	49.8	63.5	66.0	44.9	62.1
SPPA [94]	75.3	48.7	69.0	59.6	15.6	49.1
RBC [95]	75.1	49.7	69.9	61.7	19.5	51.6
Upper Bound	79.1	72.6	77.4	79.1	72.6	77.4
Transformer-based Methods						
MiB [92]	75.0	59.9	72.3	66.7	26.3	58.3
RBC [95]	77.7	59.1	74.0	69.0	28.4	60.5
INC [96]	81.6	62.2	77.6	81.4	57.1	76.2
MBS [97]	81.9	67.2	79.0	81.5	64.7	78.1
Task-shared Net	79.9	67.5	77.4	76.5	61.7	72.8
TC-MoA (Ours)	81.5	68.5	78.9	78.2	60.9	73.9
CKAA (Ours)	81.7	71.9	79.9	81.3	68.5	78.7
Upper Bound	84.8	80.7	83.8	84.8	80.7	83.8

shown in Tab.XXIV, which exhibit a performance degradation with a self-supervised pre-trained backbone compared to utilizing a backbone pre-trained in a supervised manner. However, our CKAA still outperforms existing PEFT-based methods. The results demonstrate that CKAA can be adapted to models with various pre-training schemes, highlighting its generalizability.

Performances under various model sizes. We evaluate CKAA with ViT backbones of different scales, including ViT-Tiny (5.5M), ViT-Small (22M), ViT-Base (86M), and the CLIP-pretrained ViT-Base backbone [100]. For each backbone, we compare CKAA with our $\mathcal{P}_{sh}$ baseline. As shown in Tab.XXII, CKAA consistently outperforms the baseline, demonstrating the effectiveness of CKAA across different model capacities.

Performances under imbalanced settings. We verify the effectiveness of CKAA under class-imbalanced setting by building two long-tailed variants of 10S-ImageNetA and 10S-ImageNetR, denoted as 10S-ImageNetA-LT and 10S-ImageNetR-LT. Specifically, we resample the training set of each class according to a long-tailed distri-

表 XXII：不同规模 ViT 模型的实验。

车型	参数	方法	10s-ImageNetA		10s-ImageNetR	
			最后到达个	Avg-acc ↑	最后到达个	Avg-acc ↑
ViT-Tiny	5.5M	P_{sh} (基线)	21.02 $\pm$ 1.27	31.20 $\pm$ 1.66	47.82 $\pm$ 0.68	56.76 $\pm$ 0.77
		CKAA (我们)	27.48 $\pm$ 0.42	39.54 $\pm$ 0.54	58.49 $\pm$ 0.14	67.65 $\pm$ 1.08
ViT-Small	22M	P_{sh} (基线)	37.85 $\pm$ 0.60	48.37 $\pm$ 0.99	63.61 $\pm$ 0.34	71.08 $\pm$ 0.20
		CKAA (我们)	47.07 $\pm$ 0.40	57.60 $\pm$ 1.21	73.77 $\pm$ 0.15	81.05 $\pm$ 1.09
ViT-Base	86M	P_{sh} (基线)	53.83 $\pm$ 0.66	63.93 $\pm$ 0.66	77.95 $\pm$ 0.66	83.44 $\pm$ 0.66
		CKAA (我们)	63.53 $\pm$ 0.66	71.11 $\pm$ 1.03	82.15 $\pm$ 0.47	87.21 $\pm$ 0.13
ViT-Base (CLIP)	86M	P_{sh} (基线)	51.07 $\pm$ 0.69	63.08 $\pm$ 0.88	80.55 $\pm$ 0.62	86.78 $\pm$ 0.32
		CKAA (我们)	59.25 $\pm$ 0.33	69.63 $\pm$ 1.36	83.94 $\pm$ 0.46	89.42 $\pm$ 0.40

表 XXIII：在 10S-CIFAR100 和 10S-ImageNetR 数据集上与代表性传统 CL 方法的对比结果。为确保公平性，我们报告了基于 ImageNet-21K 监督预训练 ViT-B/16 基础模型所得的基线结果。

方法	缓冲区	10S-CIFAR100		10S-ImageNetR	
		最后到达个	Avg-acc ↑	最后到达个	Avg-acc ↑
GDumb [86]	1,000	81.92 $\pm$ 0.15	89.46 $\pm$ 0.94	24.23 $\pm$ 0.35	43.48 $\pm$ 0.49
DER++ [87]	1,000	84.50 $\pm$ 1.67	91.49 $\pm$ 0.61	67.75 $\pm$ 0.93	78.13 $\pm$ 1.14
iCARL [88]	1,000	85.68 $\pm$ 1.33	92.45 $\pm$ 0.98	63.75 $\pm$ 1.58	71.93 $\pm$ 0.45
BiC [89]	1,000	88.45 $\pm$ 0.57	93.37 $\pm$ 0.32	64.89 $\pm$ 0.80	73.66 $\pm$ 1.61
EWC [4]	0	89.30 $\pm$ 0.23	92.31 $\pm$ 1.66	70.27 $\pm$ 1.99	76.27 $\pm$ 2.13
LwF [5]	0	87.99 $\pm$ 0.05	92.13 $\pm$ 1.16	67.29 $\pm$ 1.67	74.47 $\pm$ 1.48
CKAA	0	92.26 $\pm$ 0.39	95.18 $\pm$ 0.86	82.41 $\pm$ 0.13	87.20 $\pm$ 0.35

B. 在不同模型与场景下的普适性

表 XXIV：现有方法在不同预训练模型下的实验结果。其中，DINO-1k 和 iBOT-1k 表示：其冻结主干网络分别通过 DINO 和 iBOT 预训练算法在 ImageNet-1k 数据集上进行预训练。

PTM	方法	最后到达个	Avg-acc ↑
DINO-1k	L2P [20]	56.71 $\pm$ 0.12	63.59 $\pm$ 0.21
	DualPrompt [23]	60.23 $\pm$ 0.42	66.57 $\pm$ 0.25
	CODAPrompt [24]	64.02 $\pm$ 0.68	71.50 $\pm$ 0.42
	C-LoRA [90]	63.07 $\pm$ 0.36	68.09 $\pm$ 0.41
	LAE [62]	61.03 $\pm$ 0.27	69.89 $\pm$ 0.15
	HiDe-Prompt [84]	68.11 $\pm$ 0.18	71.70 $\pm$ 0.01
	InfLoRA [21]	68.31 $\pm$ 0.28	76.15 $\pm$ 0.05
	VPS-NSP [†] [3]	68.96 $\pm$ 0.94	76.22 $\pm$ 0.56
	CKAA (我们)	75.19 $\pm$ 0.29	81.17 $\pm$ 0.21
iBOT-1k	L2P [20]	60.80 $\pm$ 0.35	66.58 $\pm$ 0.28
	DualPrompt [23]	63.78 $\pm$ 0.38	68.88 $\pm$ 0.16
	CODAPrompt [24]	68.02 $\pm$ 0.48	74.28 $\pm$ 0.47
	C-LoRA [90]	68.60 $\pm$ 0.07	73.47 $\pm$ 0.28
	LAE [62]	64.14 $\pm$ 0.29	72.59 $\pm$ 0.22
	HiDe-Prompt [84]	71.33 $\pm$ 0.21	73.62 $\pm$ 0.13
	InfLoRA [21]	71.84 $\pm$ 0.09	78.29 $\pm$ 0.09
	VPS-NSP [†] [3]	73.25 $\pm$ 0.78	79.65 $\pm$ 0.63
	CKAA (我们)	77.96 $\pm$ 0.22	83.26 $\pm$ 0.15

表 XXV：类别不平衡数据集下的实验结果。

方法	10S-ImageNetA-TI		10S-ImageNetR-TI	
	最后到达个	Avg-acc ↑	最后到达个	Avg-acc ↑
仅 P_{sh} (基线)	46.57 $\pm$ 0.74	56.99 $\pm$ 0.51	70.54 $\pm$ 0.12	76.03 $\pm$ 0.74
P_{sh} + TC-MoA	48.45 $\pm$ 0.99	58.34 $\pm$ 0.05	72.17 $\pm$ 0.29	77.50 $\pm$ 0.95
P_{sh} + TC-MoA+FA	52.60 $\pm$ 0.99	60.44 $\pm$ 0.23	73.22 $\pm$ 0.52	78.03 $\pm$ 0.73
CKAA	58.18 $\pm$ 0.38	64.93 $\pm$ 0.15	75.24 $\pm$ 0.21	79.60 $\pm$ 0.80

在多种预训练模型下的性能评估。我们在 10s-ImageNetR 数据集上，采用 ViT-B/16 模型，针对多种自监督预训练方法（包括 DINO[98]和 iBOT[99]）进行实验，具体方法参照

[21]。[84]结果见表 XXVI：任务不平衡设置下的实验结果。

方法	10S-ImageNetA-TI		10S-ImageNetR-TI	
	最后到达个	Avg-acc ↑	最后到达个	Avg-acc ↑
仅 P_{sh} (基线)	52.86 $\pm$ 0.92	62.91 $\pm$ 1.07	77.16 $\pm$ 0.24	81.99 $\pm$ 0.44
P_{sh} + TC-MoA	55.69 $\pm$ 0.69	64.39 $\pm$ 0.45	79.37 $\pm$ 0.21	83.97 $\pm$ 0.24
P_{sh} + TC-MoA+FA	57.71 $\pm$ 0.83	66.29 $\pm$ 0.58	80.39 $\pm$ 0.37	84.39 $\pm$ 0.46
CKAA	62.58 $\pm$ 0.42	70.77 $\pm$ 0.79	82.15 $\pm$ 0.17	86.34 $\pm$ 0.31

表 XXVII：在 Pascal VOC 数据集上，针对 15-5 与 15-1 两种互斥场景，与现有类增量语义分割 (CISS) 方法进行对比。

方法	15-5 (2步)			15-1 (6步)		
	1-15、16-20、全部	1-15、16-20、全部	1-15、16-20、全部	1-15、16-20、全部	1-15、16-20、全部	1-15、16-20、全部
基于 CNN 的方法						
EWC [4]	26.7	37.7	29.4	0.3	4.3	1.3
ILT [91]	63.2	39.5	57.3	3.7	5.7	4.2
MiB [92]	71.8	43.3	64.7	46.2	12.9	37.9
召回[93]	66.3	49.8	63.5	66.0	44.9	62.1
SPPA [94]	75.3	48.7	69.0	59.6	15.6	49.1
RBC [95]	75.1	49.7	69.9	61.7	19.5	51.6
上界	79.1	72.6	77.4	79.1	72.6	77.4
基于变压器的方法						
MiB [92]	75.0	59.9	72.3	66.7	26.3	58.3
RBC [95]	77.7	59.1	74.0	69.0	28.4	60.5
INC [96]	81.6	62.2	77.6	81.4	57.1	76.2
MBS [97]	81.9	67.2	79.0	81.5	64.7	78.1
任务共享网络	79.9	67.5	77.4	76.5	61.7	72.8
TC-MoA (我们开发)	81.5	68.5	78.9	78.2	60.9	73.9
CKAA (我们)	81.7	71.9	79.9	81.3	68.5	78.7
上界	84.8	80.7	83.8	84.8	80.7	83.8

如表. XXIV所示，与采用监督方式预训练的骨干网络相比，使用自监督预训练骨干网络时模型性能会出现下降。然而，我们的 CKAA 方法仍优于现有的基于 PEFT 的方法。实验结果表明，CKAA 可适配于采用不同预训练方案的模型，充分体现了其良好的泛化能力。

在不同模型规模下的性能评估。我们使用不同规模的 ViT 基础架构对 CKAA 进行评估，包括 ViT-Tiny (5.5 M)、ViT-Small (22M)、ViT-Base (86M) 以及 CLIP 预训练的 ViT-Base 基础架构[100]。针对每种基础架构，我们将 CKAA 与我们的 P_{sh} 基线模型进行对比。如表. XXII所示，CKAA 在所有实验场景下均优于基线模型，这证明了 CKAA 在不同模型容量下的有效性。

在数据分布不均衡的场景下进行性能测试。我们通过构建 10S-ImageNetA 和 10S-ImageNetR 的两个长尾变体（分别记为 10S-ImageNetA-LT 和 10S-ImageNetR-LT）来验证 CKAA 在类别分布不均衡场景下的有效性。具体而言，我

bution. For the i -th class among C classes, the number of retained training samples is defined as $n_i = \max\left(1, \min\left(\left\lfloor N_{\max} \cdot \rho^{-\frac{i}{C-1}} \right\rfloor, N_i^{\text{ori}}\right)\right)$. In our experiments, we set $N_{\max} = 200$ and $\rho = 20$, resulting in a head-to-tail imbalance ratio of 20 : 1. The results are shown in Tab.XXVI. We further examine the effectiveness of our method under more realistic non-uniform task distributions by building two task-imbalanced variants of 10S-ImageNetA and 10S-ImageNetR, denoted as 10S-ImageNetA-TI and 10S-ImageNetR-TI. we randomly split all classes in the original datasets into 10 non-overlapping tasks with varying class numbers. The results are shown in Tab.XXVI. These results indicate that CKAA remains effective under the challenging imbalanced scenarios.

Performances on the Class-Incremental Semantic Segmentation (CISS) task. To further verify the generality of CKAA, we conduct experiments on Pascal VOC [101] under the disjoint 15-5 and 15-1 settings [92], [97], where the model first learns classes 1–15 and then incrementally learns classes 16–20. Following prior works [92], [96], [97], we report mIoU over old classes, new classes, and all 21 semantic classes including background. Following Transformer-based CISS work [97], we adopt an ImageNet-pretrained ViT [60] as the encoder and a lightweight Transformer-based segmentation decoder. We build the task-shared network by fully fine-tuning the backbone with cross-entropy and distillation losses, while training task-specific adapters via the cross-entropy loss. For inference, to make the task confidence score more semantically discriminative, we compute it only over the predicted foreground regions. As shown in Tab. XXVII, CKAA achieves the best overall performance among Transformer-based CISS methods. The ablation variants (*i.e.*, Task-shared Net and TC-MoA) also verify the effectiveness of task-confidence-based aggregation and DKA. The improvement is relatively moderate compared to the CIL setting because CISS has strong old-new task ambiguity: new images may contain old objects and old images may contain future objects, which (1) makes task-specific structures less suitable and (2) weakens the benefit of the Gaussian sampling and feature simulation in CA.

们根据长尾分布对每个类别的训练集进行重采样。对于 C 个类别中的第 i 个类别,保留的训练样本数量定义为 $n_i = \max\left(1, \min\left(\left\lfloor N_{\max} \cdot \rho^{-\frac{i}{C-1}} \right\rfloor, N_i^{\text{ori}}\right)\right)$ 。In our experiments, we set $N_{\max} = 200$ and $\rho = 20$, resulting in a head-tail 不平衡比为 20:1。实验结果如表XXVI所示。为进一步检验本方法在更贴近实际的非均匀任务分布下的有效性,我们构建了 10S-ImageNetA 和 10S-ImageNetR 的两个任务不平衡变体,分别记作 10S-ImageNetA-TI 和 10S-ImageNetR-TI。我们将原始数据集中的所有类别随机划分为 10 个互不重叠的任务,各任务所含类别数量不等。实验结果如表XXVI所示。这些结果表明,CKAA 在具有挑战性的不平衡数据场景下依然有效。

在类增量语义分割 (CISS) 任务上的性能评估。为进一步验证 CKAA 的泛化能力,我们在 Pascal VOC[101]数据集上开展了实验,采用 disjoint 15-5 和 15-1 两种数据划分设置[92]。[97]该模型首先学习类别 1–15,随后逐步学习类别 16–20。遵循先前的研究[92], [96], [97]我们报告了针对旧类别、新类别以及包含背景在内的全部 21 个语义类别计算的 mIoU 值。借鉴基于 Transformer 的 CISS 研究[97],我们采用经过 ImageNet 预训练的 ViT[60]作为编码器,并使用一个轻量级基于 Transformer 的分割解码器。我们通过使用交叉熵损失和蒸馏损失对主干网络进行全量微调,同时利用交叉熵损失训练任务特定适配器,从而构建出任务共享网络。在推理阶段,为使任务置信度评分更具语义区分度,我们仅对预测的前景区域计算该评分。如表XXVII所示,CKAA 在所有基于 Transformer 的 CISS 方法中取得了最佳的整体性能。消融实验变体 (Task-shared Net 和 TCMoA) 也验证了基于任务置信度的聚合方法及 DKA 的有效性。与 CIL 设置相比,这一改进幅度相对有限,这是因为 CISS 具有明显的旧新任务模糊性:新图像可能包含旧物体,而旧图像可能包含未来出现的物体;这(1)使得任务特定结构不太适用,(2)削弱了 CA 中高采样与特征模拟带来的优势。